STABILITY AND EXPONENT TRANSITIONS FOR FRACTIONAL LAWSON CONES

ABSTRACT. We determine when stationary fractional Lawson cones in $\mathbb{R}^m \times \mathbb{R}^n$ are stable, for every $m, n \geq 1$ and $0 < s < 1$. We use smooth normal variations with compact support away from the vertex. Write $N = m+n$. The planar cross has a unique threshold $s_2 \in (0.9167, 0.9168)$ and is stable exactly for $s \geq s_2$. We give an explicit beta-function formula for its sharp weighted spectral gap. Every minimal Lawson cone is strictly unstable for $3 \leq N \leq 6$. In dimension seven, there are three families up to interchange of the factors. Each has a unique stability threshold. The gap has a simple zero there and changes from positive to negative. Each cone is stable exactly below or at its threshold. For $N \geq 8$, every family is stable with a positive Hardy–curvature gap and weighted coercivity. This includes cones with a one-dimensional factor. The main step is to combine two radial terms before expanding them. The resulting gamma–moment series keeps the cancellation in the gap and separates the exponent from the angular geometry. We then compare spherical moments to find its sign. In dimension seven, we also use the decrease of the minimal aperture and estimates for the derivative of the gap. We prove the bounds for the planar threshold and all finite estimates for the constants by analytic arguments.

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1. Introduction

Minimal cones describe the possible shapes of an area-minimizing hypersurface near a singular point. Their stability is important for regularity theory and the Bernstein problem. Simons proved that every stable minimal hypercone that is smooth away from its vertex is flat in ambient dimensions $3 \le N \le 7$. He also found the stable cone over $\mathbb{S}^3 \times \mathbb{S}^3$ in $\mathbb{R}^8$ [15]. Bombieri, De Giorgi, and Giusti later proved that this cone minimizes area. They also constructed nonaffine entire minimal graphs over $\mathbb{R}^{N-1}$ for $N \ge 9$ [1].

The Lawson family also allows the two Euclidean factors to have different dimensions. For positive integers m, n with $N = m + n$, consider

$$C_{m,n}(\alpha) = \{(y, z) \in \mathbb{R}^m \times \mathbb{R}^n : |z| = \alpha |y|\}, \qquad \alpha > 0. \tag{1.1}$$

For classical area and $m, n \ge 2$, the minimal aperture is $\alpha^{\mathrm{cl}}_{m,n} = \sqrt{(n-1)/(m-1)}$. Lawson used symmetry to study the Plateau problem [12]. His work and later calibration arguments [9] determine which classical cones with $m, n \ge 2$ minimize area. All such minimal product cones minimize area for $N \ge 9$. In $N = 8$, the minimizing pairs are $(3, 5)$, $(4, 4)$, and $(5, 3)$. A stable cone need not minimize area. For example, the $(2, 6)$ cone and the cone with its factors interchanged are stable but do not minimize area [9, 18]. De Philippis and Maggi [10] proved quadratic bounds for the perimeter deficit and explicit lower bounds for the first eigenvalue of the stability operator on truncated minimizing Lawson cones. These quantitative estimates excluded six pairs, up to factor interchange. Liu completed those estimates in the remaining cases [13].

For fractional perimeter, both minimality and stability also depend on the interaction exponent. We use the kernel $|x - y|^{-N-s}$, $0 < s < 1$. Dávila, del Pino, and Wei proved that there is a unique aperture $\alpha_{m,n}(s) > 0$ for which (1.1) has zero nonlocal mean curvature away from the origin [8, Theorem 3]. This result also allows a one-dimensional factor. In this paper, a minimal Lawson cone means this stationary cone. We call it stable if its second variation is nonnegative for every smooth normal velocity with compact support away from the vertex. We call it strictly unstable if one such velocity gives a negative second variation.

Regularity theory gives another reason to distinguish stationary, stable, and minimizing cones. Caffarelli, Roquejoffre, and Savin proved existence, improvement of flatness, and dimension reduction for fractional-perimeter minimizers [4]. Savin and Valdinoci proved that every nontrivial minimizing cone in $\mathbb{R}^2$ is a half-plane for every $0 < s < 1$ [16, Theorem 1]. Caffarelli and Valdinoci proved that minimizing cones are flat in ambient dimensions at most seven when s is close enough to one [5, Theorem 2]. For stable cones with smooth boundary away from the origin, Cabré, Cinti, and Serra proved flatness in $\mathbb{R}^3$ when the exponent is close to one [2, Theorem 1.2]. Chan, Dipierro, Serra, and Valdinoci proved the corresponding result in $\mathbb{R}^4$ [7]. For sufficiently small s, Caselli proved flatness of stable planar cones [6]. Tsiamis recently proved the corresponding result in dimension three for stable cones with C^2 boundary away from the vertex [17, Theorem 1.1].

The study of stable cones is also related to phase-transition equations. Cabré, Cinti, and Serra proved the following implication for $N \ge 3$ and fixed $0 < s < 1$. Suppose every stable s-minimal cone with nonempty boundary smooth away from the vertex is flat in every dimension $3, \dots, N$. Stability here uses perturbations supported away from the vertex. Then every stable entire solution in $\mathbb{R}^N$ of $(-\Delta)^{s/2} u = u - u^3$, $|u| < 1$, has the form $u(x) = g(e \cdot x)$ for some unit vector e and increasing function $g : \mathbb{R} \to (-1, 1)$ [3, Definition 1.6 and Theorem 1.8]. The cone hypothesis starts in dimension three, so it does not require flatness of stable planar crosses [3,

Remark 1.7]. Felipe-Navarro and Sanz-Perela proved stability of the symmetric Simons cone by passing from stable saddle-shaped solutions to their blow-down limit [11, Corollary 1.7]. Their result uses $0 < \gamma < 1/2$ and perimeter exponent 2γ. Thus $s = 2\gamma$ gives every $0 < s < 1$ when $N = 2m \geq 14$.

For fractional Lawson cones, Dávila, del Pino, and Wei gave a sharp Hardy criterion for stability. For sufficiently small positive s, their Theorem 4 states instability in dimensions three through six and stability in dimension seven. The sign comparisons in Section 11.3 use numerical approximations of the limiting integral constants at $s = 0$, followed by continuity [8]. For $(m, n) = (4, 3)$, they conjectured a single stability threshold [8, Remark 11.2]. These small-exponent comparisons and the symmetric examples in high dimensions do not determine the full stability intervals for asymmetric cones.

We determine the stability intervals by analytic arguments for every pair of factors. In particular, we prove the threshold conjecture for $(4, 3)$, establish the corresponding transitions for $(5, 2)$ and $(6, 1)$, and show that all three crossings are simple. The classification below also includes the planar cross and all one-dimensional factors. The coordinate swap $(y, z) \mapsto (z, y)$ takes $C_{m,n}(\alpha)$ to $C_{n,m}(\alpha^{-1})$. It preserves distances, surface measure, and normal inner products in (2.4), so it preserves the gap and stability. We therefore list the seven-dimensional pairs with $m > n$.

We use the following precise meaning of the gap. For a minimal Lawson cone Σ, let H be its sharp Hardy constant and A_{curv} its curvature coefficient, as defined in (2.2) and (2.3). The form Q_s in (2.4) is one half of the perimeter second variation, and $d\sigma$ is surface measure. Proposition 2.1 gives

$$H - A_{\mathrm{curv}} = \inf_{0 \neq \varphi \in C_c^\infty(\Sigma \setminus \{0\})} \frac{Q_s(\varphi)}{\displaystyle\int_\Sigma |x|^{-1-s} \varphi^2 \, d\sigma}.$$

Thus a negative gap gives an admissible destabilizing variation, a zero gap gives stability without a positive weighted coercivity constant, and a positive gap gives weighted coercivity. This sharp criterion includes $N = 2$.

Theorem 1.1 (Stability classification)**.** *Let $m, n \geq 1$, $N = m + n$, and $0 < s < 1$. Write $\Sigma_s = C_{m,n}(\alpha_{m,n}(s))$ for the unique s-minimal Lawson cone in (1.1), where $\alpha_{m,n}(s)$ is its minimal aperture. Stability refers to the quadratic form Q_s in (2.4) on $C_c^\infty(\Sigma_s \setminus \{0\})$, and $d\sigma$ denotes surface measure on the cone.*

(1) *If $N = 2$, the minimal cone is $C_{1,1}(1)$. There is a unique*
$$\frac{9167}{10000} < s_2 < \frac{9168}{10000}$$
such that this cone is strictly unstable for $s < s_2$ and stable for $s \geq s_2$. The threshold is the unique zero of the explicit beta-function gap in (3.2).

(2) *If $3 \leq N \leq 6$, every minimal Lawson cone is strictly unstable.*

(3) *If $N = 7$, then, up to interchanging the factors, for each $(m, n) = (4, 3), (5, 2), (6, 1)$ there is a unique*
$$\sigma_{m,n} \in (0, 7 - 2\sqrt{10})$$
such that $C_{m,n}(\alpha_{m,n}(s))$ is stable exactly for $0 < s \leq \sigma_{m,n}$. Its Hardy–curvature gap has a simple downward zero at this threshold.

(4) *If $N \geq 8$, every minimal Lawson cone is stable. More precisely, its Hardy–curvature gap is strictly positive. Equivalently, there is $\delta_{m,n}(s) > 0$ with*
$$Q_s(\varphi) \geq \delta_{m,n}(s) \int_{C_{m,n}(\alpha_{m,n}(s))} |x|^{-1-s} \varphi^2 \, d\sigma$$

for every admissible normal velocity, where Q_s is the quadratic form in (2.4). This includes cones with a one-dimensional factor.

The proof of Theorem 1.1 is given at the end of Section 6, using the four dimensional theorems. The stable planar cross also illustrates the distinction from minimization: planar minimizing cones are half-planes [16, Theorem 1].

For $N \geq 3$, we pair the radial terms by inversion before expanding them. The resulting gamma–moment series converges absolutely, although the two separate series need not converge. The gamma coefficients depend only on N and s; the angular moments depend on the factor dimensions and the aperture. On the minimal branch, the aperture also depends on s. This dependence must be included when differentiating the gap.

For $N = 2$, a direct beta-function formula and a positive integral representation show that the gap strictly increases. In dimensions three through six, we use the coefficient signs when they are all nonpositive. For the remaining exponents, we first bound them above by nonnegative coefficients and then apply upper bounds for the angular moments. The $(4, 2)$ endpoint uses an exact Jacobi recurrence. In dimension six, keeping the first-term correction in the scalar majorant closes the estimate.

In dimension seven, a one-crossing comparison of spherical exponential averages shows that the minimal aperture decreases with s. A Bessel comparison gives a positive aperture derivative for an extension of the link-normalized gap, evaluated on the minimal branch. The moving aperture therefore contributes a negative term to the derivative along that branch. Bounds at fixed aperture make a suitable normalized gap strictly decrease for $0 < s < 7-2\sqrt{10}$. Its limiting value at zero is positive. A separate Bessel estimate makes the gap negative for $7-2\sqrt{10} \leq s < 1$, so no later zero is possible. The positive normalization factors preserve zeros. At a zero, their derivatives multiply the vanishing gap, so they also preserve the negative, nonzero derivative.

For $N \geq 9$, contraction of spherical moment tensors gives nondecreasing ratios relative to spherical reference moments. The coefficients change sign at most once, from nonpositive to nonnegative. These two facts allow comparison with an explicit positive scalar hypergeometric sum. In dimension eight, the positive terms use a lower moment bound, while the negative terms use an upper moment bound. A lower bound for the scalar reference sum and a separate bound for its negative part give a strictly positive gap. The finite rational checks for these bounds and for the planar threshold are included in the appendices.

The paper is organized as follows. Section 2 gives the common Hardy criterion, the comparison for the minimal aperture, and the paired moment expansion. Section 3 treats $N = 2$. It computes the gap for the planar cross and bounds its unique threshold. Section 4 proves instability for $3 \leq N \leq 6$ by upper angular bounds and estimates that make the gap negative. Section 5 studies the minimal aperture in dimension seven and proves that each stability transition exists and is unique. Section 6 proves stability and weighted coercivity for $N \geq 8$. Dimension eight has a separate proof. Appendices A and B give the finite checks for the scalar bounds. Appendix C gives an additional angular comparison in dimension seven. Appendix D records the exact rational verification of the planar threshold bounds.

2. Preliminaries and the stability criterion

We write Γ for Euler's gamma function, $B(u, v) = \Gamma(u)\Gamma(v)/\Gamma(u + v)$ for the beta function, $\psi = \Gamma'/\Gamma$ for the digamma function, and γ for Euler's constant. The rising factorial is $(x)_0 = 1$ and $(x)_k = x(x + 1)\cdots(x + k - 1)$ for $k \geq 1$. In estimates, $K > 0$ denotes a constant that may change from line to line. We state explicitly when bounds are uniform in the parameters.

2.1. The second variation and the stability gap.

Fix integers $m, n \geq 1$ and set $N = m + n$. We study the cones (1.1) with kernel $|x-y|^{-N-s}$, $0 < s < 1$. We first allow any positive aperture.

Write $\Sigma = C_{m,n}(\alpha)$, $\Lambda = \Sigma \cap \mathbb{S}^{N-1}$, and $L_\Sigma = \mathcal{H}^{N-2}(\Lambda)$. For surface measure on Σ, we use $d\sigma$. Thus $d\sigma(r\omega) = r^{N-2}\, dr\, d\sigma_\Lambda(\omega)$. We choose the unit normal ν that points toward the phase $\{(y,z) : |z| > \alpha|y|\}$.

The correlation between the normals determines the curvature potential. For any $p \in \Lambda$, define

$$\beta = \frac{N-2-s}{2}, \qquad q = \frac{N+s}{2}, \tag{2.1}$$

$$H = \text{p. v.} \int_\Sigma \frac{1-|x|^{-\beta}}{|x-p|^{N+s}}\, d\sigma(x), \tag{2.2}$$

$$A_{\text{curv}} = \int_\Sigma \frac{1-\nu(x)\cdot\nu(p)}{|x-p|^{N+s}}\, d\sigma(x). \tag{2.3}$$

The group $O(m)\times O(n)$ acts transitively on the link, so these constants do not depend on p. At $x \neq 0$, the curvature potential is $A_{\text{curv}}|x|^{-1-s}$. We use the quadratic form

$$Q_s(\varphi) = \frac{1}{2}\iint_{\Sigma\times\Sigma} \frac{(\varphi(x)-\varphi(y))^2}{|x-y|^{N+s}}\, d\sigma(x)d\sigma(y) - A_{\text{curv}}\int_\Sigma \frac{\varphi^2}{|x|^{1+s}}\, d\sigma, \quad \varphi \in C_c^\infty(\Sigma\setminus\{0\}). \tag{2.4}$$

For a minimal cone, this is one half of the perimeter second variation in [8]. The cone is stable exactly when this form is nonnegative on the stated class of functions.

Proposition 2.1 (Sharpness and admissible variations)**.** *For every positive aperture, H is the best constant in the Hardy inequality for the first term of (2.4). Thus a minimal Lawson cone satisfies*

$$\Sigma \text{ is stable} \quad\Longleftrightarrow\quad H - A_{\text{curv}} \geq 0. \tag{2.5}$$

If $H - A_{\text{curv}} < 0$, a smooth normal velocity supported in a finite annulus has negative second variation. If $H - A_{\text{curv}} > 0$, the form is weighted coercive with constant $H - A_{\text{curv}}$. At equality, it is nonnegative. However, no positive constant gives a uniform lower bound with the same weighted norm.

Proof. Set $w(x) = |x|^{-\beta}$. In a tangent graph at a point of the link, the linear part of $w(x) - w(y)$ is odd in the tangent variables. After its cancellation, the local radial bound for the principal value is $K\int_0^\epsilon r^{-s}\, dr$. At zero and infinity, the terms containing $w(y)$ have radial orders r^{q-1} and r^{-q-1}, respectively, and the terms without $w(y)$ are also integrable. Thus the principal value exists, including when $N = 2$. The curvature integral is absolutely convergent as well: near p, $1 - \nu(x)\cdot\nu(p) = |\nu(x) - \nu(p)|^2/2 = O(|x-p|^2)$, and its radial tails are integrable because the normal is bounded. Scale (2.2) and use the transitive group action on the link. This gives

$$\text{p. v.} \int_\Sigma \frac{w(x)-w(y)}{|x-y|^{N+s}}\, d\sigma(y) = H|x|^{-1-s}w(x).$$

Write $\varphi = w\zeta$. Expand the difference of squares and symmetrize to obtain

$$Q_s(\varphi) = (H - A_{\text{curv}})\int_\Sigma \frac{\varphi^2}{|x|^{1+s}}\, d\sigma + \frac{1}{2}\iint_{\Sigma\times\Sigma} \frac{w(x)w(y)(\zeta(x)-\zeta(y))^2}{|x-y|^{N+s}}\, d\sigma(x)d\sigma(y). \tag{2.6}$$

To justify the identity, first impose the symmetric truncation $|x-y| > \epsilon$. The algebraic identity used in the expansion is

$$(w(x)\zeta(x) - w(y)\zeta(y))^2 - w(x)w(y)(\zeta(x)-\zeta(y))^2 = (w(x)-w(y))(w(x)\zeta(x)^2 - w(y)\zeta(y)^2).$$

After multiplication by $1/2$, symmetry turns its integral into the truncated operator applied to w, multiplied by φ^2/w. The truncated operator converges uniformly on the compact support of φ to the principal value above. The quadratic remainder is locally integrable because its numerator is $O(|x-y|^2)$ and $s < 1$. The tail bounds above apply when one point stays in the compact

support of ζ. They give an integrable bound for the remainder and justify the identity also when $N = 2$, where $\beta = -s/2$.

To find the best constant, choose $0 \neq \eta \in C_c^\infty(\mathbb{R})$. Set $\varphi_h(x) = w(x)\eta(\log|x|/h)$, $h > 0$. Its weighted mass is

$$\int_\Sigma |x|^{-1-s}\varphi_h^2\, d\sigma = L_\Sigma h\|\eta\|_2^2.$$

In logarithmic radial coordinates the last term of (2.6) is

$$\frac{1}{2}\iint_{\mathbb{R}^2}[\eta(t/h) - \eta(u/h)]^2 K_\Lambda(t-u)\, dt\, du, \qquad K_\Lambda(v) = \iint_{\Lambda\times\Lambda}[2(\cosh v - \omega\cdot\theta)]^{-q}\, d\sigma_\Lambda(\omega) d\sigma_\Lambda(\theta).$$

The radial powers cancel because $N - 1 - \beta - q = 0$. Put $d = N - 2$. If $d \geq 1$, compactness and smoothness of the link give uniform local coordinates with $2(\cosh v - \omega\cdot\theta) \asymp v^2 + |\omega - \theta|^2$. For $0 < |v| < 1$, the near-diagonal part is therefore bounded by

$$K\int_0^1 \frac{\rho^{d-1}}{(v^2+\rho^2)^q}\, d\rho \leq K|v|^{d-2q} = K|v|^{-2-s}.$$

Here the change of variables $\rho = |v|z$ leaves an integral bounded independently of v, since $2q > d$. The off-diagonal part is bounded. This also covers links with several components. When $d = 0$, the link is finite. Its diagonal terms are $O(|v|^{-2q}) = O(|v|^{-2-s})$, and its other terms are bounded. For $|v| \geq 1$, the kernel decays exponentially. Hence $J_\Lambda := \int_{\mathbb{R}} v^2 K_\Lambda(v)\, dv < \infty$. The translation inequality in L^2 gives

$$\int_{\mathbb{R}}[\eta(t/h) - \eta((t-v)/h)]^2\, dt \leq \frac{v^2}{h}\|\eta'\|_2^2.$$

The remainder is therefore at most $J_\Lambda\|\eta'\|_2^2/(2h)$. Divide by the weighted mass to see that the constant is best possible. If $H < A_{\mathrm{curv}}$, any h with

$$h^2 > \frac{J_\Lambda\|\eta'\|_2^2}{2(A_{\mathrm{curv}} - H)L_\Sigma\|\eta\|_2^2}$$

gives $Q_s(\varphi_h) < 0$. Use a tubular neighborhood of the annular support and an ambient cutoff to extend $\varphi_h\nu$ to a smooth compactly supported ambient vector field away from the vertex. The same extension applies to every $\varphi \in C_c^\infty(\Sigma\setminus\{0\})$, so the identity controls all the normal velocities in the definition of stability. This gives an admissible variation. The same cutoffs rule out a positive weighted coercivity constant when $H = A_{\mathrm{curv}}$. □

2.2. **Position correlations and the minimal branch.** Assume $N \geq 3$ for the rest of this subsection. We write all angular averages as weighted integrals. For $d \geq 2$, set

$$w_d(u) = \frac{\Gamma(d/2)}{\sqrt{\pi}\,\Gamma((d-1)/2)}(1-u^2)^{(d-3)/2}, \qquad -1 < u < 1, \qquad \mathcal{I}_d[f] = \int_{-1}^1 f(u)w_d(u)\,\mathrm{d}u. \tag{2.7}$$

The weight is even and its integral is one. Slicing the sphere by its first coordinate shows that this is the normalized surface integral of a function of that coordinate. In dimension one, set

$$\mathcal{I}_1[f] = \frac{f(-1) + f(1)}{2}.$$

For a function of two coordinates, apply the corresponding integral or half-sum in each coordinate. When $m, n \geq 2$, this is

$$\mathcal{I}_{m,n}[F] = \int_{-1}^1\int_{-1}^1 F(u,v)w_m(u)w_n(v)\,\mathrm{d}v\,\mathrm{d}u. \tag{2.8}$$

For example, a one-dimensional second factor gives

$$\mathcal{I}_{m,1}[F] = \frac{1}{2}\mathcal{I}_m[F(u,-1) + F(u,1)].$$

The same rule applies to a one-dimensional first factor. The beta integral gives, for every $d \geq 1$,

$$\mathcal{I}_d[u^{2j}] = \frac{(1/2)_j}{(d/2)_j}, \qquad \mathcal{I}_d[u^{2j+1}] = 0, \qquad j \geq 0. \tag{2.9}$$

For $d \geq 2$, the first identity follows by substituting $x = u^2$; for $d = 1$, it follows from the half-sum.

Set $b = \lambda = \alpha^2/(1+\alpha^2)$ and $a = 1 - \lambda$. On the minimal branch, we write $\lambda(s) = \alpha_{m,n}(s)^2/(1+\alpha_{m,n}(s)^2)$. We use u and v for the first coordinates in the two sphere factors. For $p = (\sqrt{a}\, e_1, \sqrt{b}\, e_1)$, the functions

$$Z = au + bv, \qquad T = bu + av \tag{2.10}$$

give the position correlation and the normal correlation, respectively. Indeed, a point on the link and its normal are $\omega = (\sqrt{a}\,\omega_1, \sqrt{b}\,\omega_2)$ and $\nu(\omega) = (-\sqrt{b}\,\omega_1, \sqrt{a}\,\omega_2)$. In particular $|r\omega - p|^2 = 1 + r^2 - 2rZ$. The link area is

$$L_\Sigma = |\mathbb{S}^{m-1}|\,|\mathbb{S}^{n-1}| a^{(m-1)/2} b^{(n-1)/2}, \qquad |\mathbb{S}^0| = 2.$$

Proposition 2.2 (Paired integral representation)**.** *Minimality is equivalent to*

$$G(s,\lambda) := \int_0^\infty \mathcal{I}_{m,n}[(v-u)\cosh(\beta t)(\cosh t - Z)^{-q}]\, dt = 0. \tag{2.11}$$

Define the auxiliary function

$$\Phi(s,\lambda) = \int_0^\infty \mathcal{I}_{m,n}[(Z\cosh(\beta t) - 1)(\cosh t - Z)^{-q}]\, dt. \tag{2.12}$$

On the minimal branch, the normalized gap satisfies

$$\Delta(s) := \frac{H - A_{\mathrm{curv}}}{L_\Sigma} = \Delta_{\mathrm{sp}}(s, \lambda(s)), \qquad \Delta_{\mathrm{sp}}(s,\lambda) = 2^{1-q}\Phi(s,\lambda). \tag{2.13}$$

The last function extends the normalized gap to other apertures. It need not equal the gap of a nonminimal cone.

Proof. Let $E = \{(y,z) : |z| > \alpha|y|\}$, and use the signed convention

$$\mathcal{H}_s[E](p) = \mathrm{p.\,v.} \int_{\mathbb{R}^N} \frac{\chi_E(x) - \chi_{E^c}(x)}{|x-p|^{N+s}}\, \mathrm{d}x.$$

Its zero set does not depend on this sign convention. Put $X(x) = (x-p)|x-p|^{-N-s}$, so $\operatorname{div} X = -s|x-p|^{-N-s}$. To apply the divergence theorem, first work on

$$D_{\epsilon,\delta,R} = B_R(p) \setminus (\overline{B_\epsilon(p)} \cup \overline{B_\delta(0)}), \qquad 0 < \epsilon, \delta < 1/4, \quad R > 2.$$

Subtract the formula for $E^c \cap D_{\epsilon,\delta,R}$ from the formula for $E \cap D_{\epsilon,\delta,R}$. Since ν points into E, the two interface terms give

$$-s\int_{D_{\epsilon,\delta,R}} (\chi_E - \chi_{E^c})|x-p|^{-N-s}\, \mathrm{d}x = -2\int_{\Sigma\cap D_{\epsilon,\delta,R}} X\cdot\nu\, \mathrm{d}\sigma + \int_{\partial D_{\epsilon,\delta,R}} (\chi_E - \chi_{E^c}) X\cdot n_D\, \mathrm{d}S,$$

where n_D is the outer unit normal to the truncated domain. The spherical terms at infinity and at the vertex are $O(R^{-s})$ and $O(\delta^{N-1})$, respectively. On $\partial B_\epsilon(p)$, the flux factor is $-\epsilon^{1-N-s}$. A tangent half-space cuts this sphere into two equal parts. Since Σ is C^2 near p, its signed area imbalance is $O(\epsilon^N)$. Thus the remaining spherical term is $O(\epsilon^{1-s})$ and also tends to zero. The same tangent-half-space comparison proves existence of the volume principal value. Moreover, $(x-p)\cdot\nu(x) = O(|x-p|^2)$ near p, so the surface integral is absolutely convergent there. Since $x\cdot\nu(x) = 0$ and $|p\cdot\nu(x)| \leq 1$, the omitted surface pieces at the vertex and infinity are $O(\delta^{N-1})$ and $O(R^{-1-s})$. Taking the limits gives

$$\mathcal{H}_s[E](p) = \frac{2}{s}\int_\Sigma \frac{(x-p)\cdot\nu(x)}{|x-p|^{N+s}}\, \mathrm{d}\sigma(x).$$

Finally, $p \cdot \nu(x) = \sqrt{ab}(v-u)$. In the part of the radial integral above one, substitute $r \mapsto 1/r$. The weight on $(0,1)$ becomes $r^{N-2} + r^s$. For the Hardy integral the same substitution produces

$$r^{N-2} + r^s - 2r^{(N-2+s)/2} = r^s(1-r^\beta)^2;$$

For the curvature integral, it gives $(r^{N-2}+r^s)(1-T)$. Their difference is therefore $(r^{N-2}+r^s)T - 2r^{(N-2+s)/2}$. This pairing keeps the principal value in the Hardy integral. Indeed, the inversion $I(x) = x/|x|^2$ satisfies $|I(x)-p| = |x-p|/|x|$ when $|p|=1$. Near p, the two truncations differ only in a shell of radius $O(\epsilon)$ and thickness $O(\epsilon^2)$. The Hardy numerator is $O(\epsilon)$ there. Thus the absolute error on the shell is $O(\epsilon^{1-s}) \to 0$. After pairing, the quadratic numerator above is absolutely integrable. Putting $r = e^{-t}$ and using

$$1 + r^2 - 2rZ = 2r(\cosh t - Z), \qquad r^{N-2} + r^s = 2r^{(N-2+s)/2}\cosh(\beta t)$$

cancels all radial powers against dr. In particular,

$$\mathcal{H}_s[E](p) = -\frac{2^{2-q}\sqrt{ab}L_\Sigma}{s} G(s,\lambda),$$

so minimality is equivalent to (2.11). The paired difference gives the gap with T in place of Z. Since $T - Z = (b-a)(u-v)$, it gives the exact identity

$$\frac{H - A_{\mathrm{curv}}}{L_\Sigma} = 2^{1-q}(\Phi + (a-b)G).$$

Thus (2.13) holds on the zero set of G. □

Lemma 2.3 (Parameter regularity)**.** *The paired integrals G and Φ are real analytic for $-1 < s < 1$, $0 < \lambda < 1$. They converge absolutely and locally uniformly in these parameters. Moreover,*

$$G_\lambda = q\int_0^\infty \mathcal{I}_{m,n}[(v-u)^2\cosh(\beta t)(\cosh t - Z)^{-q-1}]\,dt > 0. \tag{2.14}$$

Proof. Use local angular coordinates ξ at p. Then $1 - Z \asymp |\xi|^2$, $v - u = O(|\xi|^2)$, and $\cosh t - Z \asymp t^2 + |\xi|^2$. Both combined numerators vanish to second order. Integration over the $N-1$ radial and angular variables gives the bound $K\int_0^\epsilon R^{-s}\,dR$. An aperture derivative adds a quadratic factor to the numerator and one power to the denominator, so the same bound holds. Derivatives in s add logarithmic factors. At infinity, the absolute integrands are bounded by a constant times $e^{-(q-|\beta|)t}$, where $q - |\beta| > 0$. Differentiation adds polynomial factors. All these estimates are uniform on compact parameter sets. A discrete sphere factor adds no local integration variable, so the same argument includes it.

To prove analyticity, use $|u-v| \le (1-u) + (1-v) \le K(\cosh t - Z)$ on each compact aperture interval. A small enough complex change in λ then changes the denominator by less than a fixed fraction of its real value. Small complex changes in s keep the integrable bounds above, with an exponent strictly below one. Dominated integration for holomorphic functions now applies. Finally, $Z_\lambda = v - u$ gives (2.14). The integrand is positive on a set of positive measure. □

We will also use a simple identity for weighted integrals. Let $w \ge 0$ be integrable on an interval J and let $W = \int_J w(t)\,\mathrm{d}t > 0$. Whenever the integrals converge absolutely,

$$\begin{aligned} W\int_J f(t)g(t)w(t)\,\mathrm{d}t - \left(\int_J f(t)w(t)\,\mathrm{d}t\right)\left(\int_J g(t)w(t)\,\mathrm{d}t\right) \\ = \frac{1}{2}\iint_{J\times J}[f(t)-f(r)][g(t)-g(r)]w(t)w(r)\,\mathrm{d}t\,\mathrm{d}r. \end{aligned} \tag{2.15}$$

Expanding the right side proves the identity. Its sign is nonnegative when both functions increase, and nonpositive when one increases and the other decreases. The sign is strict if the product of their differences is strict on a set of positive weighted area.

2.2.1. *A one-crossing comparison of spherical averages.* For $d \geq 1$, define

$$\phi_d(y) = \mathcal{I}_d[e^{yu}], \qquad r_d(y) = \frac{\phi_d'(y)}{\phi_d(y)}. \tag{2.16}$$

Thus $\phi_1 = \cosh$ and $r_1 = \tanh$. For $y > 0$, evenness gives $\phi_d'(y) > 0$, and averaging $1 - u$ against the positive weight e^{yu} gives $r_d(y) < 1$. For $d \geq 2$, $\big((1-u^2)w_d(u)\big)' = -(d-1)uw_d(u)$, and $(1-u^2)w_d(u)$ vanishes at both endpoints, including when $d = 2$. Integration by parts therefore gives $y(\phi_d - \phi_d'') = (d-1)\phi_d'$. Dividing by $y\phi_d$ gives the following identity. For $d = 1$, direct differentiation gives the same identity:

$$r_d' = 1 - r_d^2 - \frac{d-1}{y} r_d. \tag{2.17}$$

Differentiate the quotient and expand the products of integrals. This gives

$$r_d'(y) = \frac{\mathcal{I}_{d,d}[(u-v)^2 e^{y(u+v)}]}{2\phi_d(y)^2} > 0.$$

The integrand is nonnegative and is positive off the diagonal. For a one-dimensional factor, the same formula is a finite sum with positive terms at the two unequal coordinate pairs.

Lemma 2.4. *The function $\chi_d(y) = y(1 - r_d(y)^2)/r_d(y)$ strictly decreases from d to $d-1$. If $m > n$ and $(n-1)/(N-2) < b < n/N$, then $f_b(x) = r_n(bx) - r_m((1-b)x)$ has exactly one positive zero. It is negative before this zero and positive after it. For $0 < b \leq (n-1)/(N-2)$, it is negative everywhere. At $b = n/N$, it is positive everywhere.*

Proof. Put $S = (\chi_d^2 + 4y^2)^{1/2}$. Equation (2.17) implies

$$\chi_d - d + 1 = \frac{yr_d'}{r_d} > 0, \qquad y\chi_d' = \chi_d - (\chi_d - d + 1)S.$$

At a critical point, this gives $y\chi_d'' = -4y(\chi_d - d + 1)/S < 0$. The coordinate moments also give

$$r_d(y) = \frac{y}{d} - \frac{y^3}{d^2(d+2)} + O(y^5), \qquad \chi_d(y) = d - \frac{2y^2}{d(d+2)} + O(y^4).$$

The derivative is negative near zero. If it reached a first zero, the second derivative there would be nonnegative. This contradicts the strict negative sign just proved. The function is therefore strictly decreasing.

To find the limit at infinity for $d \geq 2$, let $h_d = (d-1)/2 > 0$ and write $v = 1 - u$ in the spherical integral. Up to a constant factor,

$$\phi_d(y) = e^y \int_0^2 e^{-yv}(2v - v^2)^{h_d - 1}\, dv.$$

For $j = 0, 1$, splitting the integral at $v = 1$ gives

$$\int_0^2 e^{-yv} v^j (2v - v^2)^{h_d-1}\, dv = 2^{h_d-1}\Gamma(h_d + j) y^{-h_d-j}(1 + O(y^{-1})).$$

Indeed, the part on $[1, 2]$ is exponentially small, even when $d = 2$, and on $[0, 1]$ the factor $(1 - v/2)^{h_d-1}$ is $1 + O(v)$. The gamma integrals after scaling v by $1/y$ give the formula. Taking the ratio for $j = 1$ and $j = 0$ yields $1 - r_d(y) = h_d/y + O(y^{-2})$. The limit is therefore $d - 1$. For $d = 1$, use $\chi_1(y) = 2y/\sinh(2y)$.

Write $a = 1 - b$, $p = r_n(bx)$, and $\ell_b(x) = (m-n) - (a-b)\chi_n(bx)/b$. Subtract the two Riccati equations to obtain

$$f_b' = \frac{p}{x}\ell_b - \left(2ap + \frac{m-1}{x}\right) f_b + af_b^2.$$

In the stated interval, $a > b$ and ℓ_b strictly increases. Its endpoint limits are

$$\ell_b(0+) = N - \frac{n}{b} < 0, \qquad \ell_b(\infty) = N - 2 - \frac{n-1}{b} > 0.$$

The small-argument expansion above gives $f_b(x) = (b/n - a/m)x + O(x^3) < 0$ near zero. The large-argument estimates for r_d give

$$f_b(x) = \frac{1}{2x}\left(\frac{m-1}{a} - \frac{n-1}{b}\right) + O(x^{-2}) > 0.$$

For a one-dimensional factor, use $r_1(y) = 1 + O(e^{-2y})$ in this last formula. Thus a first positive zero x_0 exists, and $f_b'(x_0) \geq 0$. If this derivative were zero, the differential equation would give $\ell_b(x_0) = 0$ and $f_b''(x_0) = (p(x_0)/x_0)\ell_b'(x_0) > 0$. Taylor's formula would then make f_b positive immediately before x_0, a contradiction. Hence the first zero is simple and upward. At any later zero, ℓ_b is larger, so the differential equation again gives a positive derivative. The function cannot return to zero from positive values.

Below or at the lower endpoint, $\ell_b < (N-2) - (n-1)/b \leq 0$, so the initial negative sign cannot change. At $b = n/N$, we have $\ell_b > 0$, and the sign near zero follows from

$$f_b(x) = \frac{1}{N^3}\left(\frac{m}{m+2} - \frac{n}{n+2}\right)x^3 + O(x^5) > 0.$$

This positive sign cannot change either. □

Proposition 2.5 (Location and motion of the minimal aperture)**.** *If $m > n$, then*

$$\frac{n-1}{N-2} < \lambda(s) < \frac{n}{N}, \qquad \lambda'(s) < 0 \quad (0 < s < 1). \tag{2.18}$$

If $m = n$, then $\lambda(s) = 1/2$. The minimal branch is real analytic.

Proof. If $m = n$, interchanging the coordinates gives $G(s, 1/2) = 0$. Since $G_\lambda > 0$, this zero is unique, so $\lambda(s) = 1/2$ and is real analytic. We now assume $m > n$. In particular, $N \geq 3$ and $\beta > 0$. Use $K_\beta(x) = \int_0^\infty e^{-x\cosh t}\cosh(\beta t)\,dt$ and $P_b(x) = \phi_m(ax)\phi_n(bx)$. The Laplace formula for a negative power rewrites (2.11) as

$$G(s,b) = \frac{1}{\Gamma(q)}\int_0^\infty x^{q-1}K_\beta(x)P_b(x)f_b(x)\,dx. \tag{2.19}$$

The absolute convergence in Lemma 2.3, applied with $|v-u|$, justifies the change in the order of integration by Tonelli's theorem. For positive aperture, the integrand is $O(x^{s+1})$ at zero and $O(x^{(s-3)/2})$ at infinity. For the first bound, use $f_b = O(x)$, $P_b = O(1)$, and $K_\beta = O(x^{-\beta})$, where $\beta > 0$ and $q - \beta = 1 + s$. For the second, the preceding spherical asymptotics give $P_b = O(e^x x^{-(N-2)/2})$ and $f_b = O(x^{-1})$, including a discrete factor. Also $K_\beta = O(e^{-x}x^{-1/2})$, by scaling t by $x^{-1/2}$ near the minimum of $\cosh t$. These estimates are uniform on compact exponent and aperture intervals. The endpoint signs in Lemma 2.4, together with (2.14), place the unique zero strictly inside the stated interval. When $n = 1$, the lower endpoint is zero. The aperture is positive, so the lower bound still holds.

We next prove the strict decrease. Put $w_{x,\beta}(t) = e^{-x\cosh t}\cosh(\beta t)$ and $F_\beta(t) = t\tanh(\beta t)$. Differentiate the integral defining K_β and use (2.15). We obtain

$$\partial_x\partial_\beta \log K_\beta(x) = -\frac{1}{2K_\beta(x)^2}\iint_{(0,\infty)^2}[F_\beta(t) - F_\beta(r)][\cosh t - \cosh r]w_{x,\beta}(t)w_{x,\beta}(r)\,\mathrm{d}t\,\mathrm{d}r < 0.$$

The integral converges absolutely for $x > 0$. Both functions strictly increase, and the weight is positive. The double integral is therefore strictly positive. Hence $S_s(x) = \frac{1}{2}(\log x - \partial_\beta \log K_\beta(x))$

strictly increases with x. At a minimal aperture, let x_0 be the zero of f_b. Differentiate (2.19) with b fixed and subtract $S_s(x_0)G = 0$ to obtain

$$G_s = \frac{1}{\Gamma(q)} \int_0^\infty x^{q-1} K_\beta P_b f_b [S_s(x) - S_s(x_0)]\, dx > 0.$$

The differentiated integrals satisfy the bounds above with logarithmic factors. More explicitly, $|\partial_\beta K_\beta| \le K x^{-\beta}(1 + |\log x|)$ for $0 < x \le 1$, and $|\partial_\beta K_\beta| \le K e^{-x} x^{-3/2}$ for $x \ge 1$; the second estimate follows from $t \sinh(\beta t) = O(t^2)$ near zero in the differentiated integral. Thus differentiation under the integral is valid. The derivative of $1/\Gamma(q)$ gives zero on $G = 0$. The implicit function theorem now gives $\lambda' = -G_s/G_\lambda < 0$. Lemma 2.3 gives local analytic branches. Uniqueness joins these branches into one. Interchanging the two factors gives the same analyticity assertion when $m < n$. □

2.3. **A single series for the sign problem.** Assume $N \ge 3$ throughout this subsection. We first combine the two radial terms so that their cancellation is kept in the expansion. Set

$$A = \frac{N-2}{2}, \qquad B = \frac{s}{2}, \qquad C = \frac{N+s}{4}, \qquad A + B + 1 = 2C, \tag{2.20}$$

and let $M_{2k} = \mathcal{I}_{m,n}[Z^{2k}]$, $M_0 = 1$. Expand the even power and apply (2.9) to each coordinate. The odd terms integrate to zero. We obtain

$$M_{2k} = \frac{(2k)!}{4^k} \sum_{j=0}^{k} \frac{a^{2j} b^{2k-2j}}{j!(k-j)!(m/2)_j (n/2)_{k-j}}. \tag{2.21}$$

Proposition 2.6 (The paired moment formula)**.** *At every positive aperture,*

$$\Delta_{\mathrm{sp}} = \frac{\Gamma(C)^2}{2\Gamma(2C)} \Theta(s,\lambda), \qquad \Theta(s,\lambda) = -1 + \sum_{k=1}^{\infty} a_k M_{2k}(R_k - 1), \quad a_k = \frac{(C)_k^2}{k!(1/2)_k}, \tag{2.22}$$

$$R_k = \frac{k\Gamma(k+A)\Gamma(k+B)}{\Gamma(k+C)^2}, \qquad k \ge 1. \tag{2.23}$$

The combined series converges absolutely. On the minimal branch, its sign is the sign of the physical gap. As an identity for Δ_{sp}, the formula extends through $s = 0$ and holds for all $-1 < s < 1$. The convergence is locally uniform in this range. Equivalently, set $R_0 = 0$ to obtain

$$\Delta_{\mathrm{sp}} = \sum_{k=0}^{\infty} c_k M_{2k}(R_k - 1), \qquad c_k = \frac{2^{2k-1}\Gamma(k+C)^2}{\Gamma(2C)\Gamma(2k+1)}. \tag{2.24}$$

Proof. For $0 < \rho < 1$, let Φ_ρ be the expression in (2.12) with both occurrences of Z replaced by ρZ, and put $\Delta_\rho = 2^{1-q}\Phi_\rho$. We first expand Δ_ρ and then let $\rho \uparrow 1$. Expand $(\cosh t - \rho Z)^{-q}$ as a power series. For fixed ρ, the expansion is absolutely summable under the integral: its absolute sum is bounded by a constant times $(1 + \cosh(\beta t))(\cosh t)^{-q}$, and $q > |\beta|$. By evenness of the coordinate weights, the even term $2k$ pairs with the odd term $2k-1$ multiplied by the numerator. To compute these terms, use

$$\int_0^\infty \frac{\cosh(\beta t)}{\cosh^p t}\, dt = \frac{2^{p-2}}{\Gamma(p)} \Gamma\left(\frac{p+\beta}{2}\right) \Gamma\left(\frac{p-\beta}{2}\right), \qquad p > |\beta|. \tag{2.25}$$

To prove this identity, first extend the integral to the real line. Replace the even hyperbolic numerator by $e^{\beta t}$. Then use $u = e^{2t}$ in Euler's beta integral. The constant term in the expansion

of Δ_ρ is $-\Gamma(C)^2/(2\Gamma(2C))$. For $k \geq 1$ the paired coefficient is

$$\rho^{2k} M_{2k} \frac{2^{2k-2}}{(2k-1)!\Gamma(2C)} \left[\Gamma(k+A)\Gamma(k+B) - \frac{\Gamma(k+C)^2}{k} \right].$$

Using $(2k)! = 4^k k!(1/2)_k$ gives (2.22) with the damping factor still present.

We first justify the limit. Fix positive a, b. The bounds below are uniform on compact parameter sets. Use a change of variables in the product integral to write

$$\mathcal{I}_{m,n}[F(Z)] = \int_{-1}^{1} F(z) f_Z(z)\, \mathrm{d}z. \tag{2.26}$$

Here f_Z is a nonnegative even weight of integral one. When both factors have dimension at least two, it is the convolution of the two scaled coordinate weights. For a one-dimensional factor, it is the half-sum of two translated coordinate weights. Since $1 - Z = a(1-u) + b(1-v)$, the beta convolution at the endpoint gives

$$f_Z(z) \leq K(1 - |z|)^{(N-4)/2} \quad \text{near } |z| = 1.$$

Indeed, a factor of dimension $d \geq 2$ has endpoint power $(d-3)/2$. Convolution adds the two powers and one. A discrete factor only translates the other weight and gives the same power. The beta integral therefore bounds the endpoint part of the moment by a constant times $B(2k+1, (N-2)/2) = O(k^{-(N-2)/2})$. Away from the endpoints, the powers decay exponentially. Thus $M_{2k} = O(k^{-(N-2)/2})$. The gamma-ratio expansion

$$\frac{\Gamma(k+c)}{\Gamma(k+d)} = k^{c-d} \left(1 + \frac{(c-d)(c+d-1)}{2k} + O(k^{-2}) \right),$$

which follows from Stirling's formula and is uniform for c, d in compact sets, together with $A + B + 1 = 2C$, gives

$$R_k = 1 + \frac{D}{k} + O(k^{-2}), \qquad D = C^2 - AB - A - B,$$

locally uniformly in the exponent. The same expansion gives $a_k = O(k^{(N+s-3)/2})$ and $R_k - 1 = O(k^{-1})$. The paired summands are therefore $O(k^{(s-3)/2})$. This bound is summable. It is locally uniform away from $s = 1$ and away from zero aperture or infinite aperture.

For the integral limit, set $h = t^2 + 1 - Z$ and $\delta = 1 - \rho$ near the diagonal. The damped numerator is at most $K(h + \delta)$ in absolute value. The denominator is comparable to $(h + \delta)^q$. Since $q > 1$, the quotient is bounded by Kh^{1-q}. Its radial integral is $K \int_0^\epsilon R^{-s}\, dR$. Away from the diagonal, compactness and exponential decay give an integrable bound. We can now use dominated convergence for the integral and absolute Abel convergence for the series. This proves the claim. The two series are always paired before setting $\rho = 1$. □

Lemma 2.7 (One-crossing moment comparison)**.** *Let f, g be nonnegative even integrable weights on $[-1, 1]$, each with integral one. Suppose that, for some $z_0 \in [0, 1]$, $f - g \geq 0$ on $(0, z_0)$ and $f - g \leq 0$ on $(z_0, 1)$, up to null sets. Then $\int_{-1}^{1} z^{2k} f(z)\, \mathrm{d}z \leq \int_{-1}^{1} z^{2k} g(z)\, \mathrm{d}z$ for every integer $k \geq 0$.*

Proof. For $k = 0$, the conclusion is equality. For $k \geq 1$, both weights have integral one half on $(0, 1)$, so

$$\int_{-1}^{1} z^{2k} (f - g)(z)\, \mathrm{d}z = 2 \int_0^1 (z^{2k} - z_0^{2k})(f - g)(z)\, \mathrm{d}z \leq 0. \tag{2.27}$$

The same argument allows an integrable singularity at the crossing. □

Lemma 2.8 (Gamma comparisons used below)**.** *For $x > 0$,*

$$\psi'(x) \geq \frac{1}{x} + \frac{1}{2x^2}, \qquad \psi(x) \leq \log x - \frac{1}{2x}, \qquad \frac{\Gamma(x+1/2)}{\Gamma(x)} \geq \frac{x}{\sqrt{x+1/2}}.$$

Proof. The trigamma expansion is $\psi'(x) = \sum_{j\geq 0}(x+j)^{-2}$. The summand is convex. The trapezoidal inequality shows that the sum is at least the integral of the summand plus half its value at zero. This proves the first bound. Since $\psi(x) - \log x \to 0$, integration gives

$$\log x - \psi(x) = \int_x^\infty \left(\psi'(t) - \frac{1}{t}\right) dt \geq \frac{1}{2x}.$$

This is the second bound. For the third bound, use log-convexity at $x+1$, the midpoint of $x+1/2$ and $x+3/2$. This gives $\Gamma(x+1)^2 \leq (x+1/2)\Gamma(x+1/2)^2$. □

Lemma 2.9 (Coefficient recurrence and a rational envelope)**.** *For the parameters in (2.20), put*

$$D = C^2 - AB - A - B = \frac{s^2 - 2Ns + (N-4)^2}{16}.$$

The ratios in (2.23) satisfy

$$\frac{R_{k+1}}{R_k} = 1 + \frac{AB - Dk}{k(k+C)^2}, \qquad R_k \longrightarrow 1, \qquad \partial_s \log R_k = \frac{1}{2}[\psi(k+B) - \psi(k+C)] < 0. \tag{2.28}$$

For $0 < s < 1$, $D \leq 0$ implies $R_k < 1$ for every k. If $3 \leq N \leq 7$, $0 \leq s < 1$, and $D \geq 0$, then

$$R_k \leq 1 + \frac{D}{k} \qquad (k \geq 1). \tag{2.29}$$

Proof. The gamma recurrence proves the first identity. The relation $A + B + 1 = 2C$ and the gamma-ratio asymptotic prove the limit. Since $C > B$ and $\psi' > 0$, the derivative has the stated strict sign. When $D \leq 0$, the recurrence shows that the sequence strictly increases to one. Thus $R_k < 1$.

For the rational bound, set $v_k = 1 + D/k$ and $\rho_k = R_{k+1}/R_k$. Use the positive denominator $k^2(k+C)^2(k+1)$ to write $\rho_k v_k - v_{k+1}$ as a fraction. Its numerator is

$$k^2\{D(2C - D - 1) + AB\} + k\{D(C^2 - D + AB) + AB\} + DAB.$$

Here $C^2 - D + AB = 2AB + A + B \geq 0$. In the stated dimensions, $D \leq (N-4)^2/16 \leq (N-2)/2 \leq 2C - 1$. Every coefficient is therefore nonnegative. It follows that R_k/v_k is nondecreasing and tends to one. This proves (2.29). □

3. The planar transition: $N = 2$

Theorem 3.1. *For $0 < s < 1$, consider the planar cross $\Sigma \subset \mathbb{R}^2$ and the phase E given by*

$$\Sigma = \{(x, y) : xy = 0\}, \qquad E = \{(x, y) : xy > 0\}. \tag{3.1}$$

Let Q_s be the quadratic form (2.4) for this phase, equal to one half of the perimeter second variation. Let $d\sigma$ be length measure on Σ. We allow every velocity in $C_c^\infty(\Sigma \setminus \{0\})$, with no symmetry condition between the four rays. Let B be Euler's beta function, and define

$$D_2(s) = \left(\sec\frac{\pi s}{2} - 1\right) B\left(1 + \frac{s}{2}, 1 + \frac{s}{2}\right) - \frac{2}{1+s} - B\left(\frac{2+s}{4}, \frac{2+s}{4}\right). \tag{3.2}$$

This function is continuous and strictly increasing on $(0, 1)$. It has a unique zero s_2, with

$$\frac{9167}{10000} < s_2 < \frac{9168}{10000}. \tag{3.3}$$

The cross is strictly unstable for $s < s_2$ and stable for $s \geq s_2$. For every $s \in (0,1)$ its sharp weighted spectral gap is

$$\inf_{0\neq\varphi\in C_c^\infty(\Sigma\setminus\{0\})} \frac{Q_s(\varphi)}{\displaystyle\int_\Sigma |x|^{-1-s}\varphi^2 \,\mathrm{d}\sigma} = D_2(s). \tag{3.4}$$

We can compute the Hardy and curvature terms exactly on the four rays. We first obtain the beta formula. We then fold the Hardy integral along the same axis at one to obtain a positive integrand. This formula proves strict increase and gives the signs at both endpoints. Exact rational bounds locate the unique zero. The ground-state identity holds for every four-ray velocity, so the sign of the gap gives the full stability statement. Finally, we find the leading coercivity term as s approaches one.

For $N = 2$, the unique minimal cone in the Lawson family is $C_{1,1}(1)$. Rotating it clockwise by $\pi/4$ gives the interface and phase in (3.1). Reflection across an axis fixes each point on that axis and exchanges the two phases. At a nonzero point p on the axis, this reflection preserves $|x-p|$ and every ball centered at p. Each truncated mean-curvature integral is therefore zero, so the cross is s-minimal away from the origin.

We write the unnormalized planar gap as $D_2(s) = H_s - A_{\mathrm{curv}}(s)$. Here H_s and $A_{\mathrm{curv}}(s)$ are the constants in (2.2) and (2.3) with $N = 2$. The normalized gap for a general link is $\Delta = (H - A_{\mathrm{curv}})/L_\Sigma$. For the planar cross, $L_\Sigma = 4$.

3.1. **The four-ray operator and a convergent gap formula.**

Lemma 3.2. *For the planar cross, the Hardy–curvature gap $H_s - A_{\mathrm{curv}}(s)$ equals the function $D_2(s)$ in (3.2).*

Proof. Put $a_2 = s/2 \in (0, 1/2)$. At $p = (1, 0)$, the normal pointing into E is $(0, 1)$. It is the same on the positive horizontal ray, opposite on the negative horizontal ray, and perpendicular on the two vertical rays. Hence

$$A_{\mathrm{curv}}(s) = \frac{2}{1+2a_2} + 2\int_0^\infty (1+r^2)^{-1-a_2}\,\mathrm{d}r = \frac{2}{1+s} + \sqrt{\pi}\frac{\Gamma((1+s)/2)}{\Gamma(1+s/2)}. \tag{3.5}$$

All four unit points have this coefficient by symmetry.

The Hardy exponent is negative: $\beta = -a_2$. Thus the ground state is $w_s(x) = |x|^{a_2}$. At a unit point, split the integral into the rays on the same axis and those on the perpendicular axis. This gives

$$H_s = J(a_2) + 2\int_0^\infty \frac{1 - r^{a_2}}{(1+r^2)^{1+a_2}}\,\mathrm{d}r, \tag{3.6}$$

$$J(a) = \mathrm{p.\,v.}\int_{\mathbb{R}} \frac{1-|t|^a}{|1-t|^{2+2a}}\,\mathrm{d}t. \tag{3.7}$$

For $w_s(x) = |x|^{s/2}$, Proposition 2.1 proves convergence and the ground-state identity. This includes the principal value at a nonzero point.

Subtracting (3.5) from (3.6) cancels the unweighted perpendicular-ray integrals and gives

$$H_s - A_{\mathrm{curv}}(s) = J(a_2) - \frac{2}{1+2a_2} - B\left(\frac{1+a_2}{2}, \frac{1+a_2}{2}\right). \tag{3.8}$$

To obtain the beta term, set $u = r^2$ in $2\int_0^\infty r^{a_2}(1+r^2)^{-1-a_2}\,\mathrm{d}r$. Every integral canceled here is finite.

To compute J, fix $0 < a < 1/2$. The integral over the negative half-line is

$$J_-(a) = \int_0^\infty \frac{1-t^a}{(1+t)^{2+2a}}\,\mathrm{d}t = \frac{1}{1+2a} - B(1+a, 1+a).$$

On the positive half-line, remove $(1-\varepsilon, 1+\varepsilon)$ and integrate by parts with the primitive $-\operatorname{sgn}(t-1)|t-1|^{-1-2a}/(1+2a)$. The endpoint at zero contributes $-1/(1+2a)$. The boundary term at infinity is $O(t^{-1-a})$ and tends to zero. The two endpoints next to one have the same leading term $a\varepsilon^{-2a}/(1+2a)$. Their difference is $O(\varepsilon^{1-2a})$. Let $J_{+,\varepsilon}(a)$ denote the truncated positive-half-line integral. Inverting the variable in the part above $1+\varepsilon$ gives

$$J_{+,\varepsilon}(a) = -\frac{1}{1+2a} + \frac{a}{1+2a}\Bigg[\int_0^{1-\varepsilon} \frac{t^{a-1}}{(1-t)^{1+2a}}\,\mathrm{d}t - \int_0^{1/(1+\varepsilon)} \frac{t^{a}}{(1-t)^{1+2a}}\,\mathrm{d}t\Bigg] + O(\varepsilon^{1-2a}).$$

The inverted cutoff is $1/(1+\varepsilon)$ instead of $1-\varepsilon$. The interval between them has length $O(\varepsilon^2)$. Its integrand is $O(\varepsilon^{-1-2a})$. Thus this change adds an error of $O(\varepsilon^{1-2a})$, which also tends to zero. The difference of the two integrands is $t^{a-1}(1-t)^{-2a}$, which is integrable at both endpoints. We may therefore pass to the limit and obtain

$$J_+(a) = -\frac{1}{1+2a} + \frac{a}{1+2a}\int_0^1 t^{a-1}(1-t)^{-2a}\,\mathrm{d}t = \frac{aB(a,1-2a)-1}{1+2a}.$$

Both arguments of the beta integral are positive.

Adding the two half-lines and using gamma reflection and recurrence,

$$\frac{aB(a,1-2a)}{(1+2a)B(1+a,1+a)} = \frac{\Gamma(1-2a)\Gamma(1+2a)}{\Gamma(1-a)\Gamma(1+a)} = \sec(\pi a),$$

we obtain

$$J(a) = B(1+a,1+a)(\sec(\pi a)-1). \tag{3.9}$$

Equations (3.8) and (3.9) prove (3.2) with $D_2 = H_s - A_{\mathrm{curv}}(s)$. □

3.2. **Strict increase and the stability change.**

Lemma 3.3. *The function D_2 is continuous and strictly increasing on $(0,1)$, with limits $-(2+\pi)$ at zero and $+\infty$ at one. The sharp weighted spectral identity (3.4) holds for arbitrary admissible four-ray velocities.*

Proof. Set $a = s/2 \in (0,1/2)$. A positive integral formula makes the signs easier to see. Invert the parts $|t|>1$ in (3.7) and add them to the parts $|t|<1$. This gives

$$J(a) = \int_0^1 (1-t^a)^2\big((1-t)^{-2-2a} + (1+t)^{-2-2a}\big)\,\mathrm{d}t. \tag{3.10}$$

The difference between the cutoffs on the positive half-line has the same $O(\varepsilon^{1-2a})$ bound as before. Thus the identity uses the original symmetric principal value. Near $t=1$, its singular summand is $O((1-t)^{-2a})$; the other summand is bounded. Both are bounded near zero. Thus the folded integral is finite for $0<a<1/2$.

For each $t\in(0,1)$, the first summand in (3.10) strictly increases with a, because both positive factors $(1-t^a)^2$ and $(1-t)^{-2-2a}$ increase. For the second, write $g_a(t) = (1-t^a)/(1+t)^{1+a}$. Since $t^a \ge t$ and $-\log t \ge 1-t \ge 1-t^a$,

$$\partial_a \log g_a(t) = \frac{(-\log t)t^a}{1-t^a} - \log(1+t) \ge t - \log(1+t) > 0.$$

Pointwise comparison of the finite integrals now shows that J increases strictly. No differentiation under the integral is needed. In contrast,

$$B\left(\frac{1+a}{2},\frac{1+a}{2}\right) = 2\int_0^\infty \frac{1}{1+r^2}\left(\frac{r}{1+r^2}\right)^a\,\mathrm{d}r$$

decreases strictly, since $0 < r/(1+r^2) \le 1/2$. The remaining term $2/(1+2a)$ also decreases. Therefore (3.8) makes D_2 strictly increasing.

The beta formula gives continuity, together with

$$\lim_{s\downarrow 0} D_2(s) = -(2+\pi).$$

At the other endpoint, concavity of t^a implies $1-t^a \ge a(1-t)$. Consequently

$$J(a) \ge a^2 \int_0^1 (1-t)^{-2a}\,\mathrm{d}t = \frac{a^2}{1-2a} \longrightarrow +\infty,$$

while the subtracted terms stay bounded. There is therefore exactly one zero, with the stated signs on its two sides.

Proposition 2.1, applied with $\beta = -s/2$ and $w_s(x) = |x|^{s/2}$, identifies the infimum in (3.4) with $H_s - A_{\mathrm{curv}}(s) = D_2(s)$. Its admissible class allows independent values on all four rays.

At $s = s_2$ the quadratic form is positive for each nonzero admissible velocity. To see this, suppose the ground-state remainder vanishes. Its nonnegative integrand must then be zero almost everywhere. Since the kernel is positive, the values of φ/w_s agree for almost every pair of points on the punctured cross. Continuity makes this quotient constant on all four rays, with the same constant on each ray. Compact support makes the constant zero, so $\varphi = 0$. Since the sharp infimum in (3.4) is zero at the threshold, no positive uniform weighted coercivity constant holds there. □

3.3. **A finite rational verification of the threshold interval.**

Lemma 3.4. *The gap has the following rational bounds:*

$$-\frac{13}{10000} < D_2\left(\frac{9167}{10000}\right) < -\frac{11}{10000}, \qquad \frac{25}{10000} < D_2\left(\frac{9168}{10000}\right) < \frac{27}{10000}. \tag{3.11}$$

Proof. We bound the special functions at the two rational exponents. Euler's product gives, for $x > 0$,

$$B(x,x) = \frac{2}{x}\prod_{n\ge 1}\left(1 - \frac{x^2}{(n+x)^2}\right).$$

Let $P_M(x)$ be the product up to M and $T_M(x)$ the remaining product. For $0 < y < 1$, use $-y/(1-y) \le \log(1-y) \le -y$ to get

$$-\sum_{n>M}\frac{x^2}{n(n+2x)} \le \log T_M(x) \le -\sum_{n>M}\frac{x^2}{(n+x)^2}.$$

Both summands decrease with n. Comparison with integrals gives

$$-\frac{x}{2}\log\left(1+\frac{2x}{M}\right) \le \log T_M(x) \le -\frac{x^2}{M+1+x}.$$

The integrals used for the lower and upper bounds are, respectively,

$$\int_M^\infty \frac{x^2}{t(t+2x)}\,\mathrm{d}t, \qquad \int_{M+1}^\infty \frac{x^2}{(t+x)^2}\,\mathrm{d}t.$$

When $z = 2x/M < 1$, define

$$\ell_M(x) = \frac{x}{2}\left(z - \frac{z^2}{2} + \frac{z^3}{3}\right), \qquad u_M(x) = \frac{x^2}{M+1+x}.$$

The alternating series gives $\log(1+z) \le z - z^2/2 + z^3/3$, so $e^{-\ell_M(x)} \le T_M(x) \le e^{-u_M(x)}$. For $v \ge 0$, Taylor's formula gives $1 - v + v^2/2 - v^3/6 \le e^{-v} \le 1 - v + v^2/2$. Multiplying by the positive factor $2P_M(x)/x$ gives

$$\frac{2P_M(x)}{x}\left(1 - \ell_M + \frac{\ell_M^2}{2} - \frac{\ell_M^3}{6}\right) \le B(x,x) \le \frac{2P_M(x)}{x}\left(1 - u_M + \frac{u_M^2}{2}\right). \tag{3.12}$$

Take $M = 300$ and set

$$p_- = 3.14159265358979323846264338327950288, \qquad p_+ = p_- + 10^{-35}.$$

We can also prove $p_- < \pi < p_+$ by rational bounds. If $A_j(x) = \sum_{k=0}^{j}(-1)^k x^{2k+1}/(2k+1)$, Machin's identity and the alternating series give

$$p_- < 16A_{25}(1/5) - 4A_8(1/239) < \pi < 16A_{26}(1/5) - 4A_7(1/239) < p_+.$$

Put $S_j(t) = \sum_{k=0}^{j}(-1)^k t^{2k}/(2k)!$. At both $s = 9167/10000$ and $s = 9168/10000$, the arguments $p_-s/2$ and $p_+s/2$ lie in $(0, 3/2)$. Cosine decreases on this interval, and the terms in each alternating tail decrease. Hence

$$S_7(p_+s/2) \le \cos(p_+s/2) \le \cos(\pi s/2) \le \cos(p_-s/2) \le S_8(p_-s/2).$$

Finite rational checks give $S_7(p_+s/2) > 0$ and $S_8(p_-s/2) < 1$ at these two values. Thus the factor $1/\cos(\pi s/2) - 1$ is positive at both values. Apply (3.12) to $x = 1 + s/2$ and $x = (2+s)/4$. For a lower bound on the first product in (3.2), use the lower beta bound and the upper cosine bound. Subtract the upper bound for the other beta term. Reverse these choices for an upper bound on $D_2(s)$. We display the rational enclosures and the final comparisons. Put $Q = 10^6$, $s_- = 9167/10000$, $s_+ = 9168/10000$, and

$$B_1(s) = B(1 + s/2, 1 + s/2), \qquad B_2(s) = B((2+s)/4, (2+s)/4).$$

Substitution in the finite products and polynomials above gives

s	$QB_1(s)$	$QB_2(s)$	$Q\cos(\pi s/2)$
s_-	$(422892, 422902)$	$(1776037, 1776048)$	$(130474, 130475)$
s_+	$(422854, 422865)$	$(1775936, 1775948)$	$(130318, 130319)$.

(3.13)

Each pair gives strict lower and upper bounds. For a row with endpoints $l_1, u_1, l_2, u_2, l_c, u_c$, set

$$L_s = \frac{l_1}{Q}\left(\frac{Q}{u_c} - 1\right) - \frac{2}{1+s} - \frac{u_2}{Q}, \qquad U_s = \frac{u_1}{Q}\left(\frac{Q}{l_c} - 1\right) - \frac{2}{1+s} - \frac{l_2}{Q}.$$

The choices of bounds just described give $L_s < D_2(s) < U_s$. The remaining comparisons use only positive integers:

$$\begin{aligned}
L_{s_-} + \frac{13}{10000} &= \frac{10668871}{147106725000} > 0,\\
-\frac{11}{10000} - U_{s_-} &= \frac{31247631781}{1250397579000000} > 0,\\
L_{s_+} - \frac{25}{10000} &= \frac{2078636269}{39030540500000} > 0,\\
\frac{27}{10000} - U_{s_+} &= \frac{1422091741}{39030241000000} > 0.
\end{aligned}$$

Appendix D gives an exact check of the finite calculations in (3.13). These exact comparisons prove (3.11). □

3.4. **Proof of the planar transition theorem.**

Proof of Theorem 3.1. Lemma 3.2 gives the exact Hardy–curvature gap. Lemma 3.3 proves that it is continuous and strictly increasing, gives its endpoint signs, and proves the sharp weighted spectral identity. Thus there is exactly one zero, and Lemma 3.4 places it in (3.3). By (3.4), a negative gap gives a velocity with negative quadratic form. A nonnegative gap makes the form nonnegative for every admissible velocity. These are the claimed stability statements. □

Numerically, $s_2 \approx 0.916731033344131$. This decimal is not used in the proof.

The threshold also satisfies the following equivalent equation in gamma functions:

$$\frac{s_2^2\sqrt{\pi}\,\Gamma((1-s_2)/2)}{8(1+s_2)\Gamma(1-s_2/4)^2} = 1 + \frac{2\Gamma(1+s_2/2)}{(1+s_2)\Gamma((2+s_2)/4)^2}. \tag{3.14}$$

It follows from reflection and duplication applied to (3.9), which give explicitly

$$J(s/2) = \frac{s^2\sqrt{\pi}\,\Gamma((1-s)/2)\Gamma((2+s)/4)^2}{8(1+s)\Gamma(1-s/4)^2\Gamma(1+s/2)}.$$

Substituting this in (3.8) and dividing by the positive beta factor gives (3.14). No closed-form value of s_2 is needed.

3.5. **Coercivity as the local exponent is approached.** We can find the leading coefficient of the divergent term in (3.2). With $h = 1 - s$, Taylor's formula and $B(3/2, 3/2) = \pi/8$ give

$$\sec(\pi s/2) = \frac{1}{\sin(\pi h/2)} = \frac{2}{\pi h} + O(h), \qquad B(1+s/2, 1+s/2) = \frac{\pi}{8} + O(h).$$

The subtracted terms in (3.2) are smooth at $s = 1$. Multiplying the two expansions therefore gives

$$D_2(s) = \frac{1}{4(1-s)} + O(1), \qquad \lim_{s\uparrow 1}(1-s)D_2(s) = \frac{1}{4}. \tag{3.15}$$

In particular there exists $\bar{s}_2 \in (s_2, 1)$ such that

$$Q_s(\varphi) \geq \frac{1}{8(1-s)} \int_\Sigma |x|^{-1-s}\varphi^2 \,\mathrm{d}\sigma, \qquad \bar{s}_2 < s < 1, \tag{3.16}$$

for every admissible velocity. This follows from (3.4) and (3.15).

The curvature coefficient in (3.5) tends to three. Thus the interactions between different rays give a bounded potential near the local limit. The Hardy term on each ray gives the positive divergence. Its coefficient agrees with the classical half-line identity

$$\int_0^\infty |v'|^2 \,\mathrm{d}r - \frac{1}{4}\int_0^\infty \frac{v^2}{r^2}\,\mathrm{d}r = \int_0^\infty r\left|\left(\frac{v}{\sqrt{r}}\right)'\right|^2 \mathrm{d}r, \qquad v \in C_c^\infty(0,\infty),$$

which follows by expanding $v = \sqrt{r}\,\zeta$ and integrating the cross term. The conclusion concerns second variation away from the vertex. It does not assert that the cross minimizes perimeter or is stable under a larger class of rearrangements.

4. Instability in dimensions $3 \leq N \leq 6$

Theorem 4.1. *Let m, n be positive integers, $N = m+n \in \{3,4,5,6\}$, and $0 < s < 1$. Let $\alpha_{m,n}(s)$ be the unique s-minimal aperture for the kernel $|x-y|^{-N-s}$, and set $\Sigma_s = \{(y,z) \in \mathbb{R}^m \times \mathbb{R}^n : |z| = \alpha_{m,n}(s)|y|\}$. Then Σ_s is strictly unstable: there is $\varphi \in C_c^\infty(\Sigma_s \setminus \{0\})$ with $Q_s(\varphi) < 0$, where Q_s is the quadratic form (2.4), equal to one half of the perimeter second variation.*

We prove instability by showing that the Hardy–curvature gap is negative. For $N = 3, 5, 6$, upper bounds for the angular moments reduce the problem to a scalar series. A rational upper bound for the gamma coefficients then gives its sign. In dimension four, the coefficients give the sign directly. In dimension six, we must also keep a negative correction from the first mode to complete the estimate. We first prove the angular and scalar bounds. We then combine them to prove Theorem 4.1.

4.1. **Upper angular comparison for the low-dimensional links.** Set $a = 1 - \lambda$, $b = \lambda$, and $Z = au + bv$. We need upper bounds for the angular integrals $\mathcal{I}_{m,n}[Z^{2k}]$ on the aperture strip (2.18). In the dimensions where we need them, these integrals are at most the coordinate integrals $\mathcal{I}_{N-1}[u^{2k}]$ on $\mathbb{S}^{N-2}$. For $N = 4$, the coefficients give the sign directly, so no angular estimate is needed.

Proposition 4.2. *Let $m \ge n \ge 1$, $N = m + n \in \{3, 5, 6\}$, and set*

$$\lambda_- = \frac{n-1}{N-2}, \qquad \lambda_+ = \frac{n}{N}.$$

For every $\lambda \in [\lambda_-, \lambda_+]$ and integer $k \ge 0$,

$$\mathcal{I}_{m,n}\big[((1-\lambda)u + \lambda v)^{2k}\big] \le \frac{(1/2)_k}{((N-1)/2)_k}. \tag{4.1}$$

For fixed u, v, the function $((1-\lambda)u+\lambda v)^{2k}$ is convex in λ. Integrating its secant-line inequality against the nonnegative coordinate weights shows that the left side is also convex in λ. We therefore only need to compare the integrals at the two endpoints. If the endpoints coincide, we only need their common value. One argument works at the lower endpoint for all $m, n > 1$. At the upper endpoint, we use three direct comparisons of weights and one Jacobi-polynomial recurrence.

4.1.1. *The lower endpoint.* For $k = 0$, both sides equal one. First suppose $n > 1$ and $k \ge 1$. We use the coordinate-weight identity from the proof of (2.17). For $j > 1$, integration by parts gives

$$\mathcal{I}_j[uF(u)] = \frac{1}{j-1}\mathcal{I}_j[(1-u^2)F'(u)] \tag{4.2}$$

for each polynomial F. The boundary term contains $(1-u^2)^{(j-1)/2}$, which vanishes at the endpoints, also when $j = 2$. At λ_- write

$$d_0 = N - 2, \qquad a = \frac{m-1}{d_0}, \quad b = \frac{n-1}{d_0}, \qquad Z = au + bv.$$

Apply (4.2) to each variable while holding the other fixed. Since $a/(m-1) = b/(n-1) = 1/d_0$, this gives

$$\begin{aligned}\mathcal{I}_{m,n}[Z^{2k}] &= \frac{2k-1}{d_0}\mathcal{I}_{m,n}\big[Z^{2k-2}(1 - au^2 - bv^2)\big] \\ &\le \frac{2k-1}{d_0}\mathcal{I}_{m,n}[Z^{2k-2}(1-Z^2)].\end{aligned}$$

Here $au^2 + bv^2 - Z^2 = ab(u-v)^2 \ge 0$ because $a + b = 1$. Move the last $2k$th moment to the left to get

$$(N + 2k - 3)\mathcal{I}_{m,n}[Z^{2k}] \le (2k-1)\mathcal{I}_{m,n}[Z^{2k-2}].$$

Starting with the zeroth moment equal to one, iteration gives

$$\mathcal{I}_{m,n}[Z^{2k}] \le \prod_{j=1}^{k} \frac{2j-1}{N+2j-3} = \frac{(1/2)_k}{((N-1)/2)_k}.$$

If $n = 1$, the lower endpoint is zero and $Z = u$. The product integral then equals $\mathcal{I}_{N-1}[u^{2k}]$. The comparison is then an equality.

For the upper endpoint we use the one-crossing moment comparison of Lemma 2.7.

4.1.2. *Upper endpoints accessible from explicit weights.* First suppose $(m, n) = (2, 1), (4, 1)$, or $(5, 1)$, and put

$$a = \frac{m}{m+1}, \qquad b = \frac{1}{m+1}, \qquad c = a - b, \qquad p_0 = \frac{m-3}{2}.$$

Set $g(z) = w_m(z)$, and extend w_m by zero outside $(-1, 1)$. For every continuous F, the changes of variables $z = au + b$ and $z = au - b$ give

$$\mathcal{I}_{m,1}[F(au + bv)] = \int_{-1}^{1} F(z) f(z)\, \mathrm{d}z, \qquad f(z) = \frac{1}{2a}\left(w_m\left(\frac{z-b}{a}\right) + w_m\left(\frac{z+b}{a}\right)\right).$$

The two shifted weights overlap when $|z| < c$. Both f and g are even and have integral one. Factor their quadratic terms to obtain, for $z \geq 0$,

$$\frac{f(z)}{g(z)} = \frac{1}{2a^{2p_0+1}} \begin{cases} \left(\dfrac{c+z}{1+z}\right)^{p_0} + \left(\dfrac{c-z}{1-z}\right)^{p_0}, & 0 \leq z < c, \\ \left(\dfrac{c+z}{1+z}\right)^{p_0}, & c < z < 1. \end{cases} \tag{4.3}$$

For example, the first shifted weight uses $a^2 - (z-b)^2 = (1-z)(c+z)$; the other uses $a^2 - (z+b)^2 = (1+z)(c-z)$.

When $m = 2$, one has $p_0 = -1/2$, $a = 2/3$, and $c = 1/3$. The central ratio in (4.3) is

$$\frac{1}{2}\left(\sqrt{\frac{1+z}{c+z}} + \sqrt{\frac{1-z}{c-z}}\right) > 1.$$

The tail ratio decreases from its right limit $1/\sqrt{2}$ at c. Thus (2.27) applies with $z_0 = c$.

For $m = 4, 5$ the exponent p_0 is respectively $1/2, 1$. Set $A_z = (c+z)/(1+z)$ and $B_z = (c-z)/(1-z)$ on $(0, c)$. Since $A_z \geq B_z > 0$,

$$\frac{d}{dz}(A_z^{p_0} + B_z^{p_0}) = p_0(1-c)\left(\frac{A_z^{p_0-1}}{(1+z)^2} - \frac{B_z^{p_0-1}}{(1-z)^2}\right) \leq 0.$$

Thus the ratio on the central interval decreases. The ratio on the tail increases, and its largest value is

$$\lim_{z \uparrow 1} \frac{f(z)}{g(z)} = \frac{1}{2a^{p_0+1}} < 1.$$

For $m = 4$ the last inequality follows from $4(4/5)^3 > 1$; for $m = 5$ the limit is $18/25$. At zero, the square of the ratio is $375/256$ for $m = 4$, and the ratio is $144/125$ for $m = 5$. Both ratios are greater than one. The central and tail formulas have the same limit at c because $p_0 > 0$. Thus the central ratio decreases from above one to below one, and the entire tail stays below one. These are the signs required in Lemma 2.7. This proves the comparison at the upper endpoint for all three pairs.

For $(m, n) = (3, 2)$, let $a = 3/5$, $b = 2/5$, and $c = 1/5$. Here $w_3(u) = 1/2$ and $w_2(v) = 1/(\pi\sqrt{1-v^2})$. Hold v fixed and change variables from u to $z = au + bv$. Then integrate in v. This gives

$$\mathcal{I}_{3,2}[F(au + bv)] = \int_{-1}^{1} F(z) f(z)\, \mathrm{d}z,$$

where

$$f(z) = \begin{cases} 1/(2a), & |z| \le c, \\ \arccos((|z|-a)/b)/(2a\pi), & c < |z| < 1. \end{cases}$$

The comparison weight is $g(z) = w_4(z) = 2\sqrt{1-z^2}/\pi$. On $[0,c]$, $f/g = \pi/(4a\sqrt{1-z^2}) > 1$, using $4a = 12/5 < 3 < \pi$. On the remaining interval put $z = a + b\cos\theta$, $0 < \theta < \pi$. Write $R(z) = f(z)/g(z)$. Then

$$R(z) = \frac{\theta}{4a\sqrt{1-z^2}}, \qquad R'(z) = \frac{bz\theta\sin\theta - (1-z^2)}{4ab\sin\theta(1-z^2)^{3/2}}.$$

Here the derivative is with respect to z, and $d\theta/dz = -1/(b\sin\theta)$. Since $1 - z = b(1-\cos\theta)$, the derivative is nonpositive if and only if

$$z\theta\sin\theta \le (1-\cos\theta)(1+z).$$

This follows from $\theta \le 2\tan(\theta/2)$ and $2z \le 1+z$. The limit of the ratio at one is $1/(4a\sqrt{b}) = \sqrt{125/288} < 1$. The two formulas for f agree at c, where $R(c) > 1$. Continuity and monotonicity on $(c,1)$ therefore give a crossing with the signs in Lemma 2.7. That lemma gives the moment bound.

4.1.3. *The remaining endpoint: the pair* $(4,2)$. Write $u_k = \mathcal{I}_{4,2}[(\frac{2}{3}u + \frac{1}{3}v)^{2k}]$. Expand the polynomial and integrate each factor using the coordinate beta integrals (2.21). This gives

$$u_k = \frac{\binom{2k}{k}}{36^k} \sum_{j=0}^{k} \binom{k}{j}^2 \frac{4^j}{j+1}. \tag{4.4}$$

The Jacobi finite-sum formula [14, Eq. 18.5.8] gives

$$P_k^{(0,1)}(5/3) = 3^{-k} \sum_{j=0}^{k} \binom{k}{j}\binom{k+1}{j+1} 4^j = 3^{-k}(k+1) \sum_{j=0}^{k} \binom{k}{j}^2 \frac{4^j}{j+1}.$$

Thus

$$u_k = \frac{\binom{2k}{k}}{12^k(k+1)} P_k^{(0,1)}(5/3). \tag{4.5}$$

Put $P_k = P_k^{(0,1)}(5/3)$. For $k \ge 1$, the three-term relation [14, Eqs. 18.9.1–18.9.2] reads

$$(k+2)(2k+1)P_{k+1} = \left(\frac{5}{3}(2k+1)(2k+3) - 1\right) P_k - k(2k+3)P_{k-1}.$$

Substitute this relation into (4.5) to get

$$u_{k+1} = p_k u_k - q_k u_{k-1}, \qquad p_k = \frac{10k^2+20k+6}{9(k+2)^2}, \quad q_k = \frac{k(2k+3)(2k-1)}{36(k+2)^2(k+1)} > 0. \tag{4.6}$$

Here $u_0 = 1$, $u_1 = 1/6$, and $u_2 = 43/648$.

The comparison moments are $g_k = \mathcal{I}_5[u^{2k}] = 3/((2k+1)(2k+3))$. Substituting these moments into the recurrence gives the positive residual

$$\frac{g_{k+1}}{g_k} - p_k + q_k \frac{g_{k-1}}{g_k} = \frac{26k^2+53k+24}{36(k+2)^2(k+1)(2k+5)} > 0. \tag{4.7}$$

Every u_k is positive. Thus the ratios $\rho_k = u_k/u_{k-1}$ satisfy $\rho_{k+1} = p_k - q_k/\rho_k$. The right side increases as its positive argument increases. Put $t_k = g_k/g_{k-1} > 0$. Start with $\rho_1 = 1/6 < 1/5 = t_1$. If $\rho_k \le t_k$, then (4.7) gives

$$\rho_{k+1} = p_k - \frac{q_k}{\rho_k} \le p_k - \frac{q_k}{t_k} < t_{k+1}.$$

Induction gives $\rho_k \le t_k$ for every k. Multiplying the inequalities gives $u_k \le g_k$.

Proof of Proposition 4.2. The pairs, up to interchange of the factors, are $(2,1)$; $(4,1),(3,2)$; and $(5,1),(4,2),(3,3)$. The lower-endpoint argument applies to every pair. For $(3,3)$, both endpoints are $\lambda = 1/2$, so it already gives the full bound. The weight comparisons and the recurrence comparison give the upper endpoint for the remaining pairs. Convexity in λ then proves (4.1) throughout the strip. □

4.2. **Negative stability gaps in dimensions three through six.** Recall the parameters A, B, C in (2.20) and the paired series Θ in (2.22). Put $d = (N-1)/2$ and define

$$D = C^2 - AB - A - B = \frac{s^2 - 2Ns + (N-4)^2}{16}. \tag{4.8}$$

The sign of D tells us which argument to use. When it is nonpositive, each term has the needed sign. For the remaining small exponents, we need the upper angular comparison.

Lemma 4.3. *For $N \in \{3,5,6\}$, suppose $0 \le M_{2k} \le (1/2)_k/(d)_k$ for every $k \ge 1$. Then the paired series Θ is strictly negative for $0 < s < 1$. For $N = 4$, it is strictly negative for any nonnegative moment sequence for which the paired series converges.*

Proof. For $N = 3,5,6$, the assumed moment bound is $O(k^{-(N-2)/2})$. The gamma estimates in the proof of Proposition 2.6 therefore bound each paired summand by $O(k^{(s-3)/2})$. Thus Θ converges absolutely for every sequence in the lemma and every $0 < s < 1$.

Lemma 2.9 gives $R_k < 1$ when $D \le 0$. Every term in $\Theta + 1$ is then nonpositive, so $\Theta \le -1$. This proves the claim for every exponent in dimension four, where $D = s(s-8)/16 < 0$.

For the other three dimensions, we still need to consider

$$0 < s < \tau_N := N - 2\sqrt{2N-4}; \qquad \tau_3 = 3 - 2\sqrt{2}, \quad \tau_5 = 5 - 2\sqrt{6}, \quad \tau_6 = 6 - 4\sqrt{2}.$$

In the bounds below we also allow the endpoints $s = 0, \tau_N$. On this closed interval $D \ge 0$, and the same lemma gives

$$R_k \le 1 + \frac{D}{k}. \tag{4.9}$$

First replace $R_k - 1$ by its nonnegative upper bound D/k. Then use the assumed moment bound. This gives

$$\Theta \le -1 + DS, \qquad S = \sum_{k\ge1} \frac{(C)_k^2}{(d)_k k!\, k}. \tag{4.10}$$

In this order, the comparison works for either sign of the original coefficient $R_k - 1$.

Introduce

$$\eta = \frac{1+s}{2}, \qquad r = 1 - \eta = \frac{1-s}{2}, \qquad K = \frac{\Gamma(d)\Gamma(\eta)}{\Gamma(C)^2}, \qquad f(r) = -\psi(r) - \gamma. \tag{4.11}$$

Since $d + \eta = 2C$, the numbers

$$U_k := \frac{(C)_k^2}{(d)_k(\eta)_k} \quad \text{satisfy} \quad U_k \nearrow K,$$

because $(k+C)^2 - (k+d)(k+\eta) = (d-\eta)^2/4 \ge 0$. The gamma-ratio asymptotic gives the limit. Expanding $(1-t)^{-\eta}$ and integrating its nonnegative terms gives

$$\sum_{k\ge1} \frac{(\eta)_k}{k!\, k} = \int_0^1 \frac{(1-t)^{-\eta} - 1}{t}\, \mathrm{d}t = \int_0^1 \frac{u^{r-1} - 1}{1-u}\, \mathrm{d}u = f(r).$$

Here $u = 1 - t$. Expanding $1/(1-u)$ in the last integral gives $\sum_{j\ge0}(1/(j+r) - 1/(j+1)) = -\psi(r) - \gamma$. The integrals are finite: the first integrand is bounded near zero and is $O((1-t)^{-\eta})$

near one, with $\eta < 1$. The bound $U_k \le K$ now gives $S \le Kf(r) < \infty$. Keeping the exact first term C^2/d of S gives the stronger bound

$$S \le Kf(r) - \left(K\eta - \frac{C^2}{d}\right), \qquad K\eta - \frac{C^2}{d} \ge 0. \tag{4.12}$$

The correction is nonnegative because $U_1 = C^2/(d\eta) \le K$. Combining (4.10) and (4.12) gives the estimate we will use in dimension six:

$$\Theta \le -1 + D\left[Kf(r) - \left(K\eta - \frac{C^2}{d}\right)\right]. \tag{4.13}$$

We will use only elementary bounds on f and K:

$$\frac{1}{r} - 1 \le f(r) \le \frac{1}{r} - r, \qquad \psi'(r) \le \frac{1}{r^2} + \frac{1}{r}, \qquad 0 < r < 1. \tag{4.14}$$

For the first two bounds, write the digamma series as

$$f(r) = \frac{1}{r} - 1 + (1-r)\sum_{j\ge 1} \frac{1}{(j+r)(j+1)}.$$

The sum lies between zero and $\sum_{j\ge1} 1/(j(j+1)) = 1$. For the last bound, compare the decreasing tail of $\psi'(r) = 1/r^2 + \sum_{j\ge1}(j+r)^{-2}$ with its integral. Also, $\eta < C$ for $N \ge 3$, $s < 1$, so

$$\frac{d}{ds}\log K = \frac{1}{2}(\psi(\eta) - \psi(C)) < 0. \tag{4.15}$$

We use elementary bounds for all constants in this proof. In particular $0 < \gamma < 1$, $\pi < 4$, and

$$e^{7/10} > 1 + \frac{7}{10} + \frac{49}{200} + \frac{343}{6000} > 2$$

gives $\log 2 < 7/10$. The estimate $j! \ge 2\cdot 3^{j-2}$ for $j \ge 2$ gives $e^{1/2} \le 3/2 + (1/8)\sum_{j\ge0} 6^{-j} = 33/20 < 5/3$. Finally, $\tau_5 < 1/5$ and $\sqrt{2} - 1 > 2/5$ follow by squaring the positive inequalities $\sqrt{6} > 12/5$ and $\sqrt{2} > 7/5$. Whenever $r > 2/5$, (4.14) gives $f(r) < 21/10$.

In dimension three, $D \le 1/16$ and $r \ge \sqrt{2} - 1 > 2/5$. The tangent inequality for the convex function $\log\Gamma$ at one gives $\log\Gamma(3/4) \ge \gamma/4 > 0$. Thus $K \le \sqrt{\pi}/\Gamma(3/4)^2 < 2$, and

$$DS \le DKf(r) < \frac{1}{16}\cdot 2 \cdot \frac{21}{10} = \frac{21}{80} < 1.$$

In dimension five, $D \le 1/16$, $\tau_5 < 1/5$, and again $r > 2/5$. The same tangent inequality gives $\Gamma(5/4)^2 \ge e^{-\gamma/2}$, whence

$$K \le \frac{\sqrt{\pi}}{\Gamma(5/4)^2} \le \sqrt{\pi}e^{\gamma/2} < \frac{10}{3}, \qquad DS < \frac{1}{16}\cdot\frac{10}{3}\cdot\frac{21}{10} = \frac{7}{16} < 1.$$

In dimension six, we need the correction to the first term in (4.13). Here

$$d = \frac{5}{2}, \qquad C = \frac{6+s}{4}, \qquad D = \frac{s^2 - 12s + 4}{16}, \qquad K \le K(0) = 3.$$

We first show that $D(s)f((1-s)/2)$ is nonincreasing for $0 \le s \le \tau_6$. For $s < \tau_6$, (4.14) implies

$$\frac{d}{ds}\log f(r) = \frac{\psi'(r)}{2f(r)} \le \frac{1+r}{2r(1-r)}.$$

Substitute $s = 1 - 2r$ into D to get

$$-\frac{D'}{D} = \frac{2(5+2r)}{4r^2 + 20r - 7}.$$

We compare the bound for the logarithmic derivative of f with $-D'/D$. Multiplying by the positive denominators shows that the required inequality is

$$12r^3 + 36r^2 - 7r - 7 = (2r-1)(6r^2 + 21r + 7) \le 0.$$

This holds because $0 < r \le 1/2$. Continuity gives the endpoint $D = 0$. Since $f(1/2) = 2\log 2$, we have

$$DKf(r) \le \frac{3}{2}\log 2 < \frac{21}{20}. \tag{4.16}$$

On $1/10 \le s \le \tau_6$, the same monotonicity and (4.14) already give

$$DS \le 3D(1/10)f(9/20) \le 3 \cdot \frac{281}{1600} \cdot \frac{319}{180} = \frac{268917}{288000} < 1.$$

On $0 \le s \le 1/10$ we keep the correction. The needed estimates are

$$D \ge \frac{281}{1600}, \qquad C \le \frac{61}{40}, \qquad \eta \ge \frac{1}{2}, \qquad K \ge \frac{5}{2}.$$

To prove the last bound, use monotonicity of ψ:

$$\frac{d}{ds}\log K \ge \tfrac{1}{2}(\psi(1/2) - \psi(2)) = -\tfrac{1}{2} - \log 2 > -\frac{6}{5}.$$

Consequently $K(s) \ge 3e^{-3/25} \ge 66/25 > 5/2$. It follows that

$$K\eta - \frac{C^2}{d} \ge \frac{5}{4} - \frac{(61/40)^2}{5/2} = \frac{1279}{4000}.$$

Combining this with (4.16) proves

$$\Theta < -1 + \frac{21}{20} - \frac{281}{1600}\frac{1279}{4000} = -\frac{39399}{6400000} < 0. \tag{4.17}$$

The bounds in all three dimensions give a negative right side in (4.10). This proves the lemma. □

4.3. **Proof of the instability theorem.**

Proof of Theorem 4.1. Interchange the factors if needed so that $m \ge n$. For $N = 4$, Lemma 4.3 gives $\Theta < 0$ directly. For $N = 3, 5, 6$, Proposition 2.5 puts the minimal aperture in the strip used in Proposition 4.2. Its moment bound and Lemma 4.3 again give $\Theta < 0$. The factors in (2.13) and (2.22) are positive, so $H - A_{\mathrm{curv}} < 0$. The sharp criterion and annular cutoffs in Proposition 2.1 give a smooth normal velocity supported away from the vertex with negative quadratic form. □

5. Stability transitions in dimension $N = 7$

Theorem 5.1. *For each $(m,n) = (4,3), (5,2), (6,1)$ and $0 < s < 1$, let $\Sigma_s = \{(y,z) \in \mathbb{R}^m \times \mathbb{R}^n : |z| = \alpha_{m,n}(s)|y|\}$, where $\alpha_{m,n}(s)$ is the unique s-minimal aperture for the kernel $|x-y|^{-7-s}$. Let H_s and $A_{\mathrm{curv}}(s)$ be its optimal Hardy constant and curvature coefficient in (2.2) and (2.3), and write $D_{m,n}(s) = H_s - A_{\mathrm{curv}}(s)$. There is a unique $\sigma_{m,n} \in (0, 7 - 2\sqrt{10})$ such that Σ_s is stable exactly when $0 < s \le \sigma_{m,n}$, with respect to the form (2.4) on $C_c^\infty(\Sigma_s \setminus \{0\})$. The gap $D_{m,n}$ is real analytic on $(0,1)$, extends analytically through $s = 0$, and has a simple downward zero at $\sigma_{m,n}$.*

In this section, we write $\lambda(s) = \alpha_{m,n}(s)^2/(1 + \alpha_{m,n}(s)^2)$, and G and Φ are the minimality function and the auxiliary gap function defined in (2.11) and (2.12). We must study the gap as the minimal aperture moves. The aperture decreases with s. A suitable extension of the gap has a positive partial derivative in aperture on the minimal branch. We combine these signs with derivative bounds at fixed aperture. They show that a normalized gap strictly decreases before the cutoff $7 - 2\sqrt{10}$. The normalization factor is positive. The endpoint signs then give

exactly one crossing, and the spatial comparison rules out later zeros. We first prove the spatial comparison and endpoint signs. We then prove the derivative bounds and use them to prove Theorem 5.1.

5.1. **Spatial order and endpoint signs in dimension seven.** Assume in this section that $N = 7$ and $m > n$. The three possible pairs are $(4, 3)$, $(5, 2)$ and $(6, 1)$. Proposition 2.5 places their minimal branches in the following strips, in the same order:

$$[2/5, 3/7], \qquad [1/5, 2/7], \qquad (0, 1/7] \tag{5.1}$$

and gives $\lambda'(s) < 0$. To use this motion to study stability, we first find an extension of the link-normalized gap whose partial derivative in aperture is positive on the minimal branch. The extension is $\Delta_{\mathrm{sp}} = 2^{1-q}\Phi$ from (2.13); it agrees with Δ on $G = 0$, as in Proposition 2.2.

For $x > 0$ and real ν, write

$$K_\nu(x) = \int_0^\infty e^{-x\cosh t}\cosh(\nu t)\,\mathrm{d}t. \tag{5.2}$$

Here the order parameter satisfies $2 < \beta < 5/2$.

Lemma 5.2. *For $2 \le \beta \le 5/2$, the function*

$$h_\beta(x) = \frac{x\{K_{\beta-1}(x) - K_0(x)\}}{K_\beta(x)} \tag{5.3}$$

is strictly increasing on $(0, \infty)$, with limits

$$h_\beta(0+) = 0, \qquad h_\beta(\infty) = \frac{(\beta-1)^2}{2}.$$

Proof. Put $c = \beta - 1 \in [1, 3/2]$. Integration by parts in (5.2) gives

$$h_\beta(x) = \frac{\displaystyle\int_0^\infty e^{-x\cosh t}\cosh((c+1)t)v_c(t)\,\mathrm{d}t}{\displaystyle\int_0^\infty e^{-x\cosh t}\cosh((c+1)t)\,\mathrm{d}t},$$

where

$$v_c(t) = \frac{c\sinh(ct)\sinh t - (\cosh(ct) - 1)\cosh t}{\sinh^2 t\cosh((c+1)t)}. \tag{5.4}$$

Indeed, before division by K_β, the numerator is

$$\int_0^\infty e^{-x\cosh t}\frac{d}{dt}\left\{\frac{\cosh(ct) - 1}{\sinh t}\right\}\mathrm{d}t.$$

The quotient in braces is $O(t)$ at zero and $O(e^{(c-1)t})$ at infinity. The factor $e^{-x\cosh t}$ therefore makes both boundary terms vanish for every $x > 0$. We will prove that v_c is strictly decreasing.

Let N_c, D_c be the numerator and denominator in (5.4), and put $A_0 = (c+1)^2$, $B_0 = (c-1)^2$, $C_0 = (c+3)^2$. The hyperbolic addition formulas give

$$N_c(t) = \frac{1}{2}\{(c-1)\cosh((c+1)t) - (c+1)\cosh((c-1)t) + 2\cosh t\},$$
$$D_c(t) = \frac{1}{4}\{\cosh((c+3)t) + \cosh((c-1)t) - 2\cosh((c+1)t)\}.$$

Both constant coefficients are zero. For $k \ge 1$, the coefficients of $t^{2k}/(2k)!$ are $n_k/2, d_k/4$, where

$$n_k = (c-1)A_0^k - (c+1)B_0^k + 2, \qquad d_k = C_0^k + B_0^k - 2A_0^k.$$

The identity $n_k = (c-1)(A_0^k - B_0^k) + 2(1 - B_0^k)$ shows $n_k > 0$. Moreover, $d_1 = 8$ and

$$n_{k+1} - A_0 n_k = 4c(c+1)B_0^k - 2c(c+2) \le c(c+1) - 2c(c+2) < 0,$$
$$d_{k+1} - A_0 d_k = (4c+8)C_0^k - 4cB_0^k > 0.$$

Starting with $d_1 = 8$, the second inequality gives $d_k > 0$. The two inequalities also give

$$\frac{n_{k+1}}{d_{k+1}} < \frac{A_0 n_k}{A_0 d_k} = \frac{n_k}{d_k}.$$

Thus the coefficient ratios strictly decrease. The double series for $N_c' D_c - N_c D_c'$ converges absolutely. Pair the terms with indices $j > k$. Each contribution is a positive factor times $(j-k)(n_j d_k - n_k d_j)$, which is negative. Hence $v_c'(t) < 0$ for $t > 0$.

Set

$$\omega_x(t) = e^{-x\cosh t}\cosh((c+1)t), \qquad J(x) = \int_0^\infty \omega_x(t)\,\mathrm{d}t = K_\beta(x).$$

The quotient above is the integral of $v_c\omega_x$ divided by $J(x)$. Differentiate this quotient and use (2.15) to get

$$h_\beta'(x) = -\frac{1}{2J(x)^2}\int_0^\infty\int_0^\infty [v_c(t) - v_c(r)](\cosh t - \cosh r)\omega_x(t)\omega_x(r)\,\mathrm{d}t\,\mathrm{d}r > 0.$$

For fixed $x > 0$, the exponential factor makes these integrals absolutely convergent and allows differentiation under the integral. The product in square brackets and parentheses is strictly negative when $t \ne r$: v_c decreases strictly, while cosh increases strictly on this interval. This proves the strict inequality.

Taylor expansion gives $v_c(0) = c^2/2$, and (5.4) gives $v_c(t) \to 0$ as $t \to \infty$. Thus v_c is bounded and continuous after we fill in its removable value at zero. We now prove the limits of the quotient by estimating its integrals. For each $\varepsilon > 0$ and $x \ge 1$,

$$J(x) \ge \frac{\varepsilon}{2}e^{-x\cosh(\varepsilon/2)}, \qquad \int_\varepsilon^\infty \omega_x(t)\,\mathrm{d}t \le e^{-(x-1)\cosh\varepsilon}\int_\varepsilon^\infty \omega_1(t)\,\mathrm{d}t.$$

The second integral divided by $J(x)$ therefore tends to zero as $x \to \infty$. Splitting the numerator at ε gives

$$|h_\beta(x) - v_c(0)| \le \sup_{0\le t\le\varepsilon}|v_c(t) - v_c(0)| + 2\|v_c\|_\infty \frac{\int_\varepsilon^\infty \omega_x(t)\,\mathrm{d}t}{J(x)}.$$

First let $x \to \infty$, then let $\varepsilon \downarrow 0$. This proves the limit at infinity.

For $0 < x < 1$, the inequality $x\cosh t \le 1$ on $0 \le t \le \log(1/x)$ gives

$$J(x) \ge e^{-1}\int_0^{\log(1/x)} \cosh(\beta t)\,\mathrm{d}t = \frac{e^{-1}}{\beta}\sinh(\beta\log(1/x)) \longrightarrow \infty.$$

For every fixed $R > 0$,

$$|h_\beta(x)| \le \frac{\|v_c\|_\infty \int_0^R \cosh(\beta t)\,\mathrm{d}t}{J(x)} + \sup_{t\ge R}|v_c(t)|.$$

Let $x \downarrow 0$, then let $R \to \infty$. This proves the limit at zero. □

Proposition 5.3. *At every minimal aperture in dimension seven, $\Phi_\lambda(s, \lambda(s)) > 0$ for $0 < s < 1$. In addition, all three minimal cones are strictly unstable for*

$$s_* \le s < 1, \qquad s_* = 7 - 2\sqrt{10}. \tag{5.5}$$

Proof. Set $P_\lambda(x) = \phi_m((1-\lambda)x)\phi_n(\lambda x)$ and $f_\lambda(x) = r_n(\lambda x) - r_m((1-\lambda)x)$. Applying the Laplace formula for a negative power to the paired numerator of (2.12) gives

$$\Phi(s,\lambda) = \frac{1}{\Gamma(q)} \int_0^\infty x^{q-1} \{K_\beta(x) P_\lambda'(x) - K_0(x) P_\lambda(x)\} \, \mathrm{d}x. \tag{5.6}$$

Here $P_\lambda'(x)$ denotes differentiation with respect to x. This is a paired integral. To apply Fubini's theorem, keep the numerator $Z\cosh(\beta t) - 1$ together when introducing the gamma integral. Its absolute value times the original kernel is integrable by Lemma 2.3. Fubini's theorem therefore applies to the signed expression. We do not claim that the two terms on the right are separately integrable at infinity.

Differentiating (5.2) under the integral and integrating the derivative of $e^{-x\cosh t}\sinh(\beta t)$ gives

$$x \int_0^\infty e^{-x\cosh t} \sinh(\beta t) \sinh t \, \mathrm{d}t = \beta K_\beta(x), \qquad K_\beta' = -K_{\beta-1} - \beta K_\beta / x.$$

The boundary terms in the first identity vanish at zero and infinity. The second identity follows from the hyperbolic addition formula. On a finite interval $[\varepsilon, R]$, integration by parts now uses

$$\frac{d}{dx}\{x^{q-1} K_\beta(x)\} = s x^{q-2} K_\beta(x) - x^{q-1} K_{\beta-1}(x).$$

The endpoint estimates in the proof of Proposition 2.5 together with evenness of the spherical averages give

$$\begin{aligned} K_\beta(x) &= O(x^{-\beta}), & P_\lambda(x) &= 1 + O(x^2), & x &\downarrow 0, \\ K_\beta(x) &= O(e^{-x} x^{-1/2}), & P_\lambda(x) &= O(e^x x^{-5/2}), & x &\to \infty. \end{aligned}$$

They are uniform on compact exponent and aperture intervals. Since $q - 1 - \beta = s$ and $q - 4 = (s-1)/2$, the boundary factor $x^{q-1} K_\beta P_\lambda$ is $O(x^s)$ at zero and $O(x^{(s-1)/2})$ at infinity. Both boundary terms vanish for $0 < s < 1$. We therefore obtain

$$\Phi(s,\lambda) = \frac{1}{\Gamma(q)} \int_0^\infty x^{q-2} K_\beta(x) P_\lambda(x) (h_\beta(x) - s) \, \mathrm{d}x. \tag{5.7}$$

The resulting integral converges absolutely. Its endpoint powers are x^{s-1} and $x^{(s-3)/2}$. Thus we have used this integration by parts only for $s > 0$.

Differentiating P_λ gives $\partial_\lambda P_\lambda = x P_\lambda f_\lambda$. At a minimal aperture, Proposition 2.5 gives the unique point x_0 where f_λ changes sign, and gives

$$\int_0^\infty x^{q-1} K_\beta P_\lambda f_\lambda \, \mathrm{d}x = 0.$$

We may differentiate (5.7) in aperture. At infinity, the extra factor x is offset by $f_\lambda = O(x^{-1})$. At zero, $f_\lambda = O(x)$, so the differentiated integrand is $O(x^{s+1})$. At infinity it is $O(x^{(s-3)/2})$. These estimates hold uniformly on each compact aperture interval, by the expansions for r_d in Lemma 2.4. Subtract $h_\beta(x_0) - s$ times the last zero integral to get

$$\Phi_\lambda = \frac{1}{\Gamma(q)} \int_0^\infty x^{q-1} K_\beta P_\lambda f_\lambda(x) (h_\beta(x) - h_\beta(x_0)) \, \mathrm{d}x > 0.$$

Both signed factors are negative below x_0 and positive above it.

For the second claim, Lemma 5.2 gives the strict bound $h_\beta(x) < (\beta-1)^2/2 = (3-s)^2/8$ at every finite x. The right side of (5.7) is therefore negative when $s \geq (3-s)^2/8$. On $(0,1)$ this is exactly $s \geq s_*$. The inequality remains strict at $s = s_*$. The sign agreement on $G = 0$ in Proposition 2.2, together with the sharp criterion in Proposition 2.1, proves strict instability. □

5.1.1. *The positive limiting gap.* An elementary angular integral gives the sign at the other endpoint. By [8, Proposition 11.2], the minimal aperture has a finite positive limit $\alpha(0) = \lim_{s\downarrow 0}\alpha_{m,n}(s)$. It is the unique positive solution of

$$\int_0^{\alpha(0)} \frac{t^{n-1}}{(1+t^2)^{7/2}}\,\mathrm{d}t = \frac{1}{2}\int_0^{\infty} \frac{t^{n-1}}{(1+t^2)^{7/2}}\,\mathrm{d}t, \tag{5.8}$$

Set $u = t^2/(1+t^2)$ and $u_0 = \alpha(0)^2/(1+\alpha(0)^2)$. We write $\lambda(0) = u_0$. The equation becomes the incomplete beta integral equation

$$\int_0^{u_0} u^{n/2-1}(1-u)^{m/2-1}\,\mathrm{d}u = \frac{1}{2}B(n/2, m/2).$$

Write $z = \sqrt{u_0}$ and integrate to get

$$\begin{array}{ll} 5z^3 - 3z^5 = 1, & (m,n) = (4,3), \\ 1 + \alpha(0)^2 = 2^{2/5}, & (m,n) = (5,2), \\ 15z - 10z^3 + 3z^5 = 4, & (m,n) = (6,1), \end{array} \qquad z = \frac{\alpha(0)}{\sqrt{1+\alpha(0)^2}}. \tag{5.9}$$

The derivatives of the two polynomials on the left are $15z^2(1-z^2)$ and $15(1-z^2)^2$, respectively. Thus the solutions of these two equations in $(0,1)$ are simple. In each case $\lambda(s) \to \lambda(0) = u_0 \in (0,1)$. The local uniform convergence in Lemma 2.3 gives $G(0,u_0) = 0$, and (2.14) gives $G_\lambda(0,u_0) > 0$. The analytic implicit function theorem therefore continues the minimal branch through $s = 0$. Uniqueness identifies this continuation with the given branch for small positive s.

Proposition 5.4. *For each of the three pairs, the limiting minimal cone has $\Delta(0) > 0$. In the pairs $(5,2)$ and $(6,1)$, respectively,*

$$\Delta(0) > \frac{337}{180075}, \qquad \Delta(0) > \frac{7}{4096}.$$

Proof. We evaluate all quantities in this proof at $s = 0$. Write $a = 1-b$, $b = \lambda$, and set

$$X = \mathcal{I}_{m,n}[(1-Z)^{-2}] = \sum_{k=0}^{\infty}(2k+1)M_{2k}.$$

The angular endpoint estimates show that the integral is finite. Symmetry and monotone convergence give the series identity. With the notation of Proposition 2.6, the duplication formula gives

$$c_k R_k = \frac{4}{15}(k+\tfrac{1}{2})(k+\tfrac{3}{2}), \qquad k \geq 1.$$

Apply strict log-convexity at the midpoint $k+7/4$ of $k+3/2, k+2$ to get

$$c_k < \frac{4}{15}(k+\tfrac{1}{2})(k+1), \qquad c_k(R_k - 1) > \frac{2k+1}{15}.$$

At $k = 0$, the same inequality gives $c_0 < 2/15$. The paired series converges absolutely, so

$$\Delta_{\mathrm{sp}}(0,\lambda) > \frac{X-3}{15}. \tag{5.10}$$

We must now bound X at the three limiting apertures.

For $(m,n) = (4,3)$, the weight in the second variable is $w_3(v) = 1/2$. Integrate first in this variable to get

$$X = \mathcal{I}_4[((1-au)^2 - b^2)^{-1}] = \frac{2}{z(1-z)(1+z)^2}, \qquad z = \sqrt{b}.$$

For the second equality, use partial fractions and the semicircle integral $\mathcal{I}_4[(t-u)^{-1}] = 2(t-\sqrt{t^2-1})$, $t \geq 1$. To check this integral for $t > 1$, set $u = \cos\theta$ and use

$$\int_0^\pi \frac{\mathrm{d}\theta}{t-\cos\theta} = \frac{\pi}{\sqrt{t^2-1}}.$$

The substitution $\tan(\theta/2)$ gives this last identity. Polynomial division handles the factor $\sin^2\theta$ in the semicircle weight. Continuity gives the value at $t = 1$. With $y = z^2$ and $z \leq (1+z^2)/2$,

$$z(1-z)(1+z)^2 = (z+z^2)(1-z^2) \leq \frac{(1+3y)(1-y)}{2} = \frac{2}{3} - \frac{3}{2}(y-\tfrac{1}{3})^2 \leq \frac{2}{3}.$$

Hence $X \geq 3$. This bound does not require minimality.

For $(m,n) = (5,2)$, first integrate in v with the arcsine weight $w_2(v) = 1/(\pi\sqrt{1-v^2})$. The same trigonometric integral gives

$$\mathcal{I}_2[(c-bv)^{-1}] = (c^2-b^2)^{-1/2}, \qquad c > b > 0.$$

Differentiate in c to integrate the inverse square. With $F(u) = \{(1-au)^2 - b^2\}^{-1/2}$ and $w_5(u) = 3(1-u^2)/4$, this gives

$$\begin{aligned} X &= \frac{3}{4}\int_{-1}^1 \frac{(1-u^2)(1-au)}{\{(1-au)^2-b^2\}^{3/2}}\,\mathrm{d}u \\ &= \frac{3}{4a}\int_{-1}^1 (1-u^2)F'(u)\,\mathrm{d}u. \end{aligned}$$

The boundary term vanishes at $u = -1$ and is $O((1-u)^{1/2})$ at $u = 1$. Integration by parts therefore gives $X = 3\int_{-1}^1 uF(u)\,\mathrm{d}u/(2a)$. The change of variables $w = 1-au$ yields

$$\int_{-1}^1 uF(u)\,\mathrm{d}u = \frac{1}{a^2}\left[\operatorname{arcosh}\frac{w}{b} - \sqrt{w^2-b^2}\right]_{w=b}^{w=2-b}.$$

Since $\sqrt{(2-b)^2-b^2} = 2\sqrt{a}$, writing $r = \sqrt{a}$ gives

$$X = \frac{3}{2a^3}\left(\operatorname{arcosh}\frac{2-b}{b} - 2\sqrt{a}\right) = \frac{3(\operatorname{arctanh} r - r)}{r^6}.$$

The limiting-aperture equation gives $r = 2^{-1/5}$. The integer inequalities $3^5 < 2^8$ and $2^{14} < 7^5$ give $r^2 > 3/4$ and $r < 7/8$. For the integral that remains after we take out the first two terms of the geometric series, Cauchy–Schwarz gives

$$\left(\int_0^1 t^6\,\mathrm{d}t\right)^2 \leq \left(\int_0^1 \frac{t^6}{1-r^2t^2}\,\mathrm{d}t\right)\left(\int_0^1 t^6(1-r^2t^2)\,\mathrm{d}t\right).$$

The two elementary integrals are $1/7$ and $1/7 - r^2/9$. Thus the remaining integral is at least $1/[7(1-7r^2/9)]$, and

$$\begin{aligned} \frac{\operatorname{arctanh} r - r}{r^3} &= \frac{1}{3} + \frac{r^2}{5} + r^4\int_0^1 \frac{t^6}{1-r^2t^2}\,\mathrm{d}t \\ &\geq \frac{1}{3} + \frac{r^2}{5} + \frac{r^4}{7(1-7r^2/9)} > \frac{1}{3} + \frac{3}{20} + \frac{27}{140} = \frac{71}{105} > \frac{343}{512} > r^3. \end{aligned}$$

Since $71\cdot 512 - 343\cdot 105 = 337$, the formula for X gives

$$X - 3 = \frac{3}{r^3}\left\{\frac{\operatorname{arctanh} r - r}{r^3} - r^3\right\} > \frac{3\cdot 512}{343}\left(\frac{71}{105} - \frac{343}{512}\right) = \frac{337}{12005}.$$

For $(m,n)=(6,1)$, the second angular integral is the half-sum over $v=-1,1$. The Cauchy integral in the first variable is

$$g_6(t)=\mathcal{I}_6[(t-u)^{-1}]=\frac{4}{3}\{3t-2t^3+2(t^2-1)^{3/2}\},\qquad t\ge 1.$$

To obtain it, for example, use the relation $w_6(u)=4(1-u^2)w_4(u)/3$ between the angular weights. Differentiate this identity and take the half-sum over $v=-1,1$. Differentiation is valid also at $t=1$ because $w_6(u)/(1-u)^2$ is integrable. For $z=\sqrt{b}$ the result is

$$X=-\frac{g_6'(1)+g_6'((1+b)/a)}{2a^2}=\frac{4(1+z^2)}{(1-z)^2(1+z)^4}.$$

The residual $15z-10z^3+3z^5-4$ of the last equation in (5.9) equals $-413/1024$ at $z=1/4$, so its root satisfies $z>1/4$. The denominator has maximum $1024/729$ at $z=1/3$ on $[0,1]$. Thus $X>12393/4096=3+105/4096$. Insert the three estimates into (5.10). On the minimal branch, (2.13) gives $\Delta_{\mathrm{sp}}=\Delta=(H-A_{\mathrm{curv}})/L_\Sigma$. Thus these estimates give the stated positive limits for the link-normalized gap. The physical gap is $L_\Sigma\Delta$. □

5.2. **Quantitative decrease before the seven-dimensional cutoff.** The aperture term in the derivative of $\Phi(s,\lambda(s))$ is negative by Proposition 5.3. We still need to bound the term at fixed aperture. We choose a normalization that lets us prove this bound by summing coefficient inequalities over the whole strips (5.1). In this subsection, set

$$A=\frac{5}{2},\quad B=\frac{s}{2},\quad C=\frac{7+s}{4},\quad D=C^2-AB-A-B=\frac{s^2-14s+9}{16},\quad r=\frac{1-s}{2}.\tag{5.11}$$

Then $D\ge 0$ on $[0,s_*]$. Define

$$\Theta(s,\lambda)=\frac{2\Gamma(q)}{\Gamma(C)^2}\Delta_{\mathrm{sp}}(s,\lambda)=\frac{2^{2-q}\Gamma(q)}{\Gamma(C)^2}\Phi(s,\lambda)=-1+\sum_{k\ge1}a_kM_{2k}(R_k-1),\qquad a_k=\frac{4^k(C)_k^2}{(2k)!}.\tag{5.12}$$

In all partial s derivatives in this subsection, λ is fixed.

Lemma 5.5. *The series* (5.12) *and the series of its termwise s derivatives converge locally uniformly for $-1<s<1$ on the strips* (5.1)*. The same holds after multiplication by a smooth function of s. For $(m,n)=(6,1)$, the bound* (5.13) *below is uniform even at $\lambda=0$.*

Proof. For $(4,3)$ and $(5,2)$, the endpoint integral estimate in the proof of Proposition 2.6 gives $M_{2k}=O(k^{-5/2})$ uniformly on their compact aperture strips. For $(m,n)=(6,1)$ we use the uniform envelope

$$M_{2k}\le\frac{5}{3}k^{-5/2},\qquad k\ge1,\quad 0\le b\le 1/7.\tag{5.13}$$

The angular weight is $w_6(u)=8(1-u^2)^{3/2}/(3\pi)$. Use $1-u^2\le 2(1-u)$ and extend the gamma integral to get $e^{-y}\phi_6(y)\le 2^{5/2}/(\sqrt{\pi}y^{5/2})$; also $e^{-y}\phi_1(y)\le 1$. For $|z|\le 1$ and $k\ge1$, the inequality $\log|z|\le|z|-1$ gives

$$z^{2k}\le e^{-2k}(e^{2kz}+e^{-2kz}).$$

Integrate this bound. Evenness and the product form of the angular integral give

$$M_{2k}\le 2e^{-2k}\phi_m(2ka)\phi_n(2kb),\qquad a+b=1.\tag{5.14}$$

For the pair under consideration, this yields

$$M_{2k}\le 2e^{-2k}\phi_6(2k(1-b))\phi_1(2kb)\le\frac{2(7/6)^{5/2}}{\sqrt{\pi}}k^{-5/2}<\frac{5}{3}k^{-5/2}.$$

Square the positive quantities in the last comparison and use $\pi>25/8>16807/5400$.

On a compact exponent interval, use the gamma-ratio expansion with a uniform remainder and the exact derivatives

$$\partial_s R_k = \frac{R_k}{2}\{\psi(k+B)-\psi(k+C)\}, \qquad \partial_s a_k = \frac{a_k}{2}\{\psi(k+C)-\psi(C)\}.$$

The trigamma series bounds the first digamma difference by $O(k^{-1})$, uniformly in the exponent. The second difference is $O(\log(k+1))$. Thus

$$R_k - 1 = O(k^{-1}), \qquad \partial_s R_k = O(k^{-1}), \qquad a_k = O(k^{2+s/2}), \qquad \partial_s a_k = O(a_k \log(k+1)).$$

Thus each differentiated paired summand is bounded by $Kk^{-(3-s_1)/2}\log(k+1)$, where K is constant on the chosen compact parameter set and $s_1 < 1$ is the upper endpoint of the exponent interval. These bounds are summable and uniform on each aperture strip. The same bound without the logarithm applies to the original series. All coefficient arguments $k+B$, $k \geq 1$, stay positive for $-1 < s < 1$, and the zeroth term is smooth. The uniform convergence theorem for differentiated series proves the claim, including continuation through $s = 0$. Multiplying by a smooth function of s preserves these conclusions. Lemma 2.3 already gives differentiability in aperture. □

5.2.1. *Coefficient inequalities for all three angular integrals.* Lemma 2.8 gives the elementary bounds

$$\psi'(x) \geq \frac{1}{x} + \frac{1}{2x^2}, \qquad \psi(x) \leq \log x - \frac{1}{2x}, \qquad \frac{\Gamma(x+1/2)}{\Gamma(x)} \geq \frac{x}{\sqrt{x+1/2}} \quad (x>0). \tag{5.15}$$

The elementary bounds for constants in Appendix B give

$$\psi(C) \geq \psi(7/4) = -\gamma + \frac{\pi}{2} - 3\log 2 + \frac{4}{3} > \frac{181}{750} > \frac{6}{25}. \tag{5.16}$$

The following bounds also hold uniformly on $[0, s_*]$:

$$B < \frac{17}{50}, \quad \frac{7}{4} \leq C < \frac{48}{25}, \quad C - B > \frac{79}{50}, \quad B + C < \frac{113}{50}, \quad 0 \leq D \leq \frac{9}{16}. \tag{5.17}$$

Here $s_* < 17/25$ follows by squaring $\sqrt{10} > 79/25$.

Lemma 5.6. *For $k \geq 1$ and $0 \leq s \leq s_*$,*

$$R_k \leq 1 + \frac{D}{k}, \qquad R_1 > \frac{7}{10}.$$

Proof. The first estimate follows from the supersolution bound in Lemma 2.9. That lemma also gives $\partial_s \log R_k < 0$. The coefficients are still defined at $s = 1$, where $R_1(1) = 15\pi/64 > 7/10$. We use no geometric integral at $s = 1$. □

Put $u_k = \psi(k+B) - \psi(C)$ and $\delta_k = \psi(k+C) - \psi(k+B)$. Direct differentiation gives

$$\partial_s\{a_k(R_k - 1)\} = \frac{a_k}{2}\{(R_k - 1)u_k - \delta_k\}. \tag{5.18}$$

For $k \geq 2$, $u_k > 0$ and

$$u_k \leq \log k - \frac{6}{25}. \tag{5.19}$$

Indeed, (5.15) gives $\psi(k+B) \leq \log k$ because $\log(1 + B/k) \leq B/k \leq 1/(2(k+B))$ on (5.17). Integrate the lower bound for ψ' and use $\log(y/x) \geq 2(y-x)/(y+x)$ to get

$$\begin{aligned}\delta_k &\geq \log\frac{k+C}{k+B} + \frac{1}{2}\left(\frac{1}{k+B} - \frac{1}{k+C}\right) \\ &\geq \frac{C-B}{k+(B+C)/2} + \frac{C-B}{2(k+B)(k+C)}.\end{aligned}$$

Inserting (5.17) gives

$$\delta_k \geq \frac{79/50}{k+113/100} + \frac{79/100}{(k+17/50)(k+48/25)}. \tag{5.20}$$

We will use three consequences several times.

First, $C-1-B > 29/50$ and (5.15) give

$$\psi(C)-\psi(1+B) \geq \frac{29}{50}\left\{\frac{25}{48}+\frac{1}{2}\left(\frac{25}{48}\right)^2\right\} = \frac{3509}{9216} > \frac{3}{8}.$$

The digamma recurrence gives $\delta_1 = -u_1 + 1/C$. Thus (5.18) at the first index becomes

$$\partial_s\{a_1(R_1-1)\} = C^2R_1u_1 - C.$$

Since $a_1 = 2C^2$, $R_1 > 7/10$, and $M_2 \geq 1/7$ for every seven-dimensional angular pair, the first summand in (5.12) satisfies

$$\partial_s\{a_1M_2(R_1-1)\} \leq -CM_2 - \frac{21}{80}C^2M_2 \leq -\frac{467}{1280}. \tag{5.21}$$

The lower bound for M_2 is Cauchy's inequality applied to $a^2/m + b^2/n$ with $a+b=1$.

Second, (5.20) at $k=2$ is at least $4241431/7177716 > 59/100$. Thus

$$\frac{1}{2}\{(R_2-1)u_2 - \delta_2\} \leq \frac{1}{2}\left\{\frac{9}{32}\left(\frac{7}{10}-\frac{6}{25}\right) - \frac{59}{100}\right\} < -\frac{11}{50}. \tag{5.22}$$

Third, multiply (5.20) by k to get

$$k\delta_k \geq \frac{79}{50}\left\{1 - \frac{113}{100k} + \frac{k}{2(k+48/25)^2}\right\} =: \ell_0(k). \tag{5.23}$$

Its derivative is at least $(79/50)(113/100-1/2)k^{-2} > 0$. For $3 \leq k \leq 5$, $\ell_0(k) \geq \ell_0(3) > 1$, while $D(\log k - 6/25) \leq (9/16)(5/3-6/25) < 1$. For $6 \leq k \leq 9$, use $\ell_0(6) > 4/3$ and $(9/16)(9/4-6/25) < 4/3$ instead. The coefficient derivatives for these modes are nonpositive. For $k \geq 10$, the bound

$$\frac{1}{2}\{(R_k-1)u_k - \delta_k\} \leq \frac{D\log k - (29/20 + 6D/25)}{2k} \tag{5.24}$$

follows from $\ell_0(10) = 1617409423/1110050000 > 29/20$.

Lemma 5.7. *Let $0 \leq s \leq s_*$ and $K > 0$. Suppose that $a_kM_{2k} \leq Kk^{-r}$ for the indices included in a positive tail that starts at an index at least ten. Their total contribution to Θ_s is less than $297K/1024$.*

Proof. Set $a_0 = 29/20$, $p_0 = 6/25$ and $A_0 = a_0 + p_0D$. By (5.24), that contribution is bounded by

$$\frac{K}{2}\sum_{k\geq 10} k^{-1-r}(D\log k - A_0)_+.$$

For $D=0$, this is zero. Otherwise, extend the nonnegative summand to a function f on $(0,\infty)$. This function is unimodal. Split the integer sum at its maximum and compare each part with the adjacent integrals. This bounds the sum by $\int f + \max f$. More explicitly,

$$I(s) := \int_0^\infty f(x)\,\mathrm{d}x = \frac{D}{r^2}e^{-rA_0/D},$$

$$\max f = \frac{D}{1+r}e^{-(1+r)A_0/D-1} \leq De^{-a_0/D-p_0-1} \leq \frac{9}{16}e^{-859/225}.$$

For the last inequality, use $D \le 9/16$ and the monotonicity of $De^{-a_0/D}$.

We claim $I(s) \le I(0) = (9/4)e^{-317/225}$. Write $u = 1-s$ and $t = D/r = (u+12-4/u)/8$. The derivative with respect to u of $\log I = \log t - \log(u/2) - a_0/t - p_0 u/2$ is nonnegative if and only if $ut_u(t+a_0) - t^2 - (p_0/2)ut^2 \ge 0$. Multiply this expression by $64u^2$ to obtain

$$P(u) = -\frac{3u^5 + 72u^4 + 418u^3 + 3112u^2 - 4712u + 800}{25}.$$

All the relevant values lie in the larger interval $[8/25, 1]$, where

$$P''(u) = -\frac{4}{25}(15u^3 + 216u^2 + 627u + 1556) < 0,$$

and $P(8/25) = 3659268896/244140625 > 0$, $P(1) = 307/25 > 0$. Concavity therefore proves the claim when $D > 0$. The continuous limit is zero when $D = 0$. Finally, $\log 2 < 7/10$ gives $e^{317/225} > 4$ and $e^{859/225} > 32$. Hence

$$\frac{K}{2}(I + \max f) < \frac{K}{2}\left(\frac{9/4}{4} + \frac{9/16}{32}\right) = \frac{297K}{1024}.$$

□

5.2.2. *Two continuous angular integrals.*

Proposition 5.8. *On $0 \le s \le s_*$ the following fixed-aperture bounds hold:*

$$\Theta_s(s,b) \le -\frac{1211}{96000}, \qquad 2/5 \le b \le 3/7, \quad (m,n) = (4,3),$$
$$\Theta_s(s,b) \le -\frac{557659163}{45465600000}, \qquad 1/5 \le b \le 2/7, \quad (m,n) = (5,2).$$

Proof. First consider $(4,3)$ and put $a = 1-b$. Changing variables to $z = au + bv$ in the angular integral gives the weight

$$p(z) = \frac{1}{\pi b a^2}\int_{[z-b,z+b]\cap[-a,a]} \sqrt{a^2 - w^2}\,\mathrm{d}w.$$

More precisely, $\mathcal{I}_{4,3}[F(Z)] = \int_{-1}^{1} F(z)p(z)\,\mathrm{d}z$ for every integrable function F. For $0 \le z \le 1$, set $v = a - w$. The integration interval lies in $[0, 1-z]$, and $\sqrt{a^2-w^2} \le \sqrt{2av}$. Hence

$$p(z) \le \frac{2\sqrt{2}}{3\pi b a^{3/2}}(1-z)^{3/2}, \qquad M_{2k} \le \frac{\sqrt{2/\pi}}{ba^{3/2}}\frac{\Gamma(2k+1)}{\Gamma(2k+7/2)}.$$

The duplication formula gives

$$a_k M_{2k} \le \frac{1}{4ba^{3/2}\Gamma(C)^2}\frac{\Gamma(k+C)^2}{\Gamma(k+7/4)\Gamma(k+9/4)}.$$

Use log-convexity to interpolate C between $7/4$ and $9/4$. The last gamma ratio is at most $\{\Gamma(k+7/4)/\Gamma(k+9/4)\}^{1-s} \le k^{-r}$, where the last step uses the half-gamma bound in Lemma 2.8. On the aperture interval, $b^2(1-b)^3$ decreases and its minimum is $576/16807 > (37/200)^2$. The tangent bound for $\log\Gamma$ at two, together with $\psi(2) = 1-\gamma < 1/2$, gives $\Gamma(C) \ge e^{-1/8} \ge 7/8$. Hence

$$a_k M_{2k} \le \frac{3200}{1813}k^{-r} < \frac{9}{5}k^{-r}. \tag{5.25}$$

The first-mode bound is (5.21). For the second mode, $a_2 \ge 5929/384$ and

$$M_4 = \frac{(1-b)^4}{8} + \frac{(1-b)^2b^2}{2} + \frac{b^4}{5} \ge \frac{1}{20}.$$

To check the last bound exactly, put $t = b - 5/12$, so $-1/60 \le t \le 1/84$. Expansion gives

$$40(M_4 - 1/20) = 33t^4 - 5t^3 + \frac{75}{8}t^2 - \frac{5}{144}t + \frac{11}{6912} \ge 9t^2 - \frac{5}{144}t + \frac{11}{6912} > 0;$$

the quadratic has discriminant $-1163/20736$. Use this with (5.22) to get

$$\partial_s\{a_2 M_4(R_2 - 1)\} \le -\frac{65219}{384000}.$$

Modes three through nine are nonpositive. For the remaining modes, Lemma 5.7 applies with $K = 9/5$ by (5.25). It gives the bound $2673/5120$. Thus

$$\Theta_s \le -\frac{467}{1280} - \frac{65219}{384000} + \frac{2673}{5120} = -\frac{1211}{96000}.$$

For $(5, 2)$, the same coefficient estimates apply. We need a different angular envelope. Here

$$M_2 = \frac{a^2}{5} + \frac{b^2}{2}, \qquad M_4 = \frac{3a^4}{35} + \frac{3a^2b^2}{5} + \frac{3b^4}{8} =: p(b).$$

The first polynomial decreases on $[1/5, 2/7]$, giving $1/7 \le M_2 \le 37/250$. On this interval, differentiation gives $p''' < 0$ and $p'' \ge p''(2/7) = 1068/1715 > 3/5$. Use $p(1/4) = 3561/71680$ and $p'(1/4) = -39/4480$ in Taylor's formula, with $h = b - 1/4$, to get

$$M_4 \ge \frac{3561}{71680} - \frac{39}{4480}h + \frac{3}{10}h^2 \ge \frac{79665}{1605632} > \frac{49}{1000}.$$

Apply Cauchy–Schwarz to Z and Z^3 in the angular integral. This gives

$$M_6 \ge M_4^2/M_2 \ge (49/1000)^2/(37/250).$$

At $k = 3$, (5.20), (5.19) and $\log 3 < 11/10$ yield

$$\begin{aligned}\frac{1}{2}\{(R_3 - 1)u_3 - \delta_3\} &\le \frac{1}{2}\left\{\frac{9}{48}\left(\frac{11}{10} - \frac{6}{25}\right) - \frac{79/50}{3 + 113/100} - \frac{79/100}{(3 + 17/50)(3 + 48/25)}\right\} \\ &= -\frac{1828289543}{13573492800} < -\frac{13}{100}.\end{aligned}$$

Since $a_3 \ge 29645/1024$, the first three contributions are bounded above by

$$-\frac{467}{1280} - \frac{11}{50}\frac{5929}{384}\frac{49}{1000} - \frac{13}{100}\frac{29645}{1024}\frac{(49/1000)^2}{37/250} = -\frac{26931259163}{45465600000}. \tag{5.26}$$

The coefficient estimates already show that modes four through nine are nonpositive. For $10 \le k \le 16$, (5.24) and $\log 16 < 4(347/500)$ give the stronger upper bound

$$D(\log k - 6/25) - 29/20 < \frac{9}{16}\left(4\frac{347}{500} - \frac{6}{25}\right) - \frac{29}{20} < 0.$$

We now prove a tail envelope for every $k \ge 17$. Elementary integration gives

$$\phi_5(x) = \frac{3(x\cosh x - \sinh x)}{x^3}, \qquad e^{-x}\phi_5(x) = \frac{3\{x - 1 + (x + 1)e^{-2x}\}}{2x^3} \le \frac{3}{2x^2}.$$

For $\phi_2(x) = I_0(x)$, split its angular integral at $\pi/2$. In the first half, use $y = \sin(\theta/2)$ and $(1 - y^2)^{-1/2} \le 1 + y^2$ for $y^2 \le 1/2$. Extend the resulting Gaussian integrals to infinity. The second half is at most $e^{-x}/2$. This proves

$$e^{-x}I_0(x) \le \frac{1}{\sqrt{2\pi x}}\left(1 + \frac{1}{4x}\right) + \frac{1}{2}e^{-x}.$$

Use (5.14) and the two bounds above:

$$\begin{aligned} M_{2k} &\le 2e^{-2k}\phi_5(2ka)I_0(2kb) \\ &\le \frac{3}{8\sqrt{\pi}a^2\sqrt{b}}\left(1+\frac{1}{8kb}\right)k^{-5/2}+\frac{3}{8a^2}e^{-2kb}k^{-2}. \end{aligned}$$

The function $(1-b)^2\sqrt{b}$ decreases on the strip, so

$$\frac{3}{8\sqrt{\pi}a^2\sqrt{b}} \le \frac{147\sqrt{7}}{200\sqrt{2\pi}} < \frac{777}{1000}.$$

For $k \ge 17$, the remaining factors satisfy

$$1+\frac{1}{8kb} \le \frac{141}{136}, \quad \frac{3}{8a^2} \le \frac{147}{200}, \quad \sqrt{k}\,e^{-2k/5} \le \sqrt{17}\,e^{-34/5} < \frac{33}{6400}.$$

For the last comparison, use $\sqrt{17} < 33/8$ and $e^{17/10} > \sum_{j=0}^{5}(17/10)^j/j! > 27/5$, then take fourth powers. These rational bounds give

$$k^{5/2}M_{2k} < \frac{777}{1000}\frac{141}{136}+\frac{4851}{1280000} = \frac{17611587}{21760000} < \frac{81}{100}. \tag{5.27}$$

For the coefficients, first use the stronger lower bound

$$\Gamma(C) \ge \Gamma(7/4) > \frac{91}{100}. \tag{5.28}$$

Indeed, $\Gamma(3/2) > 177/200$, $\psi(3/2) = 2-\gamma-2\log 2 > 4/125$, and $\psi'(x) \ge 36/49$ on $[3/2, 7/4]$. Apply Taylor's formula with integral remainder to $\log\Gamma$. It gives

$$\Gamma(7/4) > \frac{177}{200}\exp\left(\frac{1}{125}+\frac{9}{392}\right) > \frac{177}{200}\left(1+\frac{1}{125}+\frac{9}{392}\right) = \frac{8941509}{9800000} > \frac{91}{100}.$$

Monotonicity here follows from (5.16). The duplication formula and log-convexity also give

$$a_k = \frac{\sqrt{\pi}\,\Gamma(k+C)^2}{\Gamma(C)^2\Gamma(k+1/2)\Gamma(k+1)}, \quad \Gamma(k+C) \le \Gamma(k+1)(k+1)^{C-1}, \quad \frac{\Gamma(k+1)}{\Gamma(k+1/2)} \le \sqrt{k+1/2}.$$

For $k \ge 17$, it follows that

$$\frac{a_k}{k^{2+B}} \le \frac{\sqrt{\pi}}{\Gamma(C)^2}(1+1/k)^{2C-2}\sqrt{1+1/(2k)} < \frac{\sqrt{22/7}}{(91/100)^2}\left(\frac{18}{17}\right)^2\sqrt{\frac{35}{34}} < \frac{49}{20}.$$

Check the last comparison by squaring the positive rational numbers. Combine this with (5.27) to get $a_kM_{2k} < 2k^{-r}$ for all $k \ge 17$. Lemma 5.7 bounds this tail by $297/512$; (5.26) now yields

$$\Theta_s \le -\frac{26931259163}{45465600000}+\frac{297}{512} = -\frac{557659163}{45465600000}.$$

□

5.2.3. *A normalization for the discrete angular factor.* For $(m,n) = (6,1)$, we remove the same gamma factor without dividing by $\Gamma(C)^2$. Define

$$\Omega(s,\lambda) = \frac{2\Gamma(q)}{\sqrt{\pi}}\Delta_{\mathrm{sp}}(s,\lambda) = \frac{2^{2-q}\Gamma(q)}{\sqrt{\pi}}\Phi(s,\lambda) = \sum_{k\ge 0} M_{2k}d_k. \tag{5.29}$$

Since $\Omega = \Gamma(C)^2\Theta/\sqrt{\pi}$, Lemma 5.5 allows us to differentiate this series term by term. Here $d_0 = -\Gamma(C)^2/\sqrt{\pi}$ and, for $k \ge 1$,

$$T_k = (k+\tfrac{1}{2})(k+\tfrac{3}{2})\frac{\Gamma(k+B)}{\Gamma(k)}, \quad U_k = \frac{\Gamma(k+C)^2}{\Gamma(k+1/2)\Gamma(k+1)}, \quad d_k = T_k - U_k, \quad R_k = T_k/U_k. \tag{5.30}$$

The coefficient T_k is unrelated to the normal correlation T. The duplication formula identifies R_k with (2.23). In particular,

$$d_k' = \frac{U_k}{2}\{(R_k-1)\psi(k+B)-\delta_k\}, \qquad d_0' = -\frac{\Gamma(C)^2\psi(C)}{2\sqrt{\pi}} < 0. \tag{5.31}$$

Proposition 5.9. *For $0 \le s \le s_*$ and $0 < \lambda \le 1/7$,*

$$\Omega_s(s,\lambda) < -\frac{709}{70000}.$$

Proof. Use $\tau = (1-s)/2$ and $\ell = C - B$. Then

$$\sqrt{10}-3 \le \tau \le \frac{1}{2}, \quad C = 2-\frac{\tau}{2}, \quad B = \frac{1}{2}-\tau, \quad \ell = \frac{3+\tau}{2}, \quad D = \frac{\tau^2+6\tau-1}{4}. \tag{5.32}$$

We keep four negative modes. We need the following moment bounds:

$$M_2 \ge \frac{1}{7}, \qquad M_4 \ge \frac{6}{125}, \qquad M_6 \ge \frac{9}{400}, \qquad M_8 \ge \frac{1}{80}. \tag{5.33}$$

The first bound follows again from Cauchy's inequality. For the others, put $t = 7\lambda \in [0,1]$ and expand the moment polynomials in the Bernstein bases of degrees four, six and eight. In the same order, their coefficients are

$$\begin{gathered}
\frac{1}{16}, \frac{3}{56}, \frac{29}{588}, \frac{33}{686}, \frac{118}{2401};\\
\frac{1}{32}, \frac{3}{112}, \frac{19}{784}, \frac{9}{392}, \frac{325}{14406}, \frac{383}{16807}, \frac{2764}{117649};\\
\frac{7}{384}, \frac{1}{64}, \frac{11}{784}, \frac{9}{686}, \frac{487}{38416}, \frac{1689}{134456}, \frac{17903}{1411788}, \frac{3053}{235298}, \frac{77281}{5764801}.
\end{gathered}$$

Every entry in each list is above the claimed bound. Since the Bernstein basis is nonnegative and sums to one, (5.33) follows. To check the lists directly, expand

$$M_{2k} = \sum_{j=0}^{k} \binom{2k}{2j} (1-b)^{2j} b^{2k-2j} \frac{(1/2)_j}{(3)_j},$$

then use $t^j = \sum_{i=j}^{2k} \binom{i}{j}\binom{2k}{j}^{-1}\binom{2k}{i} t^i (1-t)^{2k-i}$.

For the positive tail, Lemma 5.5 gives the uniform upper envelope (5.13).

We next prove the uniform coefficient bounds

$$d_1' < -\frac{19}{10}, \qquad d_2' < -\frac{12}{5}, \qquad d_3' < -\frac{13}{5}, \qquad d_4' < -\frac{14}{5}. \tag{5.34}$$

The finite scalar estimates in Lemma B.2 give the initial values

$$\begin{gathered}
d_1'(1/2) < -39/20, \qquad 0 < d_1''(1/2) < 1/4, \qquad R_2(0) < 6/5,\\
d_2'(0) < -12/5, \qquad d_3'(0) < -13/5, \qquad d_4'(0) < -14/5.
\end{gathered} \tag{5.35}$$

To extend the first estimate to the whole interval, let $f = d_1'$, $X = 1+B$ and $Y = 1+C$. Differentiating twice gives

$$\begin{aligned}
8f'' = {} & T_1\{\psi(X)^3 + 3\psi(X)\psi'(X) + \psi''(X)\}\\
& - U_1\{\psi(Y)^3 + \tfrac{3}{2}\psi(Y)\psi'(Y) + \tfrac{1}{4}\psi''(Y)\}.
\end{aligned}$$

The first bracket is negative because $X \le 67/50$ and $\psi(X) \le 17/50 - 25/67 < 0$, whereas $\psi' > 0$, $\psi'' < 0$. For the second bracket, $\psi(Y) \ge \psi(11/4) > 6/25 + 4/7 > 4/5$, and the polygamma series gives

$$-\psi''(Y) = 2\sum_{j\ge 0}(Y+j)^{-3} \le \frac{2}{Y^3} + \frac{1}{Y^2} \le \frac{304}{1331} < \frac{1}{4}.$$

Thus the second bracket is greater than $(4/5)^3 - 1/16 > 0$. Hence $f'' < 0$. The tangent at $1/2$ lies above the graph. For $s < 1/2$, this gives $f(s) < -39/20$ because $f'(1/2) > 0$. For $s \ge 1/2$, it gives

$$f(s) < -\frac{39}{20} + \frac{1}{4}\left(\frac{17}{25} - \frac{1}{2}\right) = -\frac{381}{200} < -\frac{19}{10}.$$

For $k = 2, 3, 4$, $R_k(s) \le R_2(0) < 6/5$: R_k decreases in s, and $R_{k+1}(0)/R_k(0) = (k+5/2)(k+1)/(k+7/4)^2 < 1$. With $X = k + B$, $Y = k + C$,

$$4d_k'' = U_k\{R_k[\psi(X)^2 + \psi'(X)] - \psi(Y)^2 - \psi'(Y)/2\}.$$

The three columns below give bounds for $k = 2, 3, 4$:

	2	3	4
$\psi(X) <$	16/25	53/50	34/25
$\psi(Y) >$	117/100	36/25	33/20
$\psi'(X) <$	3/4	4/9	5/16

For the last row, compare the decreasing trigamma series with its integral to get $\psi'(k) < 1/k + 1/k^2$. The middle row follows from (5.16) and the recurrence formula. For the top row, apply $\psi(X) \le \log X - 1/(2X)$ and the bound $\log 2 < 347/500$ from Appendix B. Factor the upper values of X as $2(1 + 17/100)$, $4(1 - 33/200)$ and $4(1 + 17/200)$, and use

$$\log(1+v) \le v - v^2/2 + v^3/3, \qquad \log(1-v) \le -v - v^2/2.$$

Here $v \ge 0$ in the first inequality and $0 \le v < 1$ in the second. In the first column, we also have $\psi'(Y)/2 > 1/8$ since $Y < 4$. The quantities $\psi(X)$ are positive, so the upper bounds may be squared. The three rational inequalities

$$\begin{aligned}
\frac{6}{5}\{(16/25)^2 + 3/4\} &< (117/100)^2 + 1/8,\\
\frac{6}{5}\{(53/50)^2 + 4/9\} &< (36/25)^2,\\
\frac{6}{5}\{(34/25)^2 + 5/16\} &< (33/20)^2
\end{aligned}$$

show $d_k'' < 0$. The initial values (5.35) therefore prove the other three estimates in (5.34).

We can omit the intervening modes once we prove $d_k' < 0$ for $2 \le k \le 16$. Since $\psi(k+B) > 0$, (5.15) and Lemma 5.6 show that it is enough to prove

$$Q(s,k) := \frac{D}{\ell}\left(1 + \frac{C}{k}\right)\log(k+B) < 1.$$

When $D > 0$,

$$\partial_s \log Q = -\frac{\ell}{2D} + \frac{1}{4\ell} + \frac{1}{4(k+C)} + \frac{1}{2(k+B)\log(k+B)} < 0.$$

Indeed, the first term has magnitude at least $4/3$. The other terms sum to at most $1/6 + 1/8 + 1/2 = 19/24$, by $k \ge 2$ and $\log 2 > 1/2$. At $s = 0$,

$$Q(0,k) = \frac{9}{28}\left(1 + \frac{7}{4k}\right)\log k.$$

For $k \geq 2$, its derivative has the sign of $k + 7/4 - (7/4)\log k > 0$. At $k = 16$ it is bounded by $(639/448)(7/10) = 4473/4480 < 1$. The case $D = 0$ follows directly from (5.31).

We must still bound the terms for all $k \geq 17$. The inequalities (5.15) give

$$\psi(k+B) \leq \log k - \frac{\tau}{k} + \frac{B}{2k^2}, \qquad \delta_k \geq \frac{\ell}{k+C} \geq \frac{\ell}{k} - \frac{\ell C}{k^2}.$$

For the first bound, apply $\log(1 + B/k) \leq B/k$ and $1/(k+B) \geq 1/k - B/k^2$. Thus the bracket in (5.31) is at most

$$\frac{D\log k - \ell}{k} + \frac{E}{k^2} + \frac{F}{k^3}, \qquad E = \ell C - D\tau = 3 + \frac{2\tau - 7\tau^2 - \tau^3}{4}, \quad F = \frac{DB}{2}.$$

The lower bounds $\ell \geq 3/2$ and $C \geq 7/4$ give $E \geq (3/2)(7/4) - (9/16)(1/2) = 75/32 > 0$, and $F \geq 0$. We also have $E \leq 3$ when $\tau \geq 1/3$, $E \leq 49/16$ throughout the interval, and $F \leq 9/64$. Integrate $\psi(x) \leq \log x$ in the two gamma ratios for U_k to get

$$U_k \leq (k+C)^{2+B} \leq k^{2+B}(1 + 5/k + 8/k^2), \qquad k \geq 17.$$

For the second step, use $C \leq 2$ and $2 + B \leq 5/2$. Taylor's formula with remainder gives $(1 + 2/k)^{5/2} \leq 1 + 5/k + (15/2)\sqrt{1 + 2/k}/k^2 \leq 1 + 5/k + 8/k^2$. Replace the bracket in the coefficient formula by a nonnegative upper bound. Then multiply by this envelope and (5.13) to get

$$M_{2k} d'_k \leq \frac{5}{6} k^{-1-\tau}(1 + 5/k + 8/k^2)\left\{(D\log k - \ell)_+ + \frac{E}{k} + \frac{F}{k^2}\right\}. \tag{5.36}$$

For $j = 0, 1, 2$, the nonnegative function $f_j(x) = x^{-1-\tau-j}(D\log x - \ell)_+$ is unimodal. The same sum-integral comparison used in Lemma 5.7 gives

$$\sum_{k \geq 17} f_j(k) \leq I_j + \max f_j,$$

$$I_j = \frac{D}{(\tau+j)^2} e^{-(\tau+j)\ell/D}, \qquad \max f_j = \frac{D}{e(1+\tau+j)} e^{-(1+\tau+j)\ell/D}.$$

These expressions tend continuously to zero at $D = 0$. The function I_0 increases with τ, since

$$\frac{d}{d\tau}\log I_0 = -\frac{2(6\tau^3 + 15\tau^2 - 12\tau + 1)}{\tau(\tau^2 + 6\tau - 1)^2} > 0.$$

The cubic has only one positive critical point, at $1/3$, where it has a minimum. At the two interval endpoints, its values are $20(6(\sqrt{10} - 3) - 1) < 0$ and $-1/2$. Also, ℓ/D decreases with τ and has minimum $28/9$.

First take $1/3 \leq \tau \leq 1/2$. The exponential series has positive terms and proves $e^{-14/9} < 7/33$, since

$$\sum_{j=0}^{6} \frac{(14/9)^j}{j!} = \frac{113172727}{23914845} > \frac{33}{7}.$$

Thus the weighted integrals are at most

$$I_0 + 5I_1 + 8I_2 \leq I := \frac{21}{44}\left(1 + 5\frac{49}{9801} + 8\frac{2401}{29648025}\right).$$

For the maxima, use $e^{-1} < 3/8$, $e^{-28/9} < 1/22$ and $e^{-28/27} < 3/8$. They give

$$\max f_j \leq \frac{243}{90112}\frac{4}{4+3j}\,22^{-j},$$

$$\max f_0 + 5\max f_1 + 8\max f_2 \leq M := \frac{243}{90112}\left(1 + 5\frac{4}{7\cdot 22} + 8\frac{2}{5\cdot 22^2}\right).$$

For $j = 1, 2, 3, 4$, compare the decreasing powers with their integrals to get

$$\sum_{k\geq 17} k^{-1-\tau-j} \leq T_j := \frac{2}{5}\frac{3}{(3j+1)16^j},$$

where we used $16^{-1/3} < 2/5$. The error terms in (5.36) therefore sum to at most $E_1 = 3(T_1 + 5T_2 + 8T_3) + (9/64)(T_2 + 5T_3 + 8T_4)$. Altogether,

$$\frac{5}{6}(I + M + E_1) = \frac{725716685034431}{1555964783493120} < \frac{47}{100}. \tag{5.37}$$

For $\sqrt{10} - 3 \leq \tau \leq 1/3$, we instead have $\tau > 4/25$, $D \leq 5/18$, $\ell/D \geq 6$ and $I_0 \leq (5/2)e^{-2} < 45/128$. With $\eta = 3/8$ the corresponding bounds are

$$I := \frac{45}{128}\left(1 + \frac{5\eta^6}{16} + \frac{8\eta^{12}}{49}\right), \qquad M := \sum_{j=0}^{2} w_j \frac{5}{18}\frac{\eta}{29/25 + j}\eta^{6(j+1)}, \quad (w_0, w_1, w_2) = (1, 5, 8).$$

Here I bounds the weighted integrals, and M bounds the weighted maxima. Since $16^{-4/25} < 2/3$, or equivalently $16^4 2^{25} > 3^{25}$, the power sums are bounded by

$$T_j := \frac{2}{3}\frac{25}{16^j(25j + 4)}, \qquad j = 1, 2, 3, 4.$$

With $E_2 = (49/16)(T_1 + 5T_2 + 8T_3) + (9/64)(T_2 + 5T_3 + 8T_4)$, we get

$$\frac{5}{6}(I + M + E_2) = \frac{10265170897202285641668475}{25553509385718753096892416} < \frac{47}{100}. \tag{5.38}$$

All these bounds on constants follow from exact rational comparisons: $e > 8/3$ follows from five positive terms; the square of $e^{14/9} > 33/7$ gives $e^{-28/9} < 1/22$; and $28/27 > 1$ gives $e^{-28/27} < 3/8$. The bounds $4/25 < \sqrt{10} - 3 < 1/6$ follow by squaring. Finally, $E \leq 49/16$ follows from $2\tau - 7\tau^2 - \tau^3 \leq 2\tau - 7\tau^2 \leq 1/7 < 1/4$.

Omit $d_0' < 0$ and the nonpositive modes five through sixteen. Combine (5.33), (5.34) and either tail bound to get

$$\Omega_s < -\frac{19}{70} - \frac{72}{625} - \frac{117}{2000} - \frac{7}{200} + \frac{47}{100} = -\frac{709}{70000}.$$

□

5.3. **Proof of the transition theorem.**

Proof of Theorem 5.1. Let $W = \Theta$ for the first two pairs and $W = \Omega$ for the third. Each normalization is a positive multiple of Φ. Its factor depends only on s. Along the minimal branch, Proposition 5.3 gives $W_\lambda > 0$; Propositions 5.8 and 5.9 give $W_s < 0$ on the relevant strip. Since $\lambda' < 0$ by Proposition 2.5,

$$\frac{d}{ds}W(s, \lambda(s)) = W_s + W_\lambda \lambda' < 0, \qquad 0 < s < s_*.$$

Proposition 5.4 gives a positive value at $s = 0$, and Proposition 5.3 gives a negative value at $s = s_*$. Continuity and strict decrease give exactly one zero between them and give the signs on either side. The same spatial-order proposition rules out stability and further zeros for $s \geq s_*$. The derivative of W at its zero is strictly negative.

On the minimal branch, the physical gap is a positive real analytic multiple of W. Its factor is $L_\Sigma \Gamma(C)^2/(2\Gamma(q))$ for the first two pairs and $L_\Sigma\sqrt{\pi}/(2\Gamma(q))$ for the third. At a zero, the derivative of this factor contributes nothing. Thus the physical gap also has a simple zero and a negative derivative there. This argument transfers the derivative sign only at the zero. We do not need global monotonicity of the unnormalized physical gap. The sharp criterion (2.5) includes

equality, so the stable range includes the threshold. Analyticity follows from Lemma 2.3 and the analytic aperture branch in Proposition 2.5. The link area is real analytic for $0 < \lambda < 1$. The continuation identified after (5.8) therefore also extends the physical gap analytically through zero. Finally, each s admits only one minimal aperture. Thus the unique zero on that branch is the unique common zero of G and Φ in $(0,1) \times (0,1)$; it is the pair $(\sigma_{m,n}, \lambda(\sigma_{m,n}))$. □

6. Stability in dimensions $N \geq 8$

Theorem 6.1. *Let m, n be positive integers, $N = m + n \geq 8$, and $0 < s < 1$. Let $\alpha_{m,n}(s)$ be the unique s-minimal aperture for the kernel $|x-y|^{-N-s}$, and set $\Sigma = \{(y,z) \in \mathbb{R}^m \times \mathbb{R}^n : |z| = \alpha_{m,n}(s)|y|\}$. Let Q_s be the quadratic form (2.4), equal to one half of the perimeter second variation, and let $d\sigma$ be surface measure on Σ. Then Σ is stable and has a strictly positive Hardy–curvature gap. More precisely, there is $\delta_{m,n}(s) > 0$ such that*

$$Q_s(\varphi) \geq \delta_{m,n}(s) \int_\Sigma \frac{\varphi(x)^2}{|x|^{1+s}} \, d\sigma(x) \qquad (\varphi \in C_c^\infty(\Sigma \setminus \{0\})).$$

The optimal constant $\delta_{m,n}(s)$ is the sharp Hardy constant minus the curvature coefficient in (2.4).

We prove that the gap is positive by comparing the terms in the paired series of Proposition 2.6. In dimensions at least nine, contraction of spherical moment tensors bounds the series from below by a scalar reference sum. In dimension eight, we need upper and lower angular bounds and a separate bound for the negative coefficients. We first compute the scalar reference sum, then prove the tensor comparison and the dimension-eight bounds. We then prove Theorem 6.1. The reference moments are

$$b_k(d) = \frac{(1/2)_k}{(d)_k}, \qquad k \geq 0.$$

For $d > 1/2$, these are the even coordinate moments on a sphere of dimension $2d-1$ when that dimension is an integer. The beta integral defines them for every such d. We will use two choices: $d = N/2$ in dimensions at least nine and $d = 7/2$ in dimension eight.

6.1. A scalar reference identity.

We compute the reference value while keeping the cancellation in the paired series.

Lemma 6.2. *Let $N \geq 8$, $0 < s < 1$, and let A, B, C be as in (2.20). If $d > A$ and $-1 < d - 2C < 0$, then*

$$\begin{aligned} S_d(s) &:= -1 + \sum_{k=1}^{\infty} \frac{(C)_k^2}{k!(d)_k}(R_k - 1) \\ &= \Gamma(d)\Gamma(d-2C) \left\{ \frac{\Gamma(A+1)\Gamma(B+1)}{\Gamma(C)^2\Gamma(d-A)\Gamma(d-B)} - \frac{1}{\Gamma(d-C)^2} \right\}. \end{aligned} \tag{6.1}$$

The series is absolutely convergent.

Proof. Put $\delta = d - 2C \in (-1, 0)$. Gamma-ratio asymptotics give

$$\frac{(C)_k^2}{k!(d)_k} = O(k^{-\delta-1}), \qquad R_k - 1 = O(k^{-1}),$$

so the combined terms are summable. For $0 < x < 1$, their Abel sum is

$$\begin{aligned} S_{d,x} = &- {}_2F_1(C, C; d; x) \\ &+ \frac{\Gamma(A+1)\Gamma(B+1)}{d\Gamma(C)^2} x \, {}_2F_1(A+1, B+1; d+1; x). \end{aligned}$$

This follows by expanding the two power series and using $A + B + 1 = 2C$ and the definition of R_k.

The connection formula at $x = 1$ [14, Eq. 15.8.4] writes each hypergeometric function as a regular part plus $(1 - x)^\delta$ times a function analytic at one. After we include the factor in front of the second term, the coefficients of $(1 - x)^\delta$ in the two terms are both

$$\frac{\Gamma(d)\Gamma(-\delta)}{\Gamma(C)^2}.$$

These coefficients cancel. The difference of the remaining singular parts, including the extra factor x, is $O((1-x)^{1+\delta}) = o(1)$: their analytic factors both take the value one at $x = 1$, so their difference is $O(1 - x)$. The condition $-1 < \delta < 0$ ensures that neither connection formula has an integral exponent difference. The difference of the regular coefficients is

$$\Gamma(d)\Gamma(\delta)\left\{\frac{\Gamma(A+1)\Gamma(B+1)}{\Gamma(C)^2\Gamma(d-A)\Gamma(d-B)} - \frac{1}{\Gamma(d-C)^2}\right\}.$$

Since the series converges absolutely, this Abel limit equals S_d. □

6.2. **Tensor moments and dimensions at least nine.** The first comparison uses only geometry and does not depend on the aperture.

Lemma 6.3. *Let $N \geq 2$, and let μ be a finite nonnegative Borel measure on $\mathbb{S}^{N-1}$ with total mass one. Define*

$$m_{2k} = \iint (x \cdot y)^{2k} \, \mathrm{d}\mu(x) \, \mathrm{d}\mu(y), \qquad m_0 = 1.$$

Then

$$m_{2k} \geq \frac{2k-1}{N+2k-2} \, m_{2k-2} \qquad (k \geq 1). \tag{6.2}$$

Consequently $m_{2k}/b_k(N/2)$ is nondecreasing and at least one.

Proof. Use the Euclidean inner product on symmetric tensors and set $T_\ell = \int x^{\otimes \ell} \, \mathrm{d}\mu(x)$. Then $\|T_{2k}\|^2 = m_{2k}$ and contraction of two indices gives $\operatorname{tr} T_{2k} = T_{2k-2}$. We compute the norm of contraction to make the normalization clear.

For a symmetric tensor T of order ℓ, use the polynomial $p_T(x) = T[x, \ldots, x]$. Contraction becomes $\Delta/[\ell(\ell-1)]$. Its adjoint sends a tensor U to $\operatorname{sym}(I \otimes U)$, where I is the identity tensor and sym means the average over permutations of the indices. Its polynomial is therefore $|x|^2 p_U(x)$. Thus $\operatorname{tr}\operatorname{tr}^*$ acts on polynomials of degree $2k-2$ as

$$p \longmapsto \frac{\Delta(|x|^2 p)}{2k(2k-1)}.$$

Here $\mathcal{H}_r$ denotes the space of homogeneous harmonic polynomials of degree r. Every homogeneous polynomial of degree $2k-2$ has a unique decomposition into the spaces $|x|^{2j}\mathcal{H}_{2k-2-2j}$, $0 \leq j \leq k-1$. For $h \in \mathcal{H}_r$, the identity

$$\Delta(|x|^{2j+2}h) = (2j+2)(2r+N+2j)|x|^{2j}h$$

also proves the decomposition by induction on the degree. Indeed, decompose Δp first, lift each harmonic term using this identity, and subtract the lifts from p. The remaining polynomial is harmonic. The positive coefficients in the identity give uniqueness. The eigenvalue on the jth space is

$$\frac{(2j+2)(N+4k-4-2j)}{2k(2k-1)}, \qquad 0 \leq j \leq k-1.$$

These eigenvalues strictly increase with j. Their largest value is $(N+2k-2)/(2k-1)$. Since the operator is self-adjoint and nonnegative,

$$\|\operatorname{tr}\|^2 = \frac{N+2k-2}{2k-1}.$$

Apply this to T_{2k} to get (6.2). For the quotient, use $b_k(N/2)/b_{k-1}(N/2) = (2k-1)/(N+2k-2)$. □

For the normalized area measure of a Lawson link, the group $O(m)\times O(n)$ acts transitively. Thus the double correlation moments in this lemma equal the fixed-point angular integrals $M_{2k} = \mathcal{I}_{m,n}[Z^{2k}]$. This also holds when a factor is $\mathbb{S}^0$.

Lemma 6.4. *Suppose $N \geq 9$ and $0 < s < 1$. Set $e_0 = -1$ and $e_k = a_k(R_k - 1)$ for $k \geq 1$, where $a_k = (C)_k^2/[k!(1/2)_k]$. The sequence e_k has at most one sign change, from nonpositive to nonnegative, and its tail is positive. If a sequence of moments with $m_0 = 1$ satisfies (6.2), both paired sums converge absolutely, and $S_{N/2}(s) > 0$, then*

$$\sum_{k=0}^{\infty} e_k m_{2k} \geq S_{N/2}(s) > 0. \tag{6.3}$$

Proof. Use the recurrence and limit in Lemma 2.9. For $N \geq 9$,

$$D > D|_{s=1} = \frac{(N-5)^2 - 8}{16} \geq \frac{1}{2}.$$

The sign of $AB - Dk$ therefore changes at most once, from positive to negative. Hence R_k is first nondecreasing and then strictly decreases toward one. There is at most one equality at the turning index, and the initial nondecreasing interval may be empty. The decreasing tail lies above one. The earlier nondecreasing part can cross one only once. This proves the claim about signs.

Choose $K \geq 1$ such that $e_k \leq 0$ for $k < K$ and $e_k \geq 0$ for $k \geq K$. Write $q_k = m_{2k}/b_k(N/2)$. Lemma 6.3 gives $q_k \leq q_K$ before K and $q_k \geq q_K$ afterwards. Hence $e_k b_k(N/2) q_k \geq e_k b_k(N/2) q_K$ for every k, including zero. After summation,

$$\sum e_k m_{2k} \geq q_K S_{N/2}(s) \geq S_{N/2}(s),$$

because $q_K \geq 1$ and the reference value is positive. We may make these comparisons because the sums converge absolutely. □

Proposition 6.5. *For $N \geq 9$ and $0 < s < 1$, every minimal Lawson cone satisfies*

$$\Delta \geq \frac{\Gamma(C)^2}{2\Gamma(2C)} \frac{\Gamma(N/2)|\Gamma(-s/2)|}{\Gamma((N-s)/4)^2} \left[1 - \left(\frac{35}{36}\right)^s\right] > 0. \tag{6.4}$$

Proof. Apply Lemma 6.2 with $d = N/2$. Its hypotheses hold because $d - A = 1$ and $d - 2C = -s/2 \in (-1, 0)$. If $u = (N-s)/4$, we obtain

$$S_{N/2}(s) = \frac{\Gamma(N/2)|\Gamma(-s/2)|}{\Gamma(u)^2}(1 - F_N(s)),$$

where

$$F_N(s) = \frac{\Gamma(N/2)\Gamma(1+s/2)\Gamma((N-s)/4)^2}{\Gamma((N+s)/4)^2\Gamma((N-s)/2)}.$$

To compare dimensions, regard $F_N(s)$ as a function of real $N \geq 9$. All gamma arguments are positive in this range. The trigamma series used in Lemma 2.8 shows that ψ' is positive and

strictly decreasing. Differentiation gives

$$\partial_s^2 \log F_N = \frac{1}{4}\left[\psi'(1+s/2) - \psi'((N-s)/2)\right] + \frac{1}{8}\left[\psi'((N-s)/4) - \psi'((N+s)/4)\right] > 0.$$

Also

$$\partial_N \log F_N = \frac{1}{2}[\psi(N/2) - \psi((N-s)/2) - \psi((N+s)/4) + \psi((N-s)/4)] < 0.$$

The last derivative can be written as

$$\partial_N \log F_N = \frac{1}{2}\int_0^{s/2} \left[\psi'\left(\frac{N-s}{2}+t\right) - \psi'\left(\frac{N-s}{4}+t\right)\right] dt < 0,$$

because ψ' strictly decreases. Therefore $F_N(s) \le F_9(s)$. The endpoint values are

$$F_9(0) = 1, \qquad F_9(1) = \frac{\Gamma(9/2)\Gamma(3/2)\Gamma(2)^2}{\Gamma(5/2)^2\Gamma(4)} = \frac{35}{36}.$$

Convexity of $\log F_9$ now gives $F_N(s) \le (35/36)^s < 1$. In particular $S_{N/2}(s) > 0$.

The Lawson moments satisfy Lemma 6.3. Their paired series converges absolutely by Proposition 2.6. Lemma 6.4 compares this series with $S_{N/2}$. Multiplying by $\Gamma(C)^2/[2\Gamma(2C)]$ gives (6.4). □

6.3. **The dimension-eight comparison.** The tensor bound alone does not give the comparison needed in dimension eight. We use the coordinate moments of $\mathbb{S}^6$,

$$g_k = b_k(7/2) = \frac{15}{(2k+1)(2k+3)(2k+5)}, \qquad g(z) = \frac{15}{16}(1-z^2)^2 \mathbf{1}_{(-1,1)}(z). \tag{6.5}$$

We first bound the scalar sum. We then prove the angular bounds needed to apply this bound to every family.

6.3.1. *The reference sum and its negative coefficients.* Here $A = 3$, $B = s/2$, $C = 2 + s/4$. Write

$$w_k(s) = \frac{(C)_k^2}{k!(7/2)_k}, \qquad S(s) = -1 + \sum_{k\ge 1} w_k(s)(R_k(s) - 1).$$

Lemma 6.6. *For $0 < s < 1$,*

$$S(s) \ge \frac{3 - (12/5)s}{1-s^2} \ge \frac{12}{5}. \tag{6.6}$$

Moreover,

$$E_-(s) := \sum_{R_k(s)<1} w_k(s)(R_k(s)-1) > -\frac{3}{5}. \tag{6.7}$$

Proof. Lemma 6.2 applies with $d = 7/2$, since $d - A = 1/2$ and $d - 2C = -(1+s)/2 \in (-1, 0)$. It gives $S = P(1 - F)$, where

$$P(s) = \frac{\Gamma(7/2)|\Gamma(-(1+s)/2)|}{\Gamma(3/2 - s/4)^2},$$

$$F(s) = \frac{6\Gamma(1+s/2)\Gamma(3/2-s/4)^2}{\sqrt{\pi}\,\Gamma(7/2-s/2)\Gamma(2+s/4)^2}.$$

The endpoint values are $F(0) = 4/5$ and $F(1) = 24/25$. Since

$$(\log F)'' = \frac{1}{4}[\psi'(1 + s/2) - \psi'(7/2 - s/2)] + \frac{1}{8}[\psi'(3/2 - s/4) - \psi'(2 + s/4)] > 0,$$

convexity and the weighted arithmetic–geometric mean inequality yield

$$F(s) \le (4/5)^{1-s}(24/25)^s \le \frac{4}{5} + \frac{4s}{25}.$$

The gamma recurrence gives

$$P(s) = \frac{4\Gamma(7/2)\Gamma(3/2 - s/2)}{(1 - s^2)\Gamma(3/2 - s/4)^2}.$$

The argument of the gamma function in the denominator is the midpoint of $3/2$ and $3/2 - s/2$. Log convexity therefore gives $P(s) \ge 15/(1 - s^2)$. In particular, $1 - F(s) \ge (5 - 4s)/25 > 0$. We may therefore multiply this bound by the lower bound for P. This proves the first inequality in (6.6); the second is equivalent to

$$\frac{3 - (12/5)s}{1 - s^2} - \frac{12}{5} = \frac{3(2s - 1)^2}{5(1 - s^2)} \ge 0.$$

For the negative coefficients, Lemma 2.9 gives $\partial_s \log R_k < 0$, whereas $w_k(s)$ increases with s. If $R_k(s) < 1$, then $R_k(1) < R_k(s) < 1$ and

$$0 < w_k(s)(1 - R_k(s)) \le w_k(1)(1 - R_k(1)).$$

Thus the negative coefficient at s is at least its value at one. At that endpoint,

$$\frac{R_{k+1}(1)}{R_k(1)} = 1 + \frac{24 - k}{k(4k + 9)^2} = \frac{8(k + 1)(k + 3)(2k + 1)}{k(4k + 9)^2}. \tag{6.8}$$

Appendix A proves the exact initial bound

$$R_1(1) = \frac{4096}{675B(1/4, 1/4)} > \frac{9}{11}.$$

Start with $r_1 = 9/11$ and use the multiplier in (6.8) to define r_{k+1}. Also set

$$t_1 = \frac{81}{56}, \qquad t_{k+1} = t_k \frac{(4k + 9)^2}{8(k + 1)(2k + 7)},$$

so that $t_k = w_k(1)$. The rational computations in the appendix give

$$r_k < 1 \quad (1 \le k \le 11), \qquad r_{12} = \frac{1245421218828386304}{1245364815778312075} > 1, \tag{6.9}$$

$$\sum_{k=1}^{11} t_k(r_k - 1) = -\frac{495417332211287026643}{899408417009371709440} > -\frac{3}{5}. \tag{6.10}$$

Since $R_k(1) > r_k$, the actual sequence lies above one from index twelve onward. It increases up to the possible plateau at indices $24, 25$, then decreases to one. Thus every negative coefficient at $s < 1$ has index at most eleven. For each such index,

$$w_k(s)(R_k(s) - 1) \ge w_k(1)(R_k(1) - 1) > t_k(r_k - 1).$$

Adding the omitted negative rational terms only lowers the bound. Equation (6.10) proves (6.7). □

6.3.2. *Angular bounds for the four families.*

Proposition 6.7. *For every minimal Lawson cone with $m+n=8$,*

$$\frac{1}{2}g_k \le M_{2k} \le g_k \qquad (k \ge 0). \tag{6.11}$$

Proof. Interchange the factors and apply Proposition 2.5. It is enough to work on the closed intervals

(m,n)	$a = 1-b$
$(4,4)$	$1/2$
$(5,3)$	$[5/8, 2/3]$
$(6,2)$	$[3/4, 5/6]$
$(7,1)$	$[7/8, 1]$

These intervals include the actual minimal apertures. All angular weights below are even. We apply Lemma 2.7 whenever the difference of two weights changes sign once from positive to negative.

Equal factors. For $(m,n)=(4,4)$, the binomial moment formula gives

$$M_{2k} = \frac{(1/2)_k}{4^k} \sum_{j=0}^{k} \binom{k}{j} \frac{1}{(2)_j (2)_{k-j}} = \frac{(1/2)_k (3/2)_k}{(2)_k (3)_k}.$$

For the last equality, the terminating Vandermonde identity gives

$$\sum_{j=0}^{k} \binom{k}{j} \frac{1}{(2)_j (2)_{k-j}} = \frac{{}_2F_1(-k, -k-1; 2; 1)}{(2)_k} = \frac{(k+3)_k}{(2)_k^2}.$$

Now use $(k+3)_k = (3)_{2k}/(3)_k = 4^k (3/2)_k (2)_k / (3)_k$. The quotient $q_k = M_{2k}/g_k$ therefore satisfies

$$q_0 = 1, \qquad \frac{q_{k+1}}{q_k} = \frac{(k+3/2)(k+7/2)}{(k+2)(k+3)} < 1,$$

and

$$\lim_{k\to\infty} q_k = \frac{\Gamma(2)\Gamma(3)}{\Gamma(3/2)\Gamma(7/2)} = \frac{32}{15\pi} > \frac{1}{2}.$$

This proves both bounds for this pair.

A constant coordinate weight. For $(m,n)=(5,3)$, the coordinate weights are $3(1-u^2)/4$ and $1/2$. Scale them by a, b and take their convolution. With $c = a-b = 2a-1$, the resulting weight f associated with Z on $[0,1]$ is

$$f(z) = \begin{cases} (3a^2 - b^2 - 3z^2)/(4a^3), & 0 \le z \le c, \\ (1-z)^2 (z + 3a - 1)/(8a^3 b), & c \le z \le 1. \end{cases}$$

Put $J = 3a^2 - b^2 = 2a^2 + 2a - 1$. Dividing by g gives

$$\frac{f(z)}{g(z)} = \begin{cases} \dfrac{4}{15a^3} \dfrac{J - 3z^2}{(1-z^2)^2}, & z \le c, \\ \dfrac{2}{15a^3 b} \dfrac{z + 3a - 1}{(1+z)^2}, & z \ge c. \end{cases}$$

The derivatives on these two intervals have the signs of $z(2J - 3 - 3z^2)$ and $3 - 6a - z$, respectively. Both are negative for $z > 0$, because $2J - 3 \le -5/9$ and $a \ge 5/8$. The ratio is continuous and strictly decreasing. Also,

$$\int_0^1 f(z)\,\mathrm{d}z = \int_0^1 g(z)\,\mathrm{d}z = \frac{1}{2}.$$

Thus there is a unique $z_0 \in (0,1)$ where $f-g$ changes sign from positive to negative. Lemma 2.7 now gives $M_{2k} \le g_k$. The sign change is strict, so the inequality is strict for $k \ge 1$; for $k = 0$ there is equality. For the lower bound,

$$\frac{f(z)}{g(z)} \ge \lim_{z\uparrow 1} \frac{f(z)}{g(z)} = \frac{1}{10a^2b} \ge \frac{27}{40} > \frac{1}{2},$$

where $a^2(1-a) \le 4/27$. This proves the lower bound.

The pair $(6,2)$. Let $d = a - b \in [1/2, 2/3]$ and $x_d = (1+d^2)/(2d)$. Write $\mu_k = M_{2k}$. Its finite binomial expansion is

$$\mu_k = 2(1/2)_k k! \sum_{j=0}^{k} \frac{a^{2j} b^{2(k-j)}}{j!(j+2)!(k-j)!^2}.$$

Let $P_k^{(0,2)}$ be the Jacobi polynomial with parameters $0, 2$. The identities $d(x_d+1)/2 = a^2$ and $d(x_d-1)/2 = b^2$ give, by its finite sum formula [14, Eq. 18.5.8],

$$d^k P_k^{(0,2)}(x_d) = \sum_{j=0}^{k} \binom{k}{j}\binom{k+2}{k-j} a^{2j} b^{2(k-j)},$$

so we obtain

$$\mu_k = \frac{2(1/2)_k}{k!(k+1)(k+2)} d^k P_k^{(0,2)}(x_d). \tag{6.12}$$

These polynomial identities hold also at $x_d > 1$. For $k \ge 1$, the Jacobi recurrence [14, Eqs. 18.9.1–18.9.2] specializes to

$$P_{k+1}^{(0,2)}(x) = \frac{(2k+3)((k+1)(k+2)x-1)}{(k+1)^2(k+3)} P_k^{(0,2)}(x) - \frac{k(k+2)^2}{(k+1)^2(k+3)} P_{k-1}^{(0,2)}(x).$$

Multiplying by the prefactors in (6.12) gives

$$\mu_{k+1} = \xi_k \mu_k - \eta_k \mu_{k-1},$$

where

$$\xi_k = \frac{(2k+1)(2k+3)[(k+1)(k+2)(1+d^2) - 2d]}{4(k+1)^2(k+3)^2},$$

$$\eta_k = \frac{d^2 k(k+2)(2k-1)(2k+1)}{4(k+1)^2(k+3)^2} > 0.$$

Let $t_k = \mu_k/\mu_{k-1} > 0$ and $v_k = g_k/g_{k-1} = (2k-1)/(2k+5)$. Substituting these reference ratios into the recurrence gives the error

$$v_{k+1} - \xi_k + \frac{\eta_k}{v_k} = \frac{(2k+1)P_k^{\mathrm{def}}(d)}{4(k+1)^2(k+3)^2(2k+7)},$$

where

$$P_k^{\mathrm{def}}(d) = (-1+8d-6d^2)k^2 + (-7+40d-33d^2)k + (-6+42d-42d^2).$$

Each coefficient is concave in d. The values at $1/2$ and $2/3$ are $(3/2, 5/3)$, $(19/4, 5)$, and $(9/2, 10/3)$, respectively. All are positive, so this error is strictly positive. At the first index,

$$t_1 = \frac{1-d+d^2}{6} \le \frac{7}{54} < \frac{1}{7} = v_1.$$

The recurrence reads $t_{k+1} = \xi_k - \eta_k/t_k$, an increasing function of $t_k > 0$. If $t_k < v_k$, the positive error above gives

$$t_{k+1} = \xi_k - \frac{\eta_k}{t_k} < \xi_k - \frac{\eta_k}{v_k} < v_{k+1}.$$

The initial bound therefore proves $t_k < v_k$ for every $k \geq 1$. Thus $q_k = \mu_k/g_k$ strictly decreases from one.

We find the limit from the angular weight near $z = 1$. The two coordinate weights are

$$\frac{8}{3\pi}(1-u^2)^{3/2}, \qquad \frac{1}{\pi\sqrt{1-v^2}}.$$

After scaling by a, b, the endpoint coefficients are $8\,2^{3/2}a^{-5/2}/(3\pi)$ and $b^{-1/2}/(\pi\sqrt{2})$. Convolution multiplies their product by $B(5/2, 1/2) = 3\pi/8$. Indeed, write the two endpoint distances as tr and $t(1-r)$, $0 < r < 1$. The remaining factors in the two scaled weights are $1 + O(t)$ uniformly in r, for each fixed $a, b > 0$. Their product is therefore integrated against $r^{3/2}(1-r)^{-1/2}$ with relative error $O(t)$. This gives

$$f_Z(1-t) = \frac{2}{\pi a^{5/2} b^{1/2}} t^2 (1 + O(t)).$$

Since $g(1-t) = 15t^2(1+O(t))/4$, it follows that

$$\lim_{k\to\infty} q_k = \frac{8}{15\pi a^{5/2} b^{1/2}}.$$

To justify this limit, use evenness to write both moments as twice an integral over $[0, 1]$, and split at $1 - \varepsilon$. The interior parts are exponentially small. For the endpoint parts, use

$$\int_0^1 z^{2k}(1-z)^2\,\mathrm{d}z = B(2k+1, 3), \qquad \frac{B(2k+1,4)}{B(2k+1,3)} = \frac{3}{2k+4}.$$

The second identity bounds the relative error from the $O(t)$ terms by $O(k^{-1})$. The leading coefficients therefore give the stated limit. The function $a^{5/2}(1-a)^{1/2}$ increases on $[3/4, 5/6]$. Thus

$$q_k \geq \frac{576}{125\pi\sqrt{5}} > \frac{1}{2}.$$

The strict inequality follows from $\pi < 4$, $\sqrt{5} < 9/4$, and $1152 > 1125$. This completes this case.

A discrete coordinate factor. For $(m, n) = (7, 1)$, the definition of $\mathcal{I}_1$ and the evenness of w_7 give

$$\begin{aligned} M_{2k} &= \frac{1}{2}\int_{-1}^{1} \left[(au+b)^{2k} + (au-b)^{2k}\right] w_7(u)\,\mathrm{d}u \\ &= \int_{-1}^{1} (au+b)^{2k} w_7(u)\,\mathrm{d}u. \end{aligned}$$

For $0 \leq u \leq 1$, we have $au + b - u = b(1-u) \geq 0$. Hence

$$M_{2k} \geq \int_0^1 u^{2k} w_7(u)\,\mathrm{d}u = \frac{1}{2} g_k.$$

As a function of $b \in [0, 1/8]$, M_{2k} is convex. Its integrand is a convex even power of an affine function. At $b = 0$ it equals g_k. It remains to bound the moment at $b = 1/8$, where $a = 7/8$.

The change of variables used in (4.3) applies with $m = 7$, $p_0 = 2$, and $c = a - b = 3/4$. Since $w_7 = g$ on $(-1, 1)$, it gives the following ratio of the angular weight to g:

$$\varrho(z) = \begin{cases} \dfrac{(3/4 - z^2)^2 + z^2/16}{a^5(1 - z^2)^2}, & 0 \le z \le 3/4, \\ \dfrac{1}{2a^5}\left(\dfrac{z + 3/4}{1 + z}\right)^2, & 3/4 \le z < 1. \end{cases}$$

On the first interval,

$$\varrho'(z) = \frac{z(9z^2 - 5)}{8a^5(1 - z^2)^3}.$$

Thus it decreases to $\sqrt{5}/3$ and then increases. The two formulas agree at $z = 3/4$, and the second one also increases. The values

$$\varrho(0) = \frac{18432}{16807} > 1, \qquad \lim_{z \uparrow 1} \varrho(z) = \frac{256}{343} < 1$$

therefore show that the ratio crosses one exactly once, from above to below. Lemma 2.7 gives $M_{2k} \le g_k$ at $b = 1/8$. Convexity gives the same inequality between the endpoints. Thus all four families satisfy (6.11). □

6.4. **Proof of the stability theorem.**

Proof of Theorem 6.1. Only dimension eight remains after Proposition 6.5. Write $\theta_k = M_{2k}/g_k$. Proposition 6.7 gives $1/2 \le \theta_k \le 1$. The actual paired series and its scalar reference both converge absolutely. For the reference, this also follows from Lemma 6.2. We may therefore split the sum according to the sign of $R_k - 1$. We obtain

$$\begin{aligned} \Theta &= -1 + \sum_{k \ge 1} \theta_k w_k (R_k - 1) \\ &\ge -1 + \frac{1}{2} \sum_{R_k \ge 1} w_k (R_k - 1) + \sum_{R_k < 1} w_k (R_k - 1) \\ &= \frac{1}{2} S(s) - \frac{1}{2} + \frac{1}{2} E_-(s) \\ &> \frac{1}{2}\frac{12}{5} - \frac{1}{2} - \frac{1}{2}\frac{3}{5} = \frac{2}{5}. \end{aligned}$$

On the minimal branch, $\Delta = \Delta_{\mathrm{sp}}$. The factor $\Gamma(C)^2/[2\Gamma(2C)]$ in (2.22) therefore gives, in dimension eight,

$$\Theta > \frac{2}{5}, \qquad \Delta > \frac{\Gamma(2 + s/4)^2}{5\Gamma(4 + s/2)} > 0. \tag{6.13}$$

Together with (6.4), this proves that the gap is positive in every dimension considered here. Proposition 2.1 gives the stated bound for the quadratic form. It also shows that the optimal constant is $\delta_{m,n}(s) = H - A_{\mathrm{curv}} = L_\Sigma \Delta > 0$, since the link area is strictly positive at every positive aperture. □

Proof of Theorem 1.1. The planar result is Theorem 3.1. Theorem 4.1 gives a negative gap and admissible negative variations in dimensions three through six. Theorem 5.1 gives the three thresholds in dimension seven and proves that each crossing is simple and downward. Theorem 6.1 proves stability and a positive gap in every higher dimension. Interchanging the factors is an isometry, so these results cover both orders of each pair. □

Appendix A. An exact certificate for the dimension-eight negative part

We give the arithmetic used in Lemma 6.6, including the initial gamma bound. By symmetry, the beta integral is twice its integral on $(0, 1/2)$. Expand $(1-t)^{-3/4}$ there. All terms are nonnegative, so monotone convergence allows termwise integration. Since

$$\int_0^{1/2} t^{j-3/4}\,\mathrm{d}t = \frac{2^{-j-1/4}}{j+1/4},$$

we obtain

$$B(1/4,1/4) = 2\,2^{-1/4}\sum_{j=0}^{\infty}\frac{(3/4)_j}{j!\,2^j(j+1/4)}.$$

The coefficients $(3/4)_j/j!$ start at one and decrease, since their successive ratio is $(j+3/4)/(j+1) < 1$. Hence the remaining series satisfies

$$\sum_{j=11}^{\infty}\frac{(3/4)_j}{j!\,2^j(j+1/4)} \le \frac{1}{11+1/4}\sum_{j=11}^{\infty}2^{-j} = \frac{2^{-10}}{11+1/4}.$$

Set $q_0 = 8409/10000$. The exact comparison

$$q_0^4 - \frac{1}{2} = \frac{85260660961}{10^{16}} > 0$$

implies $2^{-1/4} < q_0$. Hence

$$\begin{aligned} B(1/4,1/4) &< 2q_0\left\{\sum_{j=0}^{10}\frac{(3/4)_j}{j!\,2^j(j+1/4)} + \frac{2^{-10}}{11+1/4}\right\} \\ &= \frac{185815145431630533315394 9}{250545706031382528000000} < \frac{45056}{6075}. \end{aligned}$$

For $N = 8$ and $s = 1$, (2.23) gives $R_1(1) = \Gamma(4)\Gamma(3/2)/\Gamma(13/4)^2$. Use $\Gamma(13/4) = 45\Gamma(1/4)/64$, $\Gamma(3/2) = \sqrt{\pi}/2$, and $B(1/4,1/4) = \Gamma(1/4)^2/\sqrt{\pi}$ to obtain

$$R_1(1) = \frac{4096}{675B(1/4,1/4)} > \frac{9}{11}.$$

The following code verifies this finite bound and the rational comparisons (6.9)–(6.10). It uses only integer arithmetic and rational fractions. The analytic bound for the beta tail was proved above. At the start of loop index k, the variables `r` and `w` equal r_k and t_k from Lemma 6.6. Thus the final value of `r` is r_{12}, and `negative` is the sum through index eleven.

```
from fractions import Fraction as F

coef, total = F(1), F(0)
for j in range(11):
    total += coef / (2**j * (j + F(1, 4)))
    coef *= (j + F(3, 4)) / (j + 1)

q = F(8409, 10000)
assert q**4 > F(1, 2)
beta_upper = 2*q*(total + F(1, 2**10)/(11 + F(1, 4)))
assert beta_upper == F(185815145431630533315394 9,
                       250545706031382528000000)
assert beta_upper < F(45056, 6075)

r, w, negative = F(9, 11), F(81, 56), F(0)
for k in range(1, 12):
    assert r < 1
    negative += w*(r - 1)
    r *= F(8*(k+1)*(k+3)*(2*k+1), k*(4*k+9)**2)
```

```
    w *= F((4*k+9)**2, 8*(k+1)*(2*k+7))

assert r == F(1245421218828386304, 1245364815778312075)
assert r > 1
assert negative == -F(495417332211287026643,
                      899408417009371709440)
assert negative > -F(3, 5)
```

Appendix B. Finite scalar bounds for the seven-dimensional estimates

We give rational bounds for the constants used in the estimates at a fixed aperture. Write $\zeta(p) = \sum_{j=1}^{\infty} j^{-p}$ for $p > 1$. All terminating decimals in this appendix mean exact rational numbers. The bounds concern finitely many scalar values. We do not sample the exponent or the aperture parameter.

Lemma B.1. *The constants satisfy*

$$\frac{1}{2} < \gamma < \frac{29}{50}, \qquad \frac{157}{50} < \pi < \frac{22}{7}, \qquad \frac{693}{1000} < \log 2 < \frac{347}{500} < \frac{7}{10},$$
$$\log 3 < \frac{11}{10}, \qquad \log 5 < \frac{5}{3}, \qquad \log 9 < \frac{9}{4}.$$

Proof. For $0 < x < 1$ and every integer $J \geq 0$, integrating the geometric series on $(0, x)$ gives

$$\log \frac{1+x}{1-x} = 2 \sum_{j=0}^{J} \frac{x^{2j+1}}{2j+1} + R_J, \qquad 0 < R_J < \frac{2x^{2J+3}}{(2J+3)(1-x^2)}. \tag{B.1}$$

At $x = 1/3$, $J = 4$, the rational lower and upper sums lie between $693/1000$ and $347/500$.

Let $H_n = \sum_{j=1}^{n} 1/j$. The identity

$$H_n - \log n - \gamma = \sum_{k=n}^{\infty} \left\{ \log(1 + 1/k) - \frac{1}{k+1} \right\}$$

combined with

$$\frac{1}{2(k+1)(k+2)} < \log(1 + 1/k) - \frac{1}{k+1} < \frac{1}{2k(k+1)}$$

gives $1/(2(n+1)) < H_n - \log n - \gamma < 1/(2n)$. For the upper bound on each summand, differentiate $t(2+t)/(2+2t) - \log(1+t)$. Its derivative is positive. For the lower bound, use $\log(1+t) \geq 2t/(2+t)$. This gives the stronger bound $1/((k+1)(2k+1))$. Taking $n = 4$ and 16, respectively, yields

$$\gamma > \frac{25}{12} - \frac{694}{500} - \frac{1}{8} > \frac{1}{2}, \qquad \gamma < \frac{2436559}{720720} - \frac{2772}{1000} - \frac{1}{34} < \frac{29}{50}.$$

Machin's identity $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$ follows from the tangent addition formula, with the angles chosen in the interval giving $\pi/4$. Integrating the alternating series for $(1 + t^2)^{-1}$ gives

$$\pi > 16 \left(\frac{1}{5} - \frac{1}{375} \right) - \frac{4}{239} > \frac{157}{50}.$$

Polynomial division and integration give the upper bound

$$0 < \int_0^1 \frac{x^4 (1-x)^4}{1+x^2} \, \mathrm{d}x = \frac{22}{7} - \pi.$$

Finally, positive exponential sums imply

$$e^{11/10} > \sum_{j=0}^{5} \frac{(11/10)^j}{j!} > 3, \qquad e^{5/3} > \sum_{j=0}^{4} \frac{(5/3)^j}{j!} = \frac{10009}{1944} > 5.$$

Likewise $\sum_{j=0}^{4} (9/8)^j / j! > 3$, which gives $\log 9 = 2 \log 3 < 9/4$. □

Lemma B.2. *The scalar bounds*

x	$\Gamma(x)$	$\psi(x)$	$\psi'(x)$
5/4	$(0.906, 0.907)$	$(-0.228, -0.227)$	$(1.197, 1.198)$
7/8	$(1.089, 1.090)$	$(-0.805, -0.804)$	$(2.005, 2.006)$
3/4	$(1.225, 1.226)$	$(-1.086, -1.085)$	*not required*

hold, together with $0.577 < \gamma < 0.578$ *and* $1.772 < \sqrt{\pi} < 1.773$. *They imply all starting estimates in* (5.35).

Proof. The following finite procedure gives all these bounds. In (B.1) use $x = 1/3$, $J = 12$ to bound $\log 2$. Set $N_0 = 8$. Signed Euler–Maclaurin remainders give

$$H_{N_0} - \log N_0 - \frac{1}{2N_0} + \frac{1}{12N_0^2} - \frac{1}{120N_0^4} < \gamma < H_{N_0} - \log N_0 - \frac{1}{2N_0} + \frac{1}{12N_0^2} - \frac{1}{120N_0^4} + \frac{1}{252N_0^6}.$$

Use $\log N_0 = 3\log 2$. For integers $p \geq 2$ put

$$Z_p = \sum_{j=1}^{N_0-1} j^{-p} + \frac{N_0^{1-p}}{p-1} + \frac{1}{2N_0^p} + \frac{p}{12N_0^{p+1}} - \frac{p(p+1)(p+2)}{720N_0^{p+3}}.$$

The corresponding zeta bounds are

$$Z_p < \zeta(p) < Z_p + \frac{p(p+1)(p+2)(p+3)(p+4)}{30240N_0^{p+5}}. \tag{B.2}$$

We now justify the signs and sizes of these remainders. Use the positive partial fractions

$$\frac{t}{e^t - 1} = 1 - \frac{t}{2} + 2t^2 \sum_{n\geq 1} \frac{1}{t^2 + 4\pi^2 n^2}, \qquad t > 0.$$

This identity follows from the logarithmic derivative of Euler's product for $\sinh(t/2)$; see [14, Eq. 4.36.3]. Set

$$K(t) = \frac{1}{e^t - 1} - \frac{1}{t} + \frac{1}{2} = 2t \sum_{n\geq 1} \frac{1}{(2\pi n)^2 + t^2}.$$

Comparing coefficients at zero gives

$$\sum_{n\geq 1} (2\pi n)^{-2} = \frac{1}{24}, \qquad \sum_{n\geq 1} (2\pi n)^{-4} = \frac{1}{1440}, \qquad \sum_{n\geq 1} (2\pi n)^{-6} = \frac{1}{60480}.$$

Indeed, division of the power series for $e^t - 1$ gives $K(t) = t/12 - t^3/720 + t^5/30240 + O(t^7)$ near zero. In each summand use $1/(1+u) = 1 - u + u^2/(1+u)$. This gives the exact positive remainder

$$\mathcal{R}(t) = 2t^5 \sum_{n\geq 1} \frac{1}{(2\pi n)^6 \{1 + t^2/(2\pi n)^2\}}.$$

All sums converge absolutely. In particular, for every $t > 0$,

$$\frac{1}{e^t - 1} - \frac{1}{t} + \frac{1}{2} = \frac{t}{12} - \frac{t^3}{720} + \mathcal{R}(t), \qquad 0 < \mathcal{R}(t) < \frac{t^5}{30240}.$$

To pass from this bound to the constants above, use

$$\gamma = H_{N_0} - \log N_0 - \frac{1}{2N_0} + \int_0^\infty e^{-N_0 t}\left(\frac{1}{e^t-1} - \frac{1}{t} + \frac{1}{2}\right)\mathrm{d}t,$$
$$\zeta(p) = \sum_{j=1}^{N_0-1} j^{-p} + \frac{1}{\Gamma(p)}\int_0^\infty \frac{t^{p-1}e^{-N_0 t}}{1-e^{-t}}\,\mathrm{d}t.$$

The zeta identity follows by expanding the geometric series under the positive integral. To check the identity for γ, put $I_N = \int_0^\infty e^{-Nt}K(t)\,\mathrm{d}t$. The positive partial fractions give $0 < K(t) < t/12$, hence $0 < I_N < 1/(12N^2)$. Using $\int_0^\infty (e^{-Nt} - e^{-(N+1)t})\,\mathrm{d}t/t = \log(1+1/N)$, direct integration gives

$$I_N - I_{N+1} = \frac{1}{2N} + \frac{1}{2(N+1)} - \log(1+1/N).$$

It follows that $H_N - \log N - 1/(2N) + I_N$ is independent of N. Its limit is γ, since $I_N \to 0$. This proves the identity. Integrating the polynomial terms and the bound for $\mathcal{R}$ gives exactly the stated remainder bounds. The coefficients are $B_2 = 1/6$, $B_4 = -1/30$, $B_6 = 1/42$. The argument does not use a Taylor expansion that diverges for large t.

For $z = 1/4, -1/8, -1/4$, the convergent gamma-product series [14, Eqs. 5.7.3–5.7.4] and their derivatives give

$$\log\Gamma(1+z) = -\gamma z + \sum_{n=2}^{14}\frac{(-1)^n\zeta(n)z^n}{n} + e_0,$$
$$\psi(1+z) = -\gamma + \sum_{n=2}^{14}(-1)^n\zeta(n)z^{n-1} + e_1,$$
$$\psi'(1+z) = \sum_{n=2}^{14}(-1)^n(n-1)\zeta(n)z^{n-2} + e_2.$$

For $t = |z|$, the elementary inequality $\zeta(n) < 2$ bounds the remainders by

$$|e_0| \le \frac{2t^{15}}{15(1-t)}, \qquad |e_1| \le \frac{2t^{14}}{1-t}, \qquad |e_2| \le \frac{2t^{13}(14-13t)}{(1-t)^2}. \tag{B.3}$$

Insert the rational endpoints from (B.2). Choose the upper or lower bound according to the sign of each coefficient. To recover Γ, exponentiate the bounds for log-gamma using $\sum_{j=0}^{14} w^j/j!$. For $|w| < 1/2$, the error is at most $2|w|^{15}/15!$: the absolute tail starts with $|w|^{15}/15!$, and the ratio of successive terms is at most $|w|/16 < 1/32$. The finite bounds for log-gamma above lie in $(-1/2, 1/2)$, so this estimate applies at both endpoints. Since the exponential is increasing, evaluate this polynomial at each rational endpoint and subtract or add the corresponding error bound. These finite rational operations give the three rows above. For example, the same procedure gives the following bounds rounded outwards:

x	$\Gamma(x)$	$\psi(x)$	$\psi'(x)$
5/4	$(0.90640247, 0.90640249)$	$(-0.227454, -0.227453)$	$(1.197328, 1.197330)$
7/8	$(1.08965235, 1.08965236)$	$(-0.804018, -0.804017)$	$(2.005740, 2.005742)$
3/4	$(1.22541669, 1.22541671)$	$(-1.085862, -1.085860)$	not required.

The bounds for $\sqrt{\pi}$ follow directly from Lemma B.1, since

$$(1.772)^2 < \frac{157}{50} < \pi < \frac{22}{7} < (1.773)^2.$$

To check the resulting coefficient bounds explicitly, at $s = 1/2$ put $X = 5/4$, $Y = 23/8$. Recurrence gives

$$\Gamma(Y) = \frac{105}{64}\Gamma(7/8), \quad \psi(Y) = \psi(7/8) + \frac{8}{7} + \frac{8}{15}, \quad \psi'(Y) = \psi'(7/8) - \frac{64}{49} - \frac{64}{225}.$$

Set $T_1 = (15/4)\Gamma(X)$, $U_1 = 2\Gamma(Y)^2/\sqrt{\pi}$. Then

$$d_1' = \frac{T_1\psi(X) - U_1\psi(Y)}{2}, \qquad d_1'' = \frac{T_1\{\psi(X)^2 + \psi'(X)\} - U_1\{\psi(Y)^2 + \psi'(Y)/2\}}{4}.$$

Even the wider bounds in the three rows of the statement give $d_1'(1/2) < -1.954 < -39/20$ and $0.186 < d_1''(1/2) < 0.194 < 1/4$. At $s = 0$, use

$$\Gamma(k + 7/4) = \Gamma(3/4)\prod_{j=0}^{k}(3/4 + j), \qquad \psi(k + 7/4) = \psi(3/4) + \sum_{j=0}^{k}(3/4 + j)^{-1},$$

$$\Gamma(k + 1/2) = \sqrt{\pi}\prod_{j=0}^{k-1}(j + 1/2), \qquad \psi(k) = H_{k-1} - \gamma.$$

Substituting these in (5.30) and (5.31) gives

$$d_2'(0) < -2.495 < -12/5, \qquad d_3'(0) < -2.717 < -13/5,$$
$$d_4'(0) < -2.884 < -14/5, \qquad R_2(0) < 1.191 < 6/5.$$

Every operation uses rational endpoints and the stated bounds for the remainders. This proves (5.35). □

Appendix C. An auxiliary angular comparison in dimension seven

The transition proof uses direct bounds for the moments. The stronger comparison between successive moments below is independent of those bounds. It gives another property of the three angular weights.

Proposition C.1. *For each pair on its full strip* (5.1),

$$\frac{M_{2k+2}}{M_{2k}} \le \frac{k+1}{k+7/2}, \qquad k \ge 0.$$

Proof. Let $\rho = f_Z$ be the even angular weight associated with $Z = au + bv$, where $b = \lambda$ and $a = 1 - b$. We show that

$$z \longmapsto \frac{\rho(z)}{z(1-z^2)^{3/2}} \quad \text{is decreasing on } (0,1). \tag{C.1}$$

To deduce the moment bound, substitute $x = z^2$ in the even moment integral. Set

$$h(x) = \frac{\rho(\sqrt{x})}{\sqrt{x}(1-x)^{3/2}}, \qquad W_k(x) = x^k(1-x)^{3/2}, \qquad 0 < x < 1.$$

The function h decreases by the displayed comparison, and

$$M_{2k} = \int_0^1 W_k(x)h(x)\,\mathrm{d}x, \qquad \int_0^1 W_k(x)\,\mathrm{d}x = B(k+1, 5/2).$$

All these integrals are finite. The double integral in (2.15), with weight W_k and functions x and $h(x)$, converges absolutely: its absolute integrand is bounded by $(h(x) + h(y))W_k(x)W_k(y)$.

Since h decreases, $(x-y)(h(x)-h(y))\le 0$. The identity therefore gives

$$\frac{M_{2k+2}}{M_{2k}}=\frac{\int_0^1 xW_k(x)h(x)\,\mathrm{d}x}{\int_0^1 W_k(x)h(x)\,\mathrm{d}x}\le\frac{\int_0^1 xW_k(x)\,\mathrm{d}x}{\int_0^1 W_k(x)\,\mathrm{d}x}$$
$$=\frac{B(k+2,5/2)}{B(k+1,5/2)}=\frac{k+1}{k+7/2}.$$

Thus it remains to prove (C.1).

For $(4,3)$ put $d=a-b\in[1/7,1/5]$. The convolution gives

$$\rho(z)\ \propto\ \int_{z-b}^{\min(z+b,a)}\sqrt{a^2-w^2}\,\mathrm{d}w.$$

Here and below, the positive factors omitted in proportionality signs do not depend on z, so they do not affect logarithmic derivatives. On $0<z<d$, $\rho'(z)\ \propto\ \sqrt{a^2-(z+b)^2}-\sqrt{a^2-(z-b)^2}<0$, while the denominator in (C.1) increases because $z<d<1/2$. On $d\le z<1$, put $t=1-z$ and $w=a-v$:

$$\rho(z)\ \propto\ t^{3/2}I(t),\qquad I(t)=\int_0^1\sqrt{r(2a-tr)}\,\mathrm{d}r.$$

Here $0<t\le 2b$ and $2a-tr\ge 2a-t=d+z\ge 2d>0$. The integrand and its t derivative are therefore bounded by a constant times $\sqrt{r}$, so differentiation under the integral gives

$$-\frac{I'(t)}{I(t)}=\frac{\displaystyle\int_0^1\sqrt{r(2a-tr)}\,\frac{r}{2(2a-tr)}\,\mathrm{d}r}{\displaystyle\int_0^1\sqrt{r(2a-tr)}\,\mathrm{d}r}\le\frac{1}{2(2a-t)}=\frac{1}{2(d+z)}.$$

Here $r/(2a-tr)\le 1/(2a-t)$ for $0\le r\le 1$. Since $t=1-z$,

$$(\log\rho)'(z)=-\frac{3}{2t}-\frac{I'(t)}{I(t)}\le-\frac{3}{2t}+\frac{1}{2(d+z)}.$$

After subtracting the logarithmic derivative of the denominator, the result is at most $1/(2(d+z))-1/z-3/(2(1+z))<0$. The weight is continuous at d, so the ratio decreases across both intervals.

For $(5,2)$ the convolution gives

$$\rho(z)\ \propto\ \int_{-b}^{b}\frac{[a^2-(z-v)^2]_+}{\sqrt{b^2-v^2}}\,\mathrm{d}v.$$

On $0<z<d=a-b$ this is a constant multiple of $c-z^2$, with $c=a^2-b^2/2\in[23/49,31/50]\subset[1/3,2/3]$. The derivative in (C.1) has the sign of $-c+(4c-1)z^2-2z^4$. As a quadratic in z^2 it has negative leading coefficient and discriminant $16c^2-16c+1\le -23/9<0$. On $d<z<1$, put $t=1-z$ and $v=b-u$ to obtain

$$\rho(z)\ \propto\ t^{3/2}I(t),\qquad I(t)=\int_0^1\frac{(1-r)(2a-t(1-r))}{\sqrt{r(2b-tr)}}\,\mathrm{d}r.$$

The logarithmic derivative of this integrand in t is $-(1-r)/(2a-t(1-r))+r/(2(2b-tr))$. Thus

$$-I'(t)=\int_0^1\frac{(1-r)(2a-t(1-r))}{\sqrt{r(2b-tr)}}\left[\frac{1-r}{2a-t(1-r)}-\frac{r}{2(2b-tr)}\right]\mathrm{d}r$$
$$\le\frac{I(t)}{2a-t}.$$

For this bound, discard the negative term in brackets and use $(1-r)/(2a-t(1-r))\le 1/(2a-t)$. It follows that $-I'/I\le 1/(2a-t)=1/(d+z)$. The logarithmic derivative of the ratio is therefore

at most $1/(d+z) - 1/z - 3/(2(1+z)) < 0$. These differentiations hold for $t < 2b$: on every compact subinterval of this range, the integrand and its t derivative are bounded by a constant times $r^{-1/2}$. For $0 < t \le 2b$, the integrand defining $I(t)$ is at most $(2a/\sqrt{2b})\sqrt{(1-r)/r}$, an integrable function on $(0,1)$. Dominated convergence therefore gives continuity at $t = 2b$. Thus the ratio decreases across both intervals.

For $(6,1)$ we may also include the limiting endpoint $b = 0$. At that endpoint, $\rho(z)$ is a positive constant times $(1-z^2)^{3/2}$, so the ratio in (C.1) is a positive constant times $1/z$. It therefore decreases. Now assume $0 < b \le 1/7$. The two terms in $\mathcal{I}_1$ give

$$\rho(z) \;\propto\; [a^2 - (z-b)^2]^{3/2} + [a^2 - (z+b)^2]_+^{3/2}.$$

After division by $z(1-z^2)^{3/2}$, this is proportional to

$$g_+(z) + g_-(z), \qquad g_+(z) = \frac{1}{z}\left(1 - \frac{2b}{1+z}\right)^{3/2}, \quad g_-(z) = \frac{1}{z}\left(1 - \frac{2b}{1-z}\right)_+^{3/2}.$$

On $0 < z < 1 - 2b$,

$$(\log g_-)' = -\frac{1}{z} - \frac{3b}{(1-z)(1-z-2b)} < 0.$$

Thus g_- decreases, vanishes continuously at $1-2b$, and equals zero afterwards. For the first term,

$$(\log g_+)' = -\frac{1}{z} + \frac{3b}{(1+z)(1+z-2b)} < 0,$$

because $(1+z)(1+z-2b) - 3bz = z^2 + (2-5b)z + 1 - 2b > 0$ for $0 < b \le 1/7$. This proves (C.1) in all three cases. □

Appendix D. Exact arithmetic for the planar threshold bounds

The following finite calculation checks (3.13) and all four positive margins in Lemma 3.4. It uses only integer arithmetic and rational fractions. The product bounds and the remainder estimates are proved in that lemma. The function `beta_interval` evaluates the two bounds in (3.12) with $M = 300$; its inputs satisfy $0 < x < 150$, as required by $2x/M < 1$. The functions `S` and `A` are the cosine and arctangent polynomials used in the same proof. Each row lists $10^4 s$ and the six integer endpoints in (3.13). The variables `L` and `U` are the resulting lower and upper bounds for $D_2(s)$; `dL` and `dU` are the two positive margins from (3.11).

```
from fractions import Fraction as F
from math import factorial

Q = 10**6

def beta_interval(x):
    assert 0 < x < 150
    p = F(1)
    for n in range(1, 301):
        p *= 1 - x*x/(n+x)**2
    z = x/150
    ell = x/2*(z-z*z/2+z**3/3)
    u = x*x/(301+x)
    return (2*p/x*(1-ell+ell**2/2-ell**3/6),
            2*p/x*(1-u+u*u/2))

def S(t, j):
    return sum(F((-1)**k)*t**(2*k)/factorial(2*k)
               for k in range(j+1))

def A(x, j):
    return sum(F((-1)**k)*x**(2*k+1)/(2*k+1)
               for k in range(j+1))
```

```
p = F("3.14159265358979323846264338327950288")
P = p + F(1, 10**35)
assert (p < 16*A(F(1,5),25)-4*A(F(1,239),8)
          < 16*A(F(1,5),26)-4*A(F(1,239),7) < P)

rows = [
    (9167, 422892, 422902, 1776037, 1776048, 130474, 130475),
    (9168, 422854, 422865, 1775936, 1775948, 130318, 130319)
]
targets = [(F(-13,10000), F(-11,10000)),
           (F(25,10000), F(27,10000))]
margins = [(F(10668871,147106725000),
            F(31247631781,1250397579000000)),
           (F(2078636269,39030540500000),
            F(1422091741,39030241000000))]

for row, (left, right), (dL, dU) in zip(rows, targets, margins):
    n, l1, u1, l2, u2, lc, uc = row
    s = F(n, 10000)
    assert 0 < p*s/2 < P*s/2 < F(3,2)
    for x, l, u in [(1+s/2, l1, u1), ((2+s)/4, l2, u2)]:
        low, up = beta_interval(x)
        assert F(l,Q) < low <= up < F(u,Q)
    assert 0 < F(lc,Q) < S(P*s/2,7)
    assert S(p*s/2,8) < F(uc,Q) < 1
    L = F(l1,Q)*(F(Q,uc)-1)-2/(1+s)-F(u2,Q)
    U = F(u1,Q)*(F(Q,lc)-1)-2/(1+s)-F(l2,Q)
    assert L < U
    assert L-left == dL > 0
    assert right-U == dU > 0
```

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